EVEN–ODD REFLECTION COMPRESSIONS OF THE LI–WEIL KERNEL: THE EVEN-FAMILY RH ROUTE, INVERSE DYNAMICS, AND DETERMINANT GEOMETRY

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ABSTRACT. We study the even and odd reflection compressions $\mathsf{E}_N = (\lambda_{m+n} - \lambda_{|m-n|})$ and $\mathsf{O}_N = (\lambda_{m+n-1} - \lambda_{|m-n|})$ obtained by reflection-antisymmetrizing Lagarias's bilateral Li–Weil Gram kernel. The underlying Li test class and Weil scalar product are classical; the matrices $\mathsf{E}_N, \mathsf{O}_N$ are formed here by the explicit antisymmetric changes of basis stated below and are then analyzed in those coordinates. Unconditionally they inherit Toeplitz–Hankel form, discrete wave equations, rank-two displacement, exact Schur recursions, and a closed two-generator inverse dynamics. Under RH they become strictly positive sine Gram matrices and admit Chebyshev moment, Hankel, Christoffel–Darboux, Jacobi, Cauchy–Binet, and determinantal-process descriptions.

A central point of the present version is that the odd family is *not needed* for the implication to the Riemann hypothesis. The even family alone satisfies

$$\mathrm{RH} \iff \mathsf{E}_N \succeq 0 \text{ for every } N \iff \mathsf{E}_N \succ 0 \text{ for every } N.$$

Thus the proof problem may be formulated entirely as propagation of the even Schur pivots $p_N = \lambda_{2N} - b_N^T \mathsf{E}_{N-1}^{-1} b_N$. The local LDL^T dynamics already reduces that propagation to one explicit 2×2 cell inequality.

We therefore separate two proof routes. Route A works directly with E_N: the local reflection dynamics, the prime–archimedean normal form, and an exact integration-by-parts formula reduce positivity to a signed prime-number-theorem discrepancy against an explicit structured polynomial test. Route B is organized around E_N^{-1}: inversion preserves exact displacement rank two, nesting adds a rank-one update, the inverse is the coefficient matrix of the Christoffel–Darboux reproducing kernel, and the unique Schur residual direction produces an adaptive prime–continuum test. Either the local-cell inequality or the adaptive inverse inequality, proved for every N, would imply RH.

Finally, the de Bruijn–Newman heat flow supplies a falsification filter for proof mechanisms. Rodgers–Tao proved $\Lambda \geq 0$; hence every negative heat time lies outside the all-real-zero region. The algebraic reflection, displacement, Schur, and inverse identities survive the heat deformation, whereas the Euler-product/prime representation does not. Consequently no argument using only the heat-flow-invariant matrix structure can supply the missing sign. Within the present framework the required zeta-specific input must enter through the arithmetic prime side. All finite computations are explicitly separated from theorem statements.

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1. Introduction

The Riemann hypothesis asserts that the nontrivial zeros of the Riemann zeta-function lie on the critical line $\Re s = 1/2$; standard accounts include [28, 17, 39, 16, 2]. Li's criterion replaces the location of the zeros by positivity of a scalar sequence $\{\lambda_n\}_{n\geq 1}$ [26]. Bombieri and Lagarias connected Li's coefficients with the Guinand–Weil explicit formula and with positivity of Weil's distribution [47, 3]. Lagarias later placed the Li coefficients inside an exact Hermitian Weil scalar product on a rational test-function class and proved, for the special functions

$$G_k(s) := 1 - \left(1 - \frac{1}{s}\right)^k, \qquad k \in \mathbb{Z}, \tag{1.1}$$

the identity

$$\langle G_r, G_s\rangle_W = \lambda_r + \lambda_{-s} - \lambda_{r-s}. \tag{1.2}$$

For the Riemann zeta-function $\lambda_{-n} = \lambda_n$, so (1.2) is the bilateral Li–Weil kernel [25, Theorem 3.1]. Bombieri studied Weil's quadratic functional directly in a different analytic framework [4]. Suzuki has developed further Hilbert-space and moment realizations of the same positivity framework [32, 33, 35].

The present paper applies a specific finite reflection-antisymmetrization to the bilateral Li–Weil kernel (1.2), which is taken directly from Lagarias [25, Theorem 3.1]. More precisely, the even test vectors are

$$F_n^E = \frac{G_n - G_{-n}}{\sqrt{2}},$$

and the odd vectors are the corresponding half-shifted reflections $F_n^O = (G_n - G_{1-n})/\sqrt{2}$. Taking their Lagarias–Weil Gram matrices produces

$$\mathsf{E}_N := \left(\lambda_{m+n} - \lambda_{|m-n|}\right)_{m,n=1}^{N}, \tag{1.3}$$

$$\mathsf{O}_N := \left(\lambda_{m+n-1} - \lambda_{|m-n|}\right)_{m,n=1}^{N}. \tag{1.4}$$

Thus the source object—Li's coefficients, the Weil form, and Lagarias's bilateral kernel—is classical [26, 3, 25]. The transformation to the two reflection matrices above, and the identities obtained from their native reflection coordinates, are the objects studied in this paper. They should not be confused with earlier matrix reformulations of RH, including Bombieri's finite Weil forms, the Hankel/Toeplitz eigenvalue constructions of Matiyasevich, the multiplicative matrices of Hilberdink, or Suzuki's screw- and moment-space realizations [4, 42, 43, 19, 20, 33]. Their diagonal splitting is

$$(\mathsf{E}_N)_{nn} = \lambda_{2n}, \qquad (\mathsf{O}_N)_{nn} = \lambda_{2n-1}. \tag{1.5}$$

Thus the pair contains the complete Li sequence while retaining nontrivial mixed entries.

The first structural point is exact and does not assume the Riemann hypothesis. If

$$F_n^E := \frac{G_n - G_{-n}}{\sqrt{2}}, \qquad F_n^O := \frac{G_n - G_{1-n}}{\sqrt{2}}, \tag{1.6}$$

then polarization of (1.2) gives

$$(\mathsf{E}_N)_{mn} = \langle F_m^E, F_n^E\rangle_W, \qquad (\mathsf{O}_N)_{mn} = \langle F_m^O, F_n^O\rangle_W. \tag{1.7}$$

Under RH, if $\rho = 1/2 + i\gamma$ and

$$1 - \frac{1}{\rho} = \mathrm{e}^{i\theta_\gamma}, \tag{1.8}$$

then these exact Weil Gram matrices become the positive sine Gram matrices

$$(\mathsf{E}_N)_{mn} = 4\sum_{\gamma>0}\sin(m\theta_\gamma)\sin(n\theta_\gamma), \tag{1.9}$$

$$(\mathsf{O}_N)_{mn} = 4\sum_{\gamma>0}\sin\big((m-\tfrac{1}{2})\theta_\gamma\big)\sin\big((n-\tfrac{1}{2})\theta_\gamma\big), \tag{1.10}$$

where zeros are counted with multiplicity. These identities imply strict positivity of every finite truncation. Conversely, positivity of all truncations recovers the even and odd Li inequalities from the diagonals and therefore recovers Li's criterion.

The remainder of the paper develops consequences of this reflection pair. Native coordinates give Toeplitz–Hankel matrices and low-rank displacement identities. Under RH, the zero-side sine representations become moment Gram matrices after Chebyshev changes of basis; the two measures are linked by a Christoffel transform at -1. Their determinants have positive Cauchy–Binet–Vandermonde expansions. We first prove an abstract endpoint-exponent law: for a positive discrete Gram measure whose support satisfies $1-x(t)\asymp t^{-\beta}$ and only coarse logarithmic node and separation conditions, the determinant obeys $\log D_N = -(\beta+o(1))N^2\log N$. The Li–Cayley transform has $\beta=2$, hence

$$\log\det\mathsf{E}_N = -(2+o(1))N^2\log N, \qquad \log\det\mathsf{O}_N = -(2+o(1))N^2\log N. \tag{1.11}$$

The coefficient -2 is therefore an endpoint-geometry constant, not a record of the logarithmic refinement in the Riemann–von Mangoldt density. The generic occurrence of $N^2\log N$ decay in discrete Hankel determinants has precedents in the Dirichlet-series setting of Monien and Haynes–Zudilin [27, 23]; the distinction here is the Li–Weil reflection measure and the endpoint-exponent mechanism. At this order no pair-correlation input enters. Schur complements are residual Gram energies, and inverse matrices represent Christoffel–Darboux kernels. Finally, the associated *infinite* Jacobi operators are unitarily equivalent to multiplication by

$$x_\gamma := \frac{4\gamma^2-1}{4\gamma^2+1} \tag{1.12}$$

on an atomic L^2 space. Their spectrum therefore contains the transformed distinct ordinates and the accumulation point 1. This is an infinite moment reconstruction from the full Li data; the eigenvalues of a finite Jacobi truncation are Gaussian-quadrature nodes, not exact zeta-zero ordinates.

A complementary route, developed in Section 26, separates the completed zeta function into its archimedean factor and the analytic arithmetic factor $(s-1)\zeta(s)$. In the Cayley variable $s=1/(1-z)$ the latter has coefficients Z_k with $\lambda_k = A_k + Z_k$. The arithmetic coefficients admit an exact regularized von Mangoldt–Laguerre formula, and reflection removes the prime main term so that the prime matrix pairs a Laguerre-difference kernel directly with $\psi(t)-t$. The associated scalar root-growth criterion RH $\Longleftrightarrow \limsup|Z_k|^{1/k}\le 1$ is useful diagnostically but belongs to the classical Keiper–Li singularity mechanism after factorization. The extra structure specific to the reflection compression appears only before scalarization: quadratic tests become Fourier transforms of nonnegative sine squares, each prime atom has rank-two displacement/Hankel structure, and first differences expose a contractive matrix-valued cosine family together with an explicit positive decomposition of the archimedean matrix up to one scalar negative constant.

Three finite-N structures specific to the reflection pair are central below. The bordering columns $\mathsf{b}_m^{(N)} = \lambda_{m+N} - \lambda_{|m-N|}$ satisfy, unconditionally, the same discrete wave equation in N as the rows do in m; together with the rank-two displacement this closes the Schur update and yields an exact local recurrence for consecutive pivot ratios in terms of the first two subdiagonals of the LDL^T factor. Independently, the Christoffel bridge of Corollary 16.18 normalizes as $\det\mathsf{E}_N = 4^N F_N \det\mathsf{O}_N$ and factorises the ratio of the even and odd pivot sequences as $\kappa_N = 4F_N/F_{N-1}$. All numerical positivity

statements are based on the arithmetic reconstruction of the Li coefficients in Subsection 26.2, which uses Stieltjes constants and no zero of ζ.

A point not visible from any single Vandermonde summand is how the *full* Cauchy–Binet determinant distributes its mass among zero subsets. Section 11 resolves this at finite N. After normalizing the nonnegative Cauchy–Binet terms one obtains a canonical probability measure on N-element subsets of positive zero ordinates. It is a rank-N projection determinantal process, and its marginal inclusion probabilities are

$$\ell_{N,\gamma}^{E,O} = (a_{N,\gamma}^{E,O})^T (\mathsf{E}_N \text{ or } \mathsf{O}_N)^{-1} a_{N,\gamma}^{E,O}. \tag{1.13}$$

These quantities have a direct matrix meaning: removing the atom at γ changes the determinant by the exact factor $1 - \ell_{N,\gamma}^{E,O}$. Thus the question of which zeros actually control the determinant is encoded by the matrix itself, without replacing the full Cauchy–Binet sum by one selected Vandermonde term.

Remark 1.1 (Exact source ledger). The construction starts from published material, and the source is recorded explicitly rather than absorbed into the notation.

(a) The coefficients λ_n and their positivity criterion are Li's [26]; the closely related power-series viewpoint predates Li in Keiper [22].

(b) The underlying explicit-formula positivity principle is Weil's quadratic criterion [47]. The general multiset extension of Li's criterion, its relation with the Guinand–Weil explicit formula, and the inverse-Mellin Li test functions are taken from Bombieri–Lagarias [3].

(c) The rational test functions $G_n(s) = 1 - (1 - 1/s)^n$, the Li test class, and especially the exact identity

$$\langle G_n, G_m \rangle_W = \lambda_n + \lambda_{-m} - \lambda_{n-m}$$

are taken directly from Lagarias [25, Section 3, Theorem 3.1]. In other words, the parent kernel used in this paper is copied, up to notation, from Lagarias's Theorem 3.1; it is not derived here. This theorem is the immediate source from which the matrices in this paper are formed.

(d) The even matrix is obtained by the elementary change of test basis

$$G_n \longmapsto F_n^E := \frac{G_n - G_{-n}}{\sqrt{2}}$$

and then taking the Lagarias–Weil Gram matrix. The odd matrix uses $F_n^O = (G_n - G_{1-n})/\sqrt{2}$. Thus E_N and O_N are not independent replacements for the Li–Weil theory: they are finite Gram matrices produced by these explicit linear transformations of Lagarias's published kernel. The present paper claims only the identities proved for these transformed coordinates; it does not claim priority for the underlying Li/Weil kernel, and it does not assert that the elementary antisymmetrization itself has never appeared elsewhere.

(e) The associated Laguerre polynomial in the Li test function is also not new. Lagarias records that the inverse-Mellin functions were explicitly determined in Bombieri–Lagarias and identifies the corresponding polynomial with the associated Laguerre family; see [25, Appendix A] and [3]. Coffey developed an explicit theta–Laguerre formulation of the Li/Keiper constants [13]. The scalar decomposition $\lambda_n = A_n + Z_n$ used below is obtained directly by splitting $\log \xi(s)$ into the archimedean factor and $\log((s-1)\zeta(s))$ before taking Cayley coefficients; no priority is claimed for that scalar factorization. What is derived here is the cancellation and matrix structure that appear after taking the reflection differences.

(f) Cauchy–Binet, Hankel moment theory, Christoffel transforms, Christoffel–Darboux kernels, Jacobi matrices, Darboux factorization, and projection determinantal processes are standard tools; see [12, 36, 18]. They are not claimed as new mechanisms here.

Other approaches include Bombieri's work on Weil's quadratic functional and Roesler's eigenvalue problems [4, 29, 30]; Matiyasevich's Taylor-coefficient Toeplitz/Hankel matrices [42, 43]; Hilberdink's multiplicative Toeplitz matrices [19, 20]; and Suzuki's screw/model-space constructions [32, 33].

2. The xi-function and Li coefficients

Define the completed zeta-function

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s). \tag{2.1}$$

It is entire of order one and satisfies

$$\xi(s) = \xi(1-s) = \overline{\xi(\overline{s})}. \tag{2.2}$$

The standard analytic theory may be found in [17, 39, 2].

Definition 2.1 (Li coefficients)**.** For $n \geq 1$, define

$$\lambda_n = \frac{1}{(n-1)!}\,\frac{\mathrm{d}^n}{\mathrm{d}s^n}\left[s^{n-1}\log\xi(s)\right]\Big|_{s=1}, \qquad \lambda_0 := 0. \tag{2.3}$$

Equivalently, with the symmetric limiting convention over nontrivial zeros,

$$\lambda_n = \sum_{\rho}\left[1-\left(1-\frac{1}{\rho}\right)^n\right]. \tag{2.4}$$

These numbers are also called the Keiper–Li coefficients: the power-series expansion of $\log\xi$ underlying (2.3) and its high-precision evaluation were studied by Keiper [22] before Li's positivity criterion appeared.

For the zeta xi-function the zero symmetry implies an integer-index extension

$$\lambda_{-n} = \lambda_n \qquad (n \geq 1), \tag{2.5}$$

as recorded, for example, in [25, Eq. (1.2)]. This identity is unconditional and will be used whenever differences of integer indices occur.

Theorem 2.2 (Li)**.** *The Riemann hypothesis is equivalent to*

$$\lambda_n \geq 0 \qquad \textit{for every } n \geq 1. \tag{2.6}$$

Proof. This is Li's positivity criterion [26]; see also [3, 25]. □

Define

$$R_\xi := \min\left\{1, \inf_{\rho}\left|1-\frac{1}{\rho}\right|\right\}. \tag{2.7}$$

The Li generating identity is the formal power-series identity

$$\Lambda(z) := \sum_{n\geq 1}\lambda_n z^n = \frac{z}{(1-z)^2}\,\frac{\xi'}{\xi}\left(\frac{1}{1-z}\right). \tag{2.8}$$

Analytically, (2.8) holds in the disk $|z| < R_\xi$. Under RH, every nontrivial zero satisfies $|1-1/\rho| = 1$, and therefore $R_\xi = 1$. If RH fails, a zero with real part $> 1/2$ produces a singularity with $|1-1/\rho| < 1$, so the unconditional radius must not be stated as 1.

Under RH we may parameterize each zero in the upper half-plane by

$$\rho = \frac{1}{2} + i\gamma, \qquad 1-\frac{1}{\rho} = \mathrm{e}^{i\theta_\gamma}, \qquad 0 < \theta_\gamma < \pi. \tag{2.9}$$

A direct calculation gives

$$\cos\theta_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}, \tag{2.10}$$

$$\sin\theta_\gamma = \frac{4\gamma}{4\gamma^2 + 1}. \tag{2.11}$$

Pairing conjugate zeros in (2.4) yields the absolutely convergent representation

$$\lambda_n = 2\sum_{\gamma>0} \left(1 - \cos(n\theta_\gamma)\right). \tag{2.12}$$

For fixed n, the summands are $O_n(\gamma^{-2})$.

3. Definition of the reflection matrices

Definition 3.1 (Even and odd reflection matrices)**.** For $N \geq 1$, define

$$\mathsf{E}_N := (e_{mn})_{m,n=1}^N, \qquad e_{mn} := \lambda_{m+n} - \lambda_{|m-n|}, \tag{3.1}$$

$$\mathsf{O}_N := (o_{mn})_{m,n=1}^N, \qquad o_{mn} := \lambda_{m+n-1} - \lambda_{|m-n|}. \tag{3.2}$$

Both are real symmetric. Their first 4×4 truncations are

$$\mathsf{E}_4 = \begin{pmatrix} \lambda_2 & \lambda_3 - \lambda_1 & \lambda_4 - \lambda_2 & \lambda_5 - \lambda_3 \\ \lambda_3 - \lambda_1 & \lambda_4 & \lambda_5 - \lambda_1 & \lambda_6 - \lambda_2 \\ \lambda_4 - \lambda_2 & \lambda_5 - \lambda_1 & \lambda_6 & \lambda_7 - \lambda_1 \\ \lambda_5 - \lambda_3 & \lambda_6 - \lambda_2 & \lambda_7 - \lambda_1 & \lambda_8 \end{pmatrix}, \tag{3.3}$$

and

$$\mathsf{O}_4 = \begin{pmatrix} \lambda_1 & \lambda_2 - \lambda_1 & \lambda_3 - \lambda_2 & \lambda_4 - \lambda_3 \\ \lambda_2 - \lambda_1 & \lambda_3 & \lambda_4 - \lambda_1 & \lambda_5 - \lambda_2 \\ \lambda_3 - \lambda_2 & \lambda_4 - \lambda_1 & \lambda_5 & \lambda_6 - \lambda_1 \\ \lambda_4 - \lambda_3 & \lambda_5 - \lambda_2 & \lambda_6 - \lambda_1 & \lambda_7 \end{pmatrix}. \tag{3.4}$$

Proposition 3.2 (Toeplitz–Hankel decomposition)**.** *Let*

$$\mathsf{T}_N := \left(\lambda_{|m-n|}\right)_{m,n=1}^N, \tag{3.5}$$

$$\mathsf{H}_N^{(0)} := \left(\lambda_{m+n}\right)_{m,n=1}^N, \qquad \mathsf{H}_N^{(-1)} := \left(\lambda_{m+n-1}\right)_{m,n=1}^N. \tag{3.6}$$

Then

$$\mathsf{E}_N = \mathsf{H}_N^{(0)} - \mathsf{T}_N, \qquad \mathsf{O}_N = \mathsf{H}_N^{(-1)} - \mathsf{T}_N. \tag{3.7}$$

Moreover,

$$\mathsf{E}_N - \mathsf{O}_N = \left(\lambda_{m+n} - \lambda_{m+n-1}\right)_{m,n=1}^N, \tag{3.8}$$

which is a pure Hankel matrix.

Proof. All statements follow directly from (3.1) and (3.2). □

4. Algebraic reconstruction and information content

The two arrays contain substantial algebraic redundancy. This can be described exactly without assuming RH.

Proposition 4.1 (Reconstruction from the infinite first rows)**.** *Since*

$$o_{1j} = \lambda_j - \lambda_{j-1}, \qquad j \geq 1, \tag{4.1}$$

one has

$$\lambda_n = \sum_{j=1}^{n} o_{1j} \qquad (n \geq 1), \tag{4.2}$$

and, for all $m, n \geq 1$*,*

$$o_{mn} = \sum_{j=|m-n|+1}^{m+n-1} o_{1j}. \tag{4.3}$$

If $r := \min(m, n)$*, then*

$$e_{mn} = \sum_{j=0}^{r-1} e_{1,\,|m-n|+1+2j}. \tag{4.4}$$

Finally,

$$e_{mn} = \sum_{j=|m-n|+1}^{m+n} o_{1j}. \tag{4.5}$$

Thus the semi-infinite odd array determines the complete Li sequence and the semi-infinite even array.

Proof. Equation (4.2) is telescoping. Substituting it into the definition of o_{mn} gives (4.3). Since

$$e_{1k} = \lambda_{k+1} - \lambda_{k-1}, \tag{4.6}$$

the right side of (4.4) telescopes from $\lambda_{|m-n|}$ to λ_{m+n}. Equation (4.5) is the same telescoping identity written in the unit-step differences o_{1j}. □

Remark 4.2 (Finite truncations versus infinite first rows). Proposition 4.1 concerns the first row of the *semi-infinite* arrays. The leading block O_N is not determined by its own N displayed first-row entries: reconstructing all of O_N from (4.3) requires the prefix o_{1j} through $j = 2N - 1$. Likewise, reconstructing E_N from its infinite first row requires e_{1j} through $j = 2N - 1$. Reconstructing E_N from the odd first row by (4.5) requires o_{1j} through $j = 2N$.

Proposition 4.3 (Injectivity of the odd array and one-dimensional kernel of the even array)**.** *On real sequences* $(\lambda_n)_{n\geq 0}$ *with* $\lambda_0 = 0$*, the map*

$$(\lambda_n)_{n\geq 0} \longmapsto (o_{mn})_{m,n\geq 1} \tag{4.7}$$

is injective. The kernel of

$$(\lambda_n)_{n\geq 0} \longmapsto (e_{mn})_{m,n\geq 1} \tag{4.8}$$

consists exactly of the one-parameter family

$$\lambda_{2k} = 0, \qquad \lambda_{2k+1} = c \qquad (k \geq 0), \tag{4.9}$$

with $c \in \mathbb{R}$*. Consequently* (E, λ_1) *determines the whole Li sequence and hence* O*.*

Proof. If the odd array vanishes, then its first row gives $\lambda_j - \lambda_{j-1} = 0$ for every j, and $\lambda_0 = 0$ gives $\lambda_j = 0$. Thus the odd map is injective.

If the even array vanishes, its diagonal gives $\lambda_{2n} = 0$ for all $n \geq 1$. Its entries $e_{1,2k} = 0$ give $\lambda_{2k+1} = \lambda_{2k-1}$, so every odd term equals λ_1. Conversely, a sequence of the form (4.9) gives $e_{mn} = 0$ because $m + n$ and $m - n$ have the same parity. This proves the kernel statement. □

Proposition 4.4 (Reflection boundary conditions)**.** *Using the even extension $\lambda_{-n} = \lambda_n$, the bilateral kernels satisfy*

$$e(-m, n) = -e(m, n), \tag{4.10}$$

$$o(1 - m, n) = -o(m, n). \tag{4.11}$$

Hence

$$e(0, n) = 0, \qquad o(0, n) = -o(1, n). \tag{4.12}$$

Together with symmetry in m, n and the discrete wave equation of Section 7, these are the natural reflection boundary conditions at 0 and 1/2.

Proof. The identities follow by direct substitution in $e(m, n) = \lambda_{m+n} - \lambda_{m-n}$ and $o(m, n) = \lambda_{m+n-1} - \lambda_{m-n}$, using $\lambda_{-k} = \lambda_k$. □

5. Exact Weil realization and critical-line Gram representations

Lagarias defines on the Li test-function class a Hermitian Weil scalar product for which the functions (1.1) satisfy (1.2); see [25, Section 3, Theorem 3.1]. In particular, for the Riemann zeta-function,

$$\langle G_r, G_s \rangle_W = \lambda_r + \lambda_s - \lambda_{|r-s|}, \qquad r, s \in \mathbb{Z}, \tag{5.1}$$

where (2.5) has been used.

Theorem 5.1 (Exact reflected Weil Gram matrices)**.** *For every $m, n \geq 1$, without assuming* RH*,*

$$e_{mn} = \langle F_m^E, F_n^E \rangle_W, \tag{5.2}$$

$$o_{mn} = \langle F_m^O, F_n^O \rangle_W, \tag{5.3}$$

where F_n^E, F_n^O are given by (1.6)*.*

Proof. Using (5.1) and bilinearity/polarization,

$$\langle F_m^E, F_n^E \rangle_W = \frac{1}{2}\big[K(m, n) - K(m, -n) - K(-m, n) + K(-m, -n)\big] \tag{5.4}$$

$$= \lambda_{m+n} - \lambda_{|m-n|} = e_{mn}, \tag{5.5}$$

where $K(r, s) := \lambda_r + \lambda_s - \lambda_{|r-s|}$. Similarly,

$$\langle F_m^O, F_n^O \rangle_W = \frac{1}{2}\big[K(m, n) - K(m, 1 - n) - K(1 - m, n) + K(1 - m, 1 - n)\big] \tag{5.6}$$

$$= \lambda_{m+n-1} - \lambda_{|m-n|} = o_{mn}. \tag{5.7}$$

□

The same reflection is transparent in the Cayley coordinate

$$q(s) := 1 - \frac{1}{s}. \tag{5.8}$$

Indeed,

$$F_n^E(s) = \frac{q(s)^{-n} - q(s)^n}{\sqrt{2}}, \tag{5.9}$$

$$F_n^O(s) = \frac{q(s)^{1-n} - q(s)^n}{\sqrt{2}}. \tag{5.10}$$

Theorem 5.2 (Integer and half-integer sine Gram representations)**.** *Assume* RH*. Enumerate the positive ordinates of the nontrivial zeros with multiplicity. Then for every $m, n \geq 1$,*

$$e_{mn} = 4 \sum_{\gamma>0} \sin(m\theta_\gamma) \sin(n\theta_\gamma), \tag{5.11}$$

$$o_{mn} = 4 \sum_{\gamma>0} \sin\big((m - \tfrac{1}{2})\theta_\gamma\big) \sin\big((n - \tfrac{1}{2})\theta_\gamma\big). \tag{5.12}$$

The two series converge absolutely for fixed m, n.

Proof. Under RH, $q(\rho) = \mathrm{e}^{i\theta_\gamma}$ for $\rho = 1/2 + i\gamma$. Equations (5.9)–(5.10) become

$$F_n^E(\rho) = -\sqrt{2}\, i \sin(n\theta_\gamma), \tag{5.13}$$

$$F_n^O(\rho) = -\sqrt{2}\, i\mathrm{e}^{i\theta_\gamma/2} \sin\big((n - \tfrac{1}{2})\theta_\gamma\big). \tag{5.14}$$

For the numerical normalization, it is safest to use the already established Li cosine formula (2.12). Subtracting the terms with indices $m + n$ and $|m - n|$ and applying

$$\cos((m-n)\theta) - \cos((m+n)\theta) = 2\sin(m\theta)\sin(n\theta) \tag{5.15}$$

gives (5.11). Likewise,

$$\cos((m-n)\theta) - \cos((m+n-1)\theta) = 2\sin\big((m - \tfrac{1}{2})\theta\big)\sin\big((n - \tfrac{1}{2})\theta\big) \tag{5.16}$$

gives (5.12). Since $\theta_\gamma = O(\gamma^{-1})$, every summand is $O_{m,n}(\gamma^{-2})$, and the zero-counting estimate gives absolute convergence. □

Remark 5.3 (Negative definite sequences and screw functions)*.* Representation (2.12) says that, under RH, the extended sequence λ is a *negative definite* function on $\mathbb{Z}$ in the sense of harmonic analysis on groups: it is real, even, vanishes at 0, and has the Lévy–Khinchine form $\int (1 - \cos n\theta)\, \mathrm{d}\sigma(\theta)$ with σ the atomic measure carried by the angles θ_γ. For any such function the kernel $\lambda_r + \lambda_s - \lambda_{r-s}$ is positive definite, and the passage from that kernel to the sine Gram matrices of Theorem 5.2 is the standard consequence of the Lévy–Khinchine representation [31, 1]. Conversely, a negative definite λ with $\lambda_0 = 0$ satisfies $\lambda_n \geq 0$ for all n, so Lagarias's kernel identity (5.1) amounts to the statement that RH is equivalent to negative definiteness of the Li sequence. The continuous counterpart of this circle of ideas is Krein's theory of screw functions, developed for the Riemann zeta-function by Suzuki [33, 34]. The positivity mechanism exploited below is therefore classical; what is specific here is the pair of reflection compressions of Definition 3.1 and the structure theory derived from it.

Theorem 5.4 (Li–Weil reflection criterion)**.** *The following are equivalent:*

(i) RH*;*
(ii) $\mathsf{E}_N \succeq 0$ *and* $\mathsf{O}_N \succeq 0$ *for every* $N \geq 1$*;*
(iii) $\mathsf{E}_N \succ 0$ *and* $\mathsf{O}_N \succ 0$ *for every* $N \geq 1$*;*
(iv) $\det \mathsf{E}_N > 0$ *and* $\det \mathsf{O}_N > 0$ *for every* $N \geq 1$*.*

Proof. Assume RH. Theorem 5.2 gives positive semidefiniteness. If $c^T \mathsf{E}_N c = 0$, then

$$\sum_{m=1}^{N} c_m \sin(m\theta_\gamma) = 0 \qquad \text{for every distinct positive zero ordinate } \gamma. \tag{5.17}$$

There are infinitely many distinct ordinates and $\theta_\gamma \to 0$. The finite trigonometric polynomial in (5.17) is analytic in θ and therefore must vanish identically; hence $c = 0$. The half-integer system is handled in the same way. Thus (i) implies (iii), and (iii) implies (ii).

Assume (ii). For $N \geq n$ the diagonal entries satisfy

$$\lambda_{2n} = (\mathsf{E}_N)_{nn} \geq 0, \qquad \lambda_{2n-1} = (\mathsf{O}_N)_{nn} \geq 0. \tag{5.18}$$

Hence every Li coefficient is nonnegative and Theorem 2.2 gives RH. Finally, because the families are nested leading principal truncations, Sylvester's criterion gives the equivalence of (iii) and (iv). □

Theorem 5.5 (The even reflection family alone is an RH criterion)**.** *The following are equivalent:*

(i) RH*;*
(ii) $\mathsf{E}_N \succeq 0$ *for every* $N \geq 1$*;*
(iii) $\mathsf{E}_N \succ 0$ *for every* $N \geq 1$*.*

In particular, the odd family is not required for a proof of RH*.*

Proof. The implication $(i) \Rightarrow (iii) \Rightarrow (ii)$ is already contained in Theorem 5.4. Assume now that $\mathsf{E}_N \succeq 0$ for every N. Then

$$\lambda_{2N} = (\mathsf{E}_N)_{NN} \geq 0 \qquad (N \geq 1).$$

We reduce the converse to the general multiset theorem of Bombieri–Lagarias [3, Theorem 1 and its symmetric corollary]; this reduction is not a new Li-type criterion.

For every nontrivial zero ρ, put

$$q_\rho := 1 - \frac{1}{\rho}.$$

Zeros with $q_\rho^2 = 1$ are already on the critical line and contribute zero to every even Li coefficient, so omit them from the following transformed multiset. For the remaining zeros define

$$\sigma_\rho := \frac{1}{1 - q_\rho^2} = \frac{\rho^2}{2\rho - 1}. \tag{5.19}$$

Then

$$1 - \frac{1}{\sigma_\rho} = q_\rho^2, \qquad \lambda_{2n} = \sum_\rho \left[1 - \left(1 - \frac{1}{\sigma_\rho}\right)^n\right]. \tag{5.20}$$

Moreover $\sigma_\rho = \rho/2 + O(1)$ as $|\rho| \to \infty$, so the Bombieri–Lagarias summability hypothesis is inherited from the zeta zero multiset. The symmetries $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \rho$ become $\sigma \mapsto \bar{\sigma}$ and $\sigma \mapsto 1 - \sigma$, respectively. Hence the symmetric Bombieri–Lagarias criterion applied to the transformed multiset gives

$$\lambda_{2n} \geq 0 \text{ for every } n \quad \Longrightarrow \quad \Re\sigma_\rho = \frac{1}{2} \text{ for every } \rho.$$

But

$$\Re\sigma_\rho = \frac{1}{2} \quad \Longleftrightarrow \quad \left|1 - \frac{1}{\sigma_\rho}\right| = 1 \quad \Longleftrightarrow \quad |q_\rho| = 1 \quad \Longleftrightarrow \quad \Re\rho = \frac{1}{2}.$$

Thus every nontrivial zero lies on the critical line and RH follows. □

Remark 5.6 (Source of the even-only converse). The converse in Theorem 5.5 is an application of the general Bombieri–Lagarias multiset criterion after the elementary quadratic Cayley transform (5.19). The paper therefore claims no priority for the underlying positivity criterion. What is used later is only that the matrix diagonal exposes precisely these even Bombieri–Lagarias sums.

Remark 5.7 (Role of the odd family). The odd matrices remain useful because they supply the Christoffel/Darboux bridge, an independent pivot sequence, and a second moment system. None of those structures is logically necessary for the implication to RH: throughout the proof routes isolated below, the target may be taken to be $\mathsf{E}_N \succ 0$ for every N.

Corollary 5.8 (Full rank)**.** *Under* RH*,*

$$\operatorname{rank} \mathsf{E}_N = \operatorname{rank} \mathsf{O}_N = N \qquad (N \geq 1). \tag{5.21}$$

Proof. Under RH, Theorem 5.4 gives $\mathsf{E}_N \succ 0$ and $\mathsf{O}_N \succ 0$. A positive definite $N \times N$ matrix is nonsingular, hence has rank N. □

Corollary 5.9 (All principal minors)**.** *Under* RH*, every principal minor of* E_N *and* O_N *is strictly positive.*

Proof. A principal submatrix is the Gram matrix of a finite set of distinct integer, respectively half-integer, sine modes. The same analytic-zero argument as in Theorem 5.4 gives linear independence. □

6. Reflection geometry inside the Li test class

The exact matrix identities above can be phrased as reflections of Lagarias's bilateral Li-class Gram kernel. Define

$$\mathsf{K}(r,s) := \langle G_r, G_s \rangle_W = \lambda_r + \lambda_s - \lambda_{|r-s|}, \qquad r, s \in \mathbb{Z}. \tag{6.1}$$

Let

$$R_0(k) = -k, \qquad R_{1/2}(k) = 1 - k. \tag{6.2}$$

Then

$$R_{1/2} R_0(k) = k + 1, \tag{6.3}$$

so the two reflections generate the infinite dihedral action on the integer label set.

If $u_m = e_m - e_{-m}$ and $v_m = e_m - e_{1-m}$ in the finitely supported bilateral coordinate space, then

$$\frac{1}{2}\, u_m^* \mathsf{K} u_n = e_{mn}, \tag{6.4}$$

$$\frac{1}{2}\, v_m^* \mathsf{K} v_n = o_{mn}. \tag{6.5}$$

Thus E_N and O_N are exact congruence compressions of the known Lagarias Gram kernel. The point of the present coordinates is the resulting even–odd diagonal split, the Toeplitz–Hankel form, and the matrix structures developed below.

7. Discrete wave equations and displacement rank

The Toeplitz–Hankel structure has a useful equivalent formulation as a low-displacement-rank identity.

Definition 7.1 (Universal tridiagonal matrices)**.** Let $\mathsf{J}_E^{(N)}$ be the $N \times N$ symmetric tridiagonal matrix with zero diagonal and $1/2$ on both adjacent diagonals:

$$\mathsf{J}_E^{(N)} = \frac{1}{2} \begin{pmatrix} 0 & 1 & & & \\ 1 & 0 & 1 & & \\ & 1 & 0 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & 0 \end{pmatrix}. \tag{7.1}$$

Let $\mathcal{J}_O^{(N)}$ be the same matrix except for the upper-left entry $-1/2$:

$$\mathcal{J}_O^{(N)} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & & & \\ \frac{1}{2} & 0 & \frac{1}{2} & & \\ & \frac{1}{2} & 0 & \ddots & \\ & & \ddots & \ddots & \frac{1}{2} \\ & & & \frac{1}{2} & 0 \end{pmatrix}. \tag{7.2}$$

Proposition 7.2 (Discrete wave relation)**.** *For $m, n \geq 2$, both arrays satisfy*

$$a_{m+1,n} + a_{m-1,n} = a_{m,n+1} + a_{m,n-1}, \tag{7.3}$$

where $a = e$ or $a = o$. The boundary relations are exactly those encoded by J_E and $\mathcal{J}_O$.

Proof. For $e_{mn} = \lambda_{m+n} - \lambda_{|m-n|}$, the four terms in (7.3) agree after matching the sum and difference indices. The same calculation applies to o_{mn} because the sum index is shifted by one. The boundary $m = 1$ is described by the identities $xU_0 = U_1/2$ and $xP_0 = (P_1 - P_0)/2$ developed in Section 8. □

Theorem 7.3 (Displacement rank at most two)**.** *Let*

$$r_E^{(N)} := (e_{1,N+1}, \ldots, e_{N-1,N+1}, 0)^T, \tag{7.4}$$

$$r_O^{(N)} := (o_{1,N+1}, \ldots, o_{N-1,N+1}, 0)^T, \tag{7.5}$$

and let e_N denote the last standard basis vector of $\mathbb{R}^N$. Then

$$\mathsf{J}_E^{(N)}\mathsf{E}_N - \mathsf{E}_N\mathsf{J}_E^{(N)} = \frac{1}{2}\left(r_E^{(N)} e_N^T - e_N (r_E^{(N)})^T\right), \tag{7.6}$$

$$\mathcal{J}_O^{(N)}\mathsf{O}_N - \mathsf{O}_N\mathcal{J}_O^{(N)} = \frac{1}{2}\left(r_O^{(N)} e_N^T - e_N (r_O^{(N)})^T\right). \tag{7.7}$$

In particular, each displacement has rank at most two.

Proof. The semi-infinite arrays satisfy the commutator relation entrywise as a consequence of (7.3) and the boundary relation; no bounded-operator assertion is needed here. Finite truncation removes only the coupling from index N to $N+1$. The omitted term is precisely the outer-product defect in (7.6) and (7.7). The final component of each boundary vector has been set to zero because it cancels identically in the skew outer product. □

Corollary 7.4 (Displacement of the inverses)**.** *Whenever E_N and O_N are invertible,*

$$\mathsf{J}_E^{(N)}\mathsf{E}_N^{-1} - \mathsf{E}_N^{-1}\mathsf{J}_E^{(N)} = -\mathsf{E}_N^{-1}\left(\mathsf{J}_E^{(N)}\mathsf{E}_N - \mathsf{E}_N\mathsf{J}_E^{(N)}\right)\mathsf{E}_N^{-1}, \tag{7.8}$$

$$\mathcal{J}_O^{(N)}\mathsf{O}_N^{-1} - \mathsf{O}_N^{-1}\mathcal{J}_O^{(N)} = -\mathsf{O}_N^{-1}\left(\mathcal{J}_O^{(N)}\mathsf{O}_N - \mathsf{O}_N\mathcal{J}_O^{(N)}\right)\mathsf{O}_N^{-1}. \tag{7.9}$$

Hence the inverse matrices also have displacement rank at most two with respect to the same tridiagonal operators. More precisely, if A denotes either E_N or O_N and A is invertible, then

$$\operatorname{rank}(\mathsf{J}^{(N)}A^{-1} - A^{-1}\mathsf{J}^{(N)}) = \operatorname{rank}(\mathsf{J}^{(N)}A - A\mathsf{J}^{(N)}), \tag{7.10}$$

with the appropriate choice of $\mathsf{J}_E^{(N)}$ or $\mathcal{J}_O^{(N)}$. Thus inversion preserves the exact displacement rank, whether it is 0 or 2.

Proof. For an invertible matrix A,

$$\mathsf{J}A^{-1} - A^{-1}\mathsf{J} = -A^{-1}(\mathsf{J}A - A\mathsf{J})A^{-1},$$

by direct expansion. Since left and right multiplication by invertible matrices preserve rank, the rank equality follows as well. □

8. Chebyshev moment representations

Let U_n denote the Chebyshev polynomial of the second kind, normalized by

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}. \tag{8.1}$$

Define

$$P_n(x) := U_n(x) + U_{n-1}(x), \qquad U_{-1} := 0. \tag{8.2}$$

Then

$$P_{m-1}(\cos\theta) = \frac{\sin((m-\frac{1}{2})\theta)}{\sin(\theta/2)}. \tag{8.3}$$

Under RH, let m_γ denote the multiplicity of the zero $1/2 + i\gamma$ and set

$$x_\gamma := \cos\theta_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}. \tag{8.4}$$

Define finite positive Borel measures on $[-1, 1]$ by

$$\mu_E := 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2\theta_\gamma \, \delta_{x_\gamma}, \tag{8.5}$$

$$\mu_O := 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2(\theta_\gamma/2) \, \delta_{x_\gamma}. \tag{8.6}$$

These are finite measures because their total masses are λ_2 and λ_1, respectively; equivalently the weights are $O(m_\gamma \gamma^{-2})$ and the standard zero-counting estimate gives convergence.

Theorem 8.1 (Moment representation)**.** *Under* RH*,*

$$e_{mn} = \int_{-1}^{1} U_{m-1}(x) U_{n-1}(x) \, \mathrm{d}\mu_E(x), \tag{8.7}$$

$$o_{mn} = \int_{-1}^{1} P_{m-1}(x) P_{n-1}(x) \, \mathrm{d}\mu_O(x). \tag{8.8}$$

Moreover,

$$\mathrm{d}\mu_E(x) = 2(1+x) \, \mathrm{d}\mu_O(x). \tag{8.9}$$

Proof. We first verify that the integrals are well defined. For fixed m, n, the polynomials $U_{m-1}U_{n-1}$ and $P_{m-1}P_{n-1}$ are bounded on $[-1, 1]$, while μ_E and μ_O are finite positive measures. Thus both Gram integrals converge absolutely.

For the even family, using the atomic definition of μ_E and $U_{k-1}(\cos\theta) = \sin(k\theta)/\sin\theta$,

$$\int U_{m-1}(x) U_{n-1}(x) \, \mathrm{d}\mu_E(x) = 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2\theta_\gamma U_{m-1}(x_\gamma) U_{n-1}(x_\gamma) \tag{8.10}$$

$$= 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin(m\theta_\gamma) \sin(n\theta_\gamma). \tag{8.11}$$

Counting each ordinate according to multiplicity turns the last sum into the right side of (5.11), hence it equals e_{mn}. For the odd family, (8.3) gives

$$\int P_{m-1}(x)P_{n-1}(x)\,\mathrm{d}\mu_O(x) = 4\sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2(\theta_\gamma/2)P_{m-1}(x_\gamma)P_{n-1}(x_\gamma) \tag{8.12}$$

$$= 4\sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin((m-\tfrac{1}{2})\theta_\gamma)\sin((n-\tfrac{1}{2})\theta_\gamma), \tag{8.13}$$

which is (5.12).

It remains to compare the atom masses. At every support point $x_\gamma = \cos\theta_\gamma$,

$$\frac{4m_\gamma \sin^2\theta_\gamma}{4m_\gamma \sin^2(\theta_\gamma/2)} = \frac{\sin^2\theta_\gamma}{\sin^2(\theta_\gamma/2)} = 4\cos^2(\theta_\gamma/2) = 2(1+x_\gamma). \tag{8.14}$$

Thus the two atomic measures have the same support and their masses satisfy $\mu_E(\{x_\gamma\}) = 2(1+x_\gamma)\mu_O(\{x_\gamma\})$ at every atom, which is exactly (8.9). □

Remark 8.2 (Christoffel transform). Equation (8.9) says that the even measure is, up to the scalar factor 2, the Christoffel transform of the odd measure at the point -1. Here -1 is the left endpoint of the ambient interval $[-1,1]$ but lies outside the actual zeta support $\{x_\gamma\}\cup\{1\}$. Thus the two matrix families are linked at the level of measures, orthogonal polynomials, Jacobi matrices, determinants, and Cholesky pivots.

The weights can be written directly in terms of zero ordinates:

$$4\sin^2\theta_\gamma = \frac{64\gamma^2}{(4\gamma^2+1)^2}, \tag{8.15}$$

$$4\sin^2(\theta_\gamma/2) = \frac{4}{4\gamma^2+1}. \tag{8.16}$$

9. Changes of basis and hidden Hankel matrices

Write

$$X_N(x) := (1, x, \dots, x^{N-1})^T. \tag{9.1}$$

There exist explicit lower-triangular matrices $B_E^{(N)}$ and $B_O^{(N)}$ satisfying

$$\big(U_0(x),\dots,U_{N-1}(x)\big)^T = B_E^{(N)} X_N(x), \tag{9.2}$$

$$\big(P_0(x),\dots,P_{N-1}(x)\big)^T = B_O^{(N)} X_N(x). \tag{9.3}$$

The coefficients of U_n are explicitly

$$U_n(x) = \sum_{j=0}^{\lfloor n/2\rfloor} (-1)^j \binom{n-j}{j} (2x)^{n-2j}. \tag{9.4}$$

Thus both change-of-basis matrices have diagonal

$$1, 2, 2^2, \dots, 2^{N-1}, \tag{9.5}$$

and hence

$$\det B_E^{(N)} = \det B_O^{(N)} = 2^{N(N-1)/2}. \tag{9.6}$$

Define moments

$$\mu_k^E := \int_{-1}^{1} x^k\,\mathrm{d}\mu_E(x), \qquad \mu_k^O := \int_{-1}^{1} x^k\,\mathrm{d}\mu_O(x), \tag{9.7}$$

and Hankel matrices

$$H_N^E := (\mu_{i+j}^E)_{i,j=0}^{N-1}, \qquad H_N^O := (\mu_{i+j}^O)_{i,j=0}^{N-1}. \tag{9.8}$$

Theorem 9.1 (Congruence to moment Hankel matrices)**.** *Under* RH*,*

$$\mathsf{E}_N = B_E^{(N)} H_N^E (B_E^{(N)})^T, \tag{9.9}$$

$$\mathsf{O}_N = B_O^{(N)} H_N^O (B_O^{(N)})^T. \tag{9.10}$$

Consequently,

$$\det \mathsf{E}_N = 2^{N(N-1)} \det H_N^E, \tag{9.11}$$

$$\det \mathsf{O}_N = 2^{N(N-1)} \det H_N^O. \tag{9.12}$$

Proof. Let $X_N(x) = (1, x, \dots, x^{N-1})^T$. For the even family put $\Phi_E(x) = (U_0(x), \dots, U_{N-1}(x))^T = B_E^{(N)} X_N(x)$. Then the moment representation gives

$$\mathsf{E}_N = \int \Phi_E(x)\Phi_E(x)^T \, \mathrm{d}\mu_E(x) \tag{9.13}$$

$$= B_E^{(N)} \left(\int X_N(x) X_N(x)^T \, \mathrm{d}\mu_E(x) \right) (B_E^{(N)})^T \tag{9.14}$$

$$= B_E^{(N)} H_N^E (B_E^{(N)})^T. \tag{9.15}$$

Every entry is a finite integral, so pulling the fixed finite matrices through the integral is purely algebraic. The same argument with $\Phi_O = (P_0, \dots, P_{N-1})^T = B_O^{(N)} X_N$ gives the odd identity.

Taking determinants and using that the basis-change matrices are triangular with diagonal $1, 2, \dots, 2^{N-1}$,

$$\det B_E^{(N)} = \det B_O^{(N)} = 2^{N(N-1)/2}. \tag{9.16}$$

Hence

$$\det \mathsf{E}_N = (\det B_E^{(N)})^2 \det H_N^E = 2^{N(N-1)} \det H_N^E, \tag{9.17}$$

and identically for O_N. □

Corollary 9.2 (Moment relation)**.** *For every $k \geq 0$,*

$$\mu_k^E = 2(\mu_k^O + \mu_{k+1}^O). \tag{9.18}$$

Equivalently,

$$H_N^E = 2\left(H_N^O + H_N^{O,(1)}\right), \qquad H_N^{O,(1)} := (\mu_{i+j+1}^O)_{i,j=0}^{N-1}. \tag{9.19}$$

Proof. The measure identity (8.9) gives, for every $k \geq 0$,

$$\mu_k^E = \int x^k \, \mathrm{d}\mu_E(x) = 2 \int x^k (1 + x) \, \mathrm{d}\mu_O(x) = 2(\mu_k^O + \mu_{k+1}^O).$$

Applying this entrywise to the Hankel matrices gives (9.19). □

The first moments may be read directly from the original matrices:

$$\mu_0^E = \lambda_2, \qquad \mu_1^E = \frac{\lambda_3 - \lambda_1}{2}, \qquad \mu_2^E = \frac{\lambda_4}{4}, \tag{9.20}$$

$$\mu_0^O = \lambda_1, \qquad \mu_1^O = \frac{\lambda_2 - 2\lambda_1}{2}, \qquad \mu_2^O = \frac{\lambda_3 - 2\lambda_2 + 3\lambda_1}{4}. \tag{9.21}$$

10. Determinants and Vandermonde expansions

Enumerate the positive zero ordinates with multiplicity as $\gamma_1, \gamma_2, \ldots$, and write $\theta_j := \theta_{\gamma_j}$ and $x_j := x_{\gamma_j}$.

Theorem 10.1 (Cauchy–Binet determinant formulas)**.** *Assume* RH. *Then*

$$\det \mathsf{E}_N = 4^N \sum_{1\le j_1<\cdots<j_N} \Big(\det\big[\sin(m\theta_{j_k})\big]_{m,k=1}^{N}\Big)^2, \tag{10.1}$$

and

$$\det \mathsf{O}_N = 4^N \sum_{1\le j_1<\cdots<j_N} \Big(\det\big[\sin((m-\tfrac12)\theta_{j_k})\big]_{m,k=1}^{N}\Big)^2. \tag{10.2}$$

Proof. We give the even proof; the odd one is identical after replacing the sine frequencies. For $M \ge 1$, let

$$A^E_{N,M} := 2\big[\sin(m\theta_j)\big]_{1\le m\le N,\ 1\le j\le M}. \tag{10.3}$$

Then the truncated Gram matrix is

$$G^E_{N,M} = A^E_{N,M}(A^E_{N,M})^T = 4\sum_{j=1}^{M} s_j s_j^T, \tag{10.4}$$

where $s_j = (\sin\theta_j, \ldots, \sin N\theta_j)^T$. For each entry (m,n), the series $4\sum_j \sin(m\theta_j)\sin(n\theta_j)$ converges absolutely by Theorem 5.2; hence $G^E_{N,M} \to \mathsf{E}_N$ entrywise. Since the dimension N is fixed, entrywise convergence is equivalent to convergence in any matrix norm, and the determinant, being a polynomial in the entries, satisfies

$$\det G^E_{N,M} \longrightarrow \det \mathsf{E}_N. \tag{10.5}$$

For each finite M, the ordinary Cauchy–Binet formula applied to the $N \times M$ matrix $A^E_{N,M}$ gives, when $M \ge N$,

$$\det G^E_{N,M} = \sum_{J\subset\{1,\ldots,M\},\ |J|=N} \det(A^E_{N,M}[:,J])^2 \tag{10.6}$$

$$= 4^N \sum_{1\le j_1<\cdots<j_N\le M} \Big(\det[\sin(m\theta_{j_k})]_{m,k=1}^{N}\Big)^2. \tag{10.7}$$

Every summand is nonnegative. Therefore the finite subset sums form an increasing sequence in M and have a limit in $[0,\infty]$. Because their values equal $\det G^E_{N,M}$ and the latter converge to the finite number $\det \mathsf{E}_N$, monotone convergence of these finite sums gives exactly (10.1). The same construction with columns $2(\sin(\theta_j/2), \ldots, \sin((N-\frac12)\theta_j))^T$ gives (10.2). □

Theorem 10.2 (Vandermonde form)**.** *Under* RH*,*

$$\det \mathsf{E}_N = 4^N 2^{N(N-1)} \sum_{1\le j_1<\cdots<j_N} \left(\prod_{a=1}^{N} \sin^2\theta_{j_a}\right)\left(\prod_{1\le a<b\le N} (x_{j_b}-x_{j_a})^2\right), \tag{10.8}$$

$$\det \mathsf{O}_N = 4^N 2^{N(N-1)} \sum_{1\le j_1<\cdots<j_N} \left(\prod_{a=1}^{N} \sin^2(\theta_{j_a}/2)\right)\left(\prod_{1\le a<b\le N} (x_{j_b}-x_{j_a})^2\right). \tag{10.9}$$

Proof. For the even family,

$$\sin(m\theta) = \sin\theta\, U_{m-1}(\cos\theta). \tag{10.10}$$

The leading coefficient of U_{m-1} is 2^{m-1}, hence

$$\det[U_{m-1}(x_j)]_{m,j=1}^{N} = 2^{N(N-1)/2} \prod_{j<k} (x_k - x_j). \tag{10.11}$$

Squaring and inserting into (10.1) yields (10.8). The proof for O_N uses (8.3); P_{m-1} has the same leading coefficient 2^{m-1}. □

In terms of the zero enumeration with multiplicity,

$$\det \mathsf{E}_N = 4^N 2^{N(N-1)} \sum_{1\le j_1<\cdots<j_N} \prod_{a=1}^{N} \frac{16\gamma_{j_a}^2}{(4\gamma_{j_a}^2+1)^2} \prod_{a<b} (x_{j_b} - x_{j_a})^2, \tag{10.12}$$

$$\det \mathsf{O}_N = 4^N 2^{N(N-1)} \sum_{1\le j_1<\cdots<j_N} \prod_{a=1}^{N} \frac{1}{4\gamma_{j_a}^2+1} \prod_{a<b} (x_{j_b} - x_{j_a})^2. \tag{10.13}$$

If two selected indices correspond to the same multiple zero, the associated Vandermonde factor vanishes; selecting one copy produces the correct multiplicity factor.

Corollary 10.3 (Arbitrary principal minors)**.** *Let $I = \{n_1 < \cdots < n_r\} \subset \{1, \dots, N\}$. Under* RH*,*

$$\det \mathsf{E}_N[I, I] = 4^r \sum_{1\le j_1<\cdots<j_r} \Big(\det\big[\sin(n_a\theta_{j_b})\big]_{a,b=1}^{r}\Big)^2, \tag{10.14}$$

and

$$\det \mathsf{O}_N[I, I] = 4^r \sum_{1\le j_1<\cdots<j_r} \Big(\det\big[\sin((n_a - \tfrac{1}{2})\theta_{j_b})\big]_{a,b=1}^{r}\Big)^2. \tag{10.15}$$

Proof. We prove the even identity. Let A_E be the full zero-column operator and let $A_E[I, :]$ denote the $r \times \infty$ operator obtained by retaining the rows indexed by $I = \{n_1, \dots, n_r\}$. Taking the corresponding principal block of the Gram identity $\mathsf{E}_N = A_E A_E^T$ gives

$$\mathsf{E}_N[I, I] = A_E[I, :] A_E[I, :]^T. \tag{10.16}$$

For a finite truncation to the first M zero indices, Cauchy–Binet gives

$$\det \mathsf{E}_{N,M}[I, I] = 4^r \sum_{1\le j_1<\cdots<j_r\le M} \Big(\det[\sin(n_a\theta_{j_b})]_{a,b=1}^{r}\Big)^2. \tag{10.17}$$

The entries of the truncated principal block converge absolutely to those of $\mathsf{E}_N[I, I]$ by Theorem 5.2, so its determinant converges. The right-hand sums increase monotonically because all terms are nonnegative. Passing to the limit $M \to \infty$ proves (10.14). The odd formula follows identically from the half-integer zero columns. □

11. Canonical Cauchy–Binet ensemble and zero influence

The determinant formulas above are sums over all N-element zero subsets. It is therefore important to distinguish the full determinant from any one Vandermonde term. In this section the complete Cauchy–Binet sum itself is normalized and used to define the canonical zero-selection law attached to each reflection matrix.

For a positive ordinate γ, counted with multiplicity, define the column vectors

$$a_{N,\gamma}^{E} := 2\big(\sin\theta_\gamma, \sin(2\theta_\gamma), \dots, \sin(N\theta_\gamma)\big)^T, \tag{11.1}$$

$$a_{N,\gamma}^{O} := 2\big(\sin(\tfrac{1}{2}\theta_\gamma), \sin(\tfrac{3}{2}\theta_\gamma), \dots, \sin((N - \tfrac{1}{2})\theta_\gamma)\big)^T. \tag{11.2}$$

Let A_E and A_O denote the $N \times \infty$ matrices having these columns. Under RH the sine Gram formulas are precisely

$$\mathsf{E}_N = A_E A_E^T, \qquad \mathsf{O}_N = A_O A_O^T. \tag{11.3}$$

The column operators are Hilbert–Schmidt because the traces of the two Gram matrices are finite.

Theorem 11.1 (Canonical projection ensemble)**.** *Assume* RH *and fix N. Let M_N denote either E_N or O_N, and let A be the corresponding column operator in* (11.3)*. For each N-element subset J of the positive zero indices define*

$$\mathbb{P}_N(J) := \frac{\det(A_J)^2}{\det M_N}. \tag{11.4}$$

Then $\mathbb{P}_N$ is a probability measure. Moreover

$$K_N := A^T M_N^{-1} A \tag{11.5}$$

is an orthogonal projection of rank N on the zero-index space, and

$$\mathbb{P}_N(J) = \det K_N[J,J], \qquad |J| = N. \tag{11.6}$$

Thus the normalized Cauchy–Binet expansion is a fixed-size projection determinantal process on the positive zero indices.

Proof. For fixed N, regard A as an operator from the zero-index Hilbert space ℓ^2 to $\mathbb{R}^N$. Its columns are $a_{N,\gamma}$. Since

$$\sum_{\gamma>0} \|a_{N,\gamma}\|_2^2 = \operatorname{tr}\left(\sum_{\gamma>0} a_{N,\gamma} a_{N,\gamma}^T\right) = \operatorname{tr} M_N < \infty, \tag{11.7}$$

A is Hilbert–Schmidt, hence bounded, and the series $AA^T = \sum a_\gamma a_\gamma^T$ converges in trace norm (in finite dimension, equivalently entrywise and in every matrix norm). Under RH, Theorem 5.4 gives $M_N \succ 0$, so $M_N^{-1/2}$ exists.

The infinite Cauchy–Binet identity is obtained by truncation. Let $A^{(M)}$ contain the first M columns. Then $A^{(M)}(A^{(M)})^T \to M_N$ in norm. For $M \geq N$, finite Cauchy–Binet gives

$$\det\big(A^{(M)}(A^{(M)})^T\big) = \sum_{J\subset\{1,\dots,M\},\ |J|=N} \det(A_J)^2. \tag{11.8}$$

The left side tends to $\det M_N$ by continuity of the determinant. The right side is monotone increasing in M because every summand is nonnegative; therefore

$$\det M_N = \sum_{|J|=N} \det(A_J)^2. \tag{11.9}$$

Since $\det M_N > 0$, (11.4) is nonnegative and has total mass one.

Now define $B := M_N^{-1/2} A$. As an operator $\ell^2 \to \mathbb{R}^N$ it satisfies

$$BB^* = M_N^{-1/2} A A^T M_N^{-1/2} = I_{\mathbb{R}^N}. \tag{11.10}$$

Thus B is a coisometry. Its adjoint product $K_N := B^*B = A^T M_N^{-1} A$ is bounded and self-adjoint, and

$$K_N^2 = B^*(BB^*)B = B^*B = K_N. \tag{11.11}$$

Hence K_N is the orthogonal projection onto $\operatorname{Ran} B^*$; because $BB^* = I_N$, this range has dimension N.

For a finite N-element index set J, the principal block is $K_N[J,J] = B_J^T B_J$. Since $B_J = M_N^{-1/2} A_J$ is square,

$$\det K_N[J,J] = \det(B_J^T B_J) = \det(B_J)^2 \tag{11.12}$$

$$= \det(M_N^{-1/2})^2 \det(A_J)^2 = \frac{\det(A_J)^2}{\det M_N}, \tag{11.13}$$

which proves (11.6). □

Definition 11.2 (Zero leverage)**.** For either reflection family define

$$\ell_{N,\gamma} := (a_{N,\gamma})^T M_N^{-1} a_{N,\gamma}. \tag{11.14}$$

If a zero has multiplicity greater than one, each copy is regarded as a separate zero index with the same column vector.

Corollary 11.3 (Exact one- and two-zero marginals)**.** *Under* RH*,*

$$\mathbb{P}_N(\gamma \in J) = \ell_{N,\gamma}, \tag{11.15}$$

$$\mathbb{P}_N(\gamma, \eta \in J) = \ell_{N,\gamma}\ell_{N,\eta} - K_N(\gamma,\eta)^2, \qquad \gamma \neq \eta, \tag{11.16}$$

$$\sum_{\gamma>0} \ell_{N,\gamma} = N. \tag{11.17}$$

In particular, the full determinant assigns an intrinsic, normalized influence profile to the zero ordinates.

Proof. We derive the marginals directly. Fix one index γ and remove its column from A. The Cauchy–Binet normalization for the reduced column set gives

$$\mathbb{P}_N(\gamma \notin J) = \frac{\det(M_N - a_\gamma a_\gamma^T)}{\det M_N}. \tag{11.18}$$

Because M_N is invertible,

$$\det(M_N - a_\gamma a_\gamma^T) = \det M_N \, \det(I - M_N^{-1} a_\gamma a_\gamma^T) \tag{11.19}$$

$$= \det M_N \, (1 - a_\gamma^T M_N^{-1} a_\gamma), \tag{11.20}$$

where the last equality follows from $\det(I - uv^T) = 1 - v^T u$ (choose a basis whose first vector is u when $u \neq 0$, or use the characteristic polynomial of the rank-one operator). Hence

$$\mathbb{P}_N(\gamma \in J) = \ell_{N,\gamma}. \tag{11.21}$$

For two distinct indices γ, η, put $A_S = (a_\gamma, a_\eta)$ and $K_S = A_S^T M_N^{-1} A_S$. The finite-rank determinant identity

$$\det(M_N - A_S A_S^T) = \det M_N \, \det(I_2 - K_S) \tag{11.22}$$

follows by factoring out M_N and using $\det(I - UV) = \det(I - VU)$ for the finite matrices $U = M_N^{-1} A_S$ and $V = A_S^T$; the latter identity follows by taking the determinant of the two Schur complements of $\left(\begin{smallmatrix} I & U \\ V & I \end{smallmatrix}\right)$. Thus

$$\mathbb{P}_N(\gamma, \eta \notin J) = (1 - \ell_\gamma)(1 - \ell_\eta) - K_N(\gamma,\eta)^2. \tag{11.23}$$

Inclusion–exclusion, together with the one-point formula, yields

$$\mathbb{P}_N(\gamma, \eta \in J) = 1 - \mathbb{P}_N(\gamma \notin J) - \mathbb{P}_N(\eta \notin J) + \mathbb{P}_N(\gamma, \eta \notin J) \tag{11.24}$$

$$= \ell_\gamma \ell_\eta - K_N(\gamma,\eta)^2. \tag{11.25}$$

Finally, absolute convergence of the leverage sum follows from positivity and the trace computation

$$\sum_{\gamma>0} \ell_{N,\gamma} = \sum_{\gamma>0} \operatorname{tr}(M_N^{-1} a_\gamma a_\gamma^T) \tag{11.26}$$

$$= \operatorname{tr}\left(M_N^{-1} \sum_{\gamma>0} a_\gamma a_\gamma^T \right) = \operatorname{tr} I_N = N, \tag{11.27}$$

where interchange of sum and trace is valid because the summands are nonnegative and the ambient matrix space is finite dimensional. □

The leverage has an equivalent deterministic interpretation that does not require probabilistic terminology.

Theorem 11.4 (Deletion law)**.** *Let $M_N^{(-\gamma)}$ be the Gram matrix obtained after removing one copy of the positive zero ordinate γ from the zero-side representation. Then*

$$\boxed{\frac{\det M_N^{(-\gamma)}}{\det M_N} = 1 - \ell_{N,\gamma}.} \tag{11.28}$$

More generally, if S is a finite set of zero indices and $K_N[S,S]$ is the corresponding principal block of (11.5)*, then*

$$\frac{\det M_N^{(-S)}}{\det M_N} = \det(I - K_N[S,S]). \tag{11.29}$$

Proof. Under RH, $M_N \succ 0$ and is therefore invertible. Removing one indexed copy of the zero column $a = a_{N,\gamma}$ changes the Gram sum by exactly

$$M_N^{(-\gamma)} = M_N - aa^T. \tag{11.30}$$

Factoring out M_N gives

$$\det(M_N - aa^T) = \det M_N \, \det(I - M_N^{-1} aa^T). \tag{11.31}$$

The rank-one matrix $M_N^{-1}aa^T$ has at most one nonzero eigenvalue, equal to its trace $a^T M_N^{-1} a = \ell_{N,\gamma}$; hence

$$\det(I - M_N^{-1}aa^T) = 1 - \ell_{N,\gamma}, \tag{11.32}$$

proving (11.28). Notice that $M_N^{(-\gamma)}$ is only positive semidefinite in general, so the ratio is allowed to be zero; no invertibility of the deleted matrix is assumed.

For a finite indexed set S of size r, let A_S be the $N \times r$ matrix formed from the corresponding columns. Then

$$M_N^{(-S)} = M_N - A_S A_S^T \tag{11.33}$$

and therefore

$$\frac{\det M_N^{(-S)}}{\det M_N} = \det(I_N - M_N^{-1} A_S A_S^T) \tag{11.34}$$

$$= \det(I_r - A_S^T M_N^{-1} A_S). \tag{11.35}$$

The second equality is Sylvester's finite-dimensional identity $\det(I_N + UV) = \det(I_r + VU)$, which follows by evaluating the determinant of $\left(\begin{smallmatrix} I_N & U \\ -V & I_r \end{smallmatrix}\right)$ using either diagonal block as the Schur pivot. By definition the $r \times r$ matrix $A_S^T M_N^{-1} A_S$ is exactly $K_N[S,S]$, proving (11.29). □

11.1. **Transport of a single zero.** The same framework gives an exact answer to how one moved zero is represented in the Li sequence and how it changes the matrix determinant. This is a sensitivity identity for the zero-side Gram representation; the perturbed zero configuration need not itself arise from an arithmetic L-function.

Let one conjugate pair $1/2 \pm i\gamma$ be moved to $1/2 \pm i\gamma'$, leaving all other zeros fixed, and write

$$\theta = 2\arctan\frac{1}{2\gamma}, \qquad \theta' = 2\arctan\frac{1}{2\gamma'}. \tag{11.36}$$

Proposition 11.5 (Single-zero transport through the Li coefficients)**.** *The contribution of the moved conjugate pair to the change of the Li coefficient is*

$$\boxed{\delta\lambda_n = 2\big(\cos(n\theta) - \cos(n\theta')\big).} \tag{11.37}$$

Consequently the reflection matrices change by

$$\delta\mathsf{E}_N = a^E_{N,\gamma'}(a^E_{N,\gamma'})^T - a^E_{N,\gamma}(a^E_{N,\gamma})^T, \tag{11.38}$$

$$\delta\mathsf{O}_N = a^O_{N,\gamma'}(a^O_{N,\gamma'})^T - a^O_{N,\gamma}(a^O_{N,\gamma})^T. \tag{11.39}$$

Thus moving one conjugate zero pair is an update of rank at most two in either reflection family.

Proof. Under RH, the pair $1/2 \pm i\gamma$ contributes

$$2\big(1 - \cos(n\theta)\big) \tag{11.40}$$

to λ_n. Subtraction before and after the move gives (11.37). Using

$$\cos((m+n)\theta) - \cos((m-n)\theta) = -2\sin(m\theta)\sin(n\theta) \tag{11.41}$$

in the definitions of E_N and O_N yields (11.38)–(11.39). □

Theorem 11.6 (Exact determinant response to a moved zero)**.** *Let M_N denote either reflection matrix, let $a = a_{N,\gamma}$ and $a' = a_{N,\gamma'}$ be the corresponding old and new zero columns, and define with respect to the* unperturbed *inverse*

$$\ell = a^T M_N^{-1} a, \qquad \ell' = (a')^T M_N^{-1} a', \qquad \kappa = (a')^T M_N^{-1} a. \tag{11.42}$$

Then

$$\boxed{\frac{\det M'_N}{\det M_N} = (1-\ell)(1+\ell') + \kappa^2.} \tag{11.43}$$

Proof. Under RH, the unperturbed matrix M_N is positive definite and hence invertible. Proposition 11.5 gives the exact finite-rank update

$$M'_N = M_N + a'(a')^T - aa^T = M_N + UCU^T, \tag{11.44}$$

where $U = (a', a)$ is $N \times 2$ and $C = \mathrm{diag}(1, -1)$. Factoring out M_N,

$$\frac{\det M'_N}{\det M_N} = \det(I_N + M_N^{-1} UCU^T). \tag{11.45}$$

Apply Sylvester's identity with $X = M_N^{-1}U$ and $Y = CU^T$:

$$\det(I_N + XY) = \det(I_2 + YX) = \det(I_2 + CU^T M_N^{-1} U). \tag{11.46}$$

No invertibility of M'_N is required for this determinant identity; if the moved configuration makes M'_N singular, both sides simply vanish.

By the definitions of ℓ, ℓ', κ and symmetry of M_N^{-1},

$$U^T M_N^{-1} U = \begin{pmatrix} \ell' & \kappa \\ \kappa & \ell \end{pmatrix}. \tag{11.47}$$

Hence

$$I_2 + CU^T M_N^{-1} U = \begin{pmatrix} 1+\ell' & \kappa \\ -\kappa & 1-\ell \end{pmatrix}, \tag{11.48}$$

and direct evaluation gives

$$\frac{\det M'_N}{\det M_N} = (1-\ell)(1+\ell') + \kappa^2. \tag{11.49}$$

□

Corollary 11.7 (Infinitesimal zero sensitivity)**.** *For the even family let*

$$s_E(\theta) := (\sin\theta, \sin 2\theta, \dots, \sin N\theta)^T, \qquad \dot{s}_E(\theta) := (\cos\theta, 2\cos 2\theta, \dots, N\cos N\theta)^T. \tag{11.50}$$

Then a motion of the conjugate pair at ordinate γ satisfies

$$\boxed{\frac{\partial}{\partial\gamma}\log\det \mathsf{E}_N = -\frac{32}{4\gamma^2+1}\,\dot{s}_E(\theta_\gamma)^T \mathsf{E}_N^{-1} s_E(\theta_\gamma).} \tag{11.51}$$

For the odd family the identical formula holds with

$$s_O(\theta) := \big(\sin(\tfrac{1}{2}\theta), \sin(\tfrac{3}{2}\theta), \dots, \sin((N-\tfrac{1}{2})\theta)\big)^T, \tag{11.52}$$

$$\dot{s}_O(\theta) := \big(\tfrac{1}{2}\cos(\tfrac{1}{2}\theta), \tfrac{3}{2}\cos(\tfrac{3}{2}\theta), \dots, (N-\tfrac{1}{2})\cos((N-\tfrac{1}{2})\theta)\big)^T. \tag{11.53}$$

Proof. Fix N and one positive ordinate $\gamma_0 > 0$ in the unperturbed zero configuration. The other zero columns are held fixed. For $t > 0$ define

$$\theta(t) := 2\arctan\frac{1}{2t}, \qquad a(t) := 2s(\theta(t)), \tag{11.54}$$

with $s = s_E$ or $s = s_O$ according to the reflection family, and define the one-parameter matrix

$$\widetilde{M}_N(t) := M_N - a(\gamma_0)a(\gamma_0)^T + a(t)a(t)^T. \tag{11.55}$$

This definition avoids any differentiation of the infinite zero sum: only one finite-rank term depends on t. Since θ is C^∞ on $(0,\infty)$ and s is a finite vector of trigonometric functions, $t \mapsto \widetilde{M}_N(t)$ is C^∞ as a map into the real symmetric $N \times N$ matrices.

At $t = \gamma_0$ one has $\widetilde{M}_N(\gamma_0) = M_N \succ 0$ under RH. The smallest eigenvalue is a continuous function of the matrix entries; hence there exists $\varepsilon > 0$, with $(\gamma_0 - \varepsilon, \gamma_0 + \varepsilon) \subset (0,\infty)$, such that $\widetilde{M}_N(t) \succ 0$ throughout this interval. Consequently $\det \widetilde{M}_N(t) > 0$ there and $\log\det \widetilde{M}_N(t)$ is well defined and differentiable.

For any differentiable curve of invertible matrices $A(t)$, Jacobi's formula follows from $\det(A + H) = \det A\,\det(I + A^{-1}H)$ and the first-order expansion $\det(I + \varepsilon B) = 1 + \varepsilon\,\mathrm{tr}\,B + o(\varepsilon)$; hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\det A(t) = \mathrm{tr}\big(A(t)^{-1}A'(t)\big). \tag{11.56}$$

Applying this at $t = \gamma_0$ to (11.55),

$$\widetilde{M}_N'(\gamma_0) = a'(\gamma_0)a(\gamma_0)^T + a(\gamma_0)a'(\gamma_0)^T, \tag{11.57}$$

and therefore, using symmetry of M_N^{-1} and cyclicity of the trace,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\log\det \widetilde{M}_N(t)\Big|_{t=\gamma_0} &= \mathrm{tr}\Big(M_N^{-1}[a'a^T + aa'^T]\Big) \\ &= 2(a')^T M_N^{-1} a, \end{aligned} \tag{11.58}$$

where $a = a(\gamma_0)$ and $a' = a'(\gamma_0)$. Finally,

$$\theta'(t) = 2\frac{1}{1+(2t)^{-2}}\left(-\frac{1}{2t^2}\right) = -\frac{4}{4t^2+1}, \tag{11.59}$$

so $a'(t) = 2\theta'(t)\dot{s}(\theta(t))$. Substitution gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\det \widetilde{M}_N(t)\Big|_{t=\gamma_0} = -\frac{32}{4\gamma_0^2+1}\dot{s}(\theta_{\gamma_0})^T M_N^{-1} s(\theta_{\gamma_0}), \tag{11.60}$$

which is (11.51) and its odd analogue. The derivative is thus a local derivative on the open domain $t > 0$ on which the perturbed matrix remains invertible; no statement is made at the boundary $t = 0$. □

Theorem 11.8 (High-zero leverage decay at fixed matrix order)**.** *Assume* RH *and fix N. Put*

$$v_E := (1, 2, \dots, N)^T, \qquad c_N^E := v_E^T \mathsf{E}_N^{-1} v_E, \tag{11.61}$$

$$v_O := (\tfrac{1}{2}, \tfrac{3}{2}, \dots, N - \tfrac{1}{2})^T, \qquad c_N^O := v_O^T \mathsf{O}_N^{-1} v_O. \tag{11.62}$$

Then, as $\gamma \to \infty$ through positive zero ordinates,

$$\ell_{N,\gamma}^E = \frac{4c_N^E}{\gamma^2} + O_N(\gamma^{-4}), \tag{11.63}$$

$$\ell_{N,\gamma}^O = \frac{4c_N^O}{\gamma^2} + O_N(\gamma^{-4}), \tag{11.64}$$

and

$$\frac{\partial}{\partial\gamma} \log\det \mathsf{E}_N = -\frac{8c_N^E}{\gamma^3} + O_N(\gamma^{-5}), \tag{11.65}$$

$$\frac{\partial}{\partial\gamma} \log\det \mathsf{O}_N = -\frac{8c_N^O}{\gamma^3} + O_N(\gamma^{-5}). \tag{11.66}$$

Consequently, if the positive ordinates are enumerated with multiplicity,

$$\ell_{N,\gamma_j}^{E,O} = O_N\left(\frac{(\log j)^2}{j^2}\right), \qquad \sum_{j>J} \ell_{N,\gamma_j}^{E,O} = O_N\left(\frac{(\log J)^2}{J}\right). \tag{11.67}$$

Proof. As $\gamma \to \infty$,

$$\theta_\gamma = 2\arctan\frac{1}{2\gamma} = \frac{1}{\gamma} + O(\gamma^{-3}). \tag{11.68}$$

For fixed N this gives

$$s_E(\theta_\gamma) = \frac{v_E}{\gamma} + O_N(\gamma^{-3}), \qquad \dot{s}_E(\theta_\gamma) = v_E + O_N(\gamma^{-2}), \tag{11.69}$$

$$s_O(\theta_\gamma) = \frac{v_O}{\gamma} + O_N(\gamma^{-3}), \qquad \dot{s}_O(\theta_\gamma) = v_O + O_N(\gamma^{-2}). \tag{11.70}$$

Substitution into (11.14) gives (11.63)–(11.64); substitution into Corollary 11.7 gives (11.65)–(11.66). Finally the Riemann–von Mangoldt formula gives $\gamma_j \asymp j/\log j$, and summation of $(\log j)^2/j^2$ gives (11.67). □

Remark 11.9 (What the leverage theorem does and does not say). The decay in Theorem 11.8 is a statement about a *fixed* matrix order N. It proves that an individual sufficiently high zero has vanishing influence on that fixed determinant and that the normalized Cauchy–Binet mass has a summable high-zero tail. It does not provide a uniform estimate when N and the zero height grow together, and therefore it does not replace the large-N determinant analysis of Section 12.

12. Universal endpoint decay and the Li–Weil determinant law

The determinant asymptotic separates naturally into a geometric leading term and lower-order information. The leading coefficient does not require the precise Riemann–von Mangoldt law. It is fixed by the rate at which the discrete support approaches its unique endpoint. The following theorem isolates that mechanism before specializing to the zeta zero measures.

12.1. **An endpoint-exponent theorem for discrete Gram determinants.** Let

$$\mu = \sum_{j\geq 1} w_j\, \delta_{x_j}, \qquad 0 < w_j, \qquad x_j = x(t_j) \uparrow 1, \tag{12.1}$$

be a finite positive measure, with distinct $t_1 < t_2 < \cdots \to \infty$. Write

$$D_N[\mu] := \det\left(\int x^{r+s}\, \mathrm{d}\mu(x)\right)_{r,s=0}^{N-1} \tag{12.2}$$

for its monomial Hankel determinant.

Theorem 12.1 (Endpoint-exponent law)**.** *Assume that there is a number $\beta > 0$ for which the following conditions hold.*

(i) There are constants $c, C, t_0 > 0$ such that, for $t \geq t_0$,

$$c\, t^{-\beta} \leq 1 - x(t) \leq C\, t^{-\beta}, \qquad c\, t^{-\beta-1} \leq x'(t) \leq C\, t^{-\beta-1}. \tag{12.3}$$

(ii) The full node sequence is not superlinearly dense on the logarithmic scale:

$$\log t_j \geq (1 - o(1)) \log j. \tag{12.4}$$

(iii) For every sufficiently large N there are N support parameters $s_{N,1} < \cdots < s_{N,N}$ and a number $\eta_N \to 0$ such that

$$s_{N,N} \leq N^{1+\eta_N}, \qquad s_{N,b} - s_{N,a} \geq (b-a) N^{-\eta_N} \quad (1 \leq a < b \leq N). \tag{12.5}$$

(iv) The weights on these selected atoms are subleading at the determinant scale:

$$\sum_{a=1}^{N} \log w(s_{N,a}) \geq -o(N^2 \log N). \tag{12.6}$$

Then

$$\boxed{\lim_{N\to\infty} \frac{\log D_N[\mu]}{N^2 \log N} = -\beta.} \tag{12.7}$$

More generally, the same limit holds for the Gram determinant in any polynomial basis $p_0, p_1, \ldots$ with $\deg p_n = n$ and leading coefficients c_n satisfying

$$\sum_{n=0}^{N-1} \log |c_n| = O(N^2). \tag{12.8}$$

Proof. We divide the proof into the upper bound, the lower bound, and the change-of-basis statement.

Upper bound. Because μ is a finite positive measure with infinite support, the monomials $1, x, \ldots, x^n$ are linearly independent in $L^2(\mu)$ for every n: a nonzero polynomial has only finitely many zeros, whereas the support contains infinitely many distinct points. Hence the monic orthogonal polynomial π_n of degree n exists uniquely, and its squared norm

$$h_n := \int |\pi_n(x)|^2\, \mathrm{d}\mu(x) > 0 \tag{12.9}$$

is the minimum of $\int |q|^2\, \mathrm{d}\mu$ among all monic polynomials q of degree n. Gram–Schmidt, or equivalently triangular change of basis from monomials to the monic orthogonal polynomials, gives the exact product identity

$$D_N[\mu] = \prod_{n=0}^{N-1} h_n. \tag{12.10}$$

For $n \geq 1$ take the monic competitor

$$q_n(x) := \prod_{j=1}^{n} (x - x_j). \tag{12.11}$$

It vanishes at the first n atoms. Since $x_j \uparrow 1$, for $j > n$ and every $1 \leq r \leq n$,

$$0 < x_j - x_r \leq 1 - x_r. \tag{12.12}$$

Therefore

$$|q_n(x_j)| \leq \prod_{r=1}^{n} (1 - x_r), \qquad j > n. \tag{12.13}$$

Writing $M := \mu(\mathbb{R}) < \infty$ and using the extremal property of π_n,

$$h_n \leq \int |q_n(x)|^2 \, \mathrm{d}\mu(x) \tag{12.14}$$

$$= \sum_{j>n} w_j |q_n(x_j)|^2 \tag{12.15}$$

$$\leq M \prod_{r=1}^{n} (1 - x_r)^2. \tag{12.16}$$

By hypothesis (i), after increasing a fixed initial index if necessary,

$$\log(1 - x_j) \leq \log C - \beta \log t_j. \tag{12.17}$$

Hypothesis (ii) means that there exists a sequence $\varepsilon_j \to 0$ such that $\log t_j \geq (1 - \varepsilon_j) \log j$ for all sufficiently large j. Hence

$$\log(1 - x_j) \leq -\beta \log j + \beta \varepsilon_j \log j + O(1). \tag{12.18}$$

The error sums to $o(n \log n)$. Indeed, for every $\epsilon > 0$ choose J so that $|\varepsilon_j| \leq \epsilon$ for $j \geq J$; then

$$\sum_{j=J}^{n} |\varepsilon_j| \log j \leq \epsilon \sum_{j=J}^{n} \log j = O(\epsilon n \log n), \tag{12.19}$$

and the finitely many $j < J$ contribute $O_J(1)$. Since

$$\sum_{j=1}^{n} \log j = \log(n!) = n \log n - n + O(\log n), \tag{12.20}$$

we obtain

$$\sum_{j=1}^{n} \log(1 - x_j) \leq -\beta n \log n + o(n \log n). \tag{12.21}$$

Thus (12.16) gives

$$\log h_n \leq -2\beta n \log n + o(n \log n). \tag{12.22}$$

Summing (12.22) for $0 \leq n < N$ is legitimate at the same scale: if $r_n = o(n \log n)$, then for every $\epsilon > 0$ one has $|r_n| \leq \epsilon n \log n$ for all sufficiently large n, so $\sum_{n<N} r_n = o(N^2 \log N)$. Finally,

$$\sum_{n=1}^{N-1} n \log n = \frac{1}{2} N^2 \log N - \frac{1}{4} N^2 + O(N \log N), \tag{12.23}$$

and therefore

$$\log D_N[\mu] \leq -\beta N^2 \log N + o(N^2 \log N). \tag{12.24}$$

Lower bound. Fix N large enough that hypothesis (iii) supplies distinct support parameters $s_{N,1} < \cdots < s_{N,N}$, and put $y_a := x(s_{N,a})$. The finite $N \times N$ moment matrix generated by these selected atoms is

$$H_N^{\mathrm{sel}} = V_N W_N V_N^T, \tag{12.25}$$

where $(V_N)_{r,a} = y_a^r$ for $0 \le r \le N-1$ and $W_N = \mathrm{diag}(w(s_{N,1}), \dots, w(s_{N,N}))$. Since the full moment matrix is the sum of this positive semidefinite matrix and the contribution of all remaining atoms, the Cauchy–Binet expansion of the full determinant contains the selected term with positive sign; equivalently, applying Cauchy–Binet first to a finite truncation containing the selected atoms and then passing monotonically to the full measure gives

$$D_N[\mu] \ge \left(\prod_{a=1}^{N} w(s_{N,a}) \right) \prod_{1 \le a < b \le N} (y_b - y_a)^2. \tag{12.26}$$

For $a < b$, monotonicity of x and hypothesis (i) give

$$y_b - y_a = \int_{s_{N,a}}^{s_{N,b}} x'(u)\, \mathrm{d}u \tag{12.27}$$

$$\ge c\, s_{N,N}^{-\beta-1} (s_{N,b} - s_{N,a}), \tag{12.28}$$

because $u \le s_{N,N}$ on the interval and $u \mapsto u^{-\beta-1}$ is decreasing. By hypothesis (iii),

$$y_b - y_a \ge c(b-a) N^{-(\beta+1)(1+\eta_N)} N^{-\eta_N} \tag{12.29}$$

$$= c(b-a) N^{-\beta-1-(\beta+2)\eta_N}. \tag{12.30}$$

Taking logarithms and summing over the $\binom{N}{2}$ pairs,

$$2 \sum_{a<b} \log(y_b - y_a) \ge 2 \sum_{a<b} \log(b-a) - 2[\beta + 1 + (\beta+2)\eta_N] \binom{N}{2} \log N + O(N^2). \tag{12.31}$$

The elementary identity

$$\prod_{1 \le a < b \le N} (b-a) = \prod_{k=1}^{N-1} k! \tag{12.32}$$

and Stirling's formula imply

$$\sum_{a<b} \log(b-a) = \frac{1}{2} N^2 \log N - \frac{3}{4} N^2 + O(N \log N). \tag{12.33}$$

Since $\eta_N \to 0$, (12.31) and (12.33) yield

$$2 \sum_{a<b} \log(y_b - y_a) \ge -\beta N^2 \log N - o(N^2 \log N). \tag{12.34}$$

Hypothesis (iv) states precisely that the selected weights do not decrease this bound at the leading scale. Inserting the result into (12.26) gives

$$\log D_N[\mu] \ge -\beta N^2 \log N - o(N^2 \log N). \tag{12.35}$$

Together with (12.24), this proves the limit.

Polynomial bases. Let $p_n(x) = c_n x^n +$ terms of degree $< n$, and let C_N be the lower triangular coefficient matrix taking the monomial vector $(1, x, \dots, x^{N-1})^T$ to $(p_0, \dots, p_{N-1})^T$. Its diagonal is $(c_0, \dots, c_{N-1})$, so $\det C_N = \prod_{n<N} c_n$. If $G_N[p]$ denotes the corresponding Gram matrix, then exactly

$$G_N[p] = C_N H_N C_N^T, \tag{12.36}$$

and hence

$$\log\det G_N[p] = \log D_N[\mu] + 2\sum_{n=0}^{N-1}\log|c_n|. \tag{12.37}$$

Hypothesis (12.8) makes the second term $O(N^2) = o(N^2\log N)$, so the leading coefficient remains $-\beta$. □

Remark 12.2 (What is universal). Theorem 12.1 shows that the leading constant is the endpoint exponent $-\beta$. The precise counting law is absent from the coefficient. In particular, $t_j \asymp j$, $t_j \asymp j/\log j$, and more generally logarithmic refinements compatible with (12.4) and (12.5) give the same $-\beta$. The counting law is used only to verify that enough nodes exist in the required logarithmic scale. Arithmetic information, if present, can therefore enter only below the $N^2\log N$ term.

Remark 12.3 (Relation with earlier discrete Hankel asymptotics). The mere occurrence of an $N^2\log N$ scale for a discrete Hankel determinant is not new. Monien [27] derived a discrete Vandermonde representation for Hankel determinants of Dirichlet series. For the reciprocal-integer zeta model his Coulomb-gas calculation leads to

$$\log H_n^{(r)}[\zeta] = -n^2\left(\log(2n) - \frac{3}{2}\right) + o(n^2) \tag{12.38}$$

at the level of the variational analysis presented there. Monien explicitly notes that the passage from the discrete optimizer to the continuous density is assumed and suggests Poissonization as a route to a fully rigorous treatment. Thus (12.38) should not be cited here as a proved general asymptotic theorem from that preprint. Its leading coefficient nevertheless provides an important external consistency check: the nodes $x_m = 1/m$ have endpoint exponent $\beta = 1$, and Theorem 12.1 gives the same rigorous leading term $-n^2\log n$ under its hypotheses. The present theorem is broader at leading order—arbitrary β and non-explicit node sets satisfying coarse separation hypotheses—whereas Monien's special-model variational calculation is finer at order n^2.

Haynes and Zudilin [23] consider $\det(\zeta(a(i+j)+b))$. They distinguish two different statements: the expected full law has coefficient $-a$, while their elementary theorem proves only the weaker upper estimate

$$\log|\det(\zeta(a(i+j)+b))_{1\le i,j\le n}| \le -\frac{a}{2}n^2\log n + O(n^2). \tag{12.39}$$

There is therefore no factor-two contradiction with the endpoint law: m^{-a} has $\beta = a$, so Theorem 12.1 predicts the conjectural/full leading coefficient $-a$ when its lower-bound hypotheses are available; Haynes–Zudilin prove a one-sided estimate with coefficient $-a/2$ by a different elementary argument.

Johansson's discrete orthogonal-polynomial framework [24] is also closely related but does not make Theorem 12.1 an immediate corollary, and the reason is structural rather than a matter of coverage. Every ensemble treated there is supported on a *uniform* lattice: the configurations range over $\mathbb{N}^N$ or $\{0,\dots,K\}^N$, the Vandermonde is formed in the lattice variable, and the asymptotic analysis proceeds through a constrained equilibrium measure whose saturation constraint $\rho \le 1$ is precisely an expression of that uniform spacing. The detailed results are then obtained for the structured Meixner, Charlier, Krawtchouk, Hahn, and Plancherel weights.

The support relevant here is of a different type. The nodes x_γ of (12.59) are nonuniform, do not lie on a lattice, and accumulate at an endpoint. Thus there is no fixed uniform saturation constraint of the form $\rho \le 1$ inherited from unit lattice spacing; any continuum occupancy constraint would instead have to reflect the local nonuniform node density. The hypotheses used above, (12.3), (12.4) and (12.5), are coarse and impose no lattice structure at all. Accordingly, Johansson supplies

the rigorous discrete-Coulomb-gas context and is directly relevant to the lower-order constrained-equilibrium questions raised in Remark 12.13, but Theorem 12.1 is logically independent of the results stated there. This is a statement about [24] only; no general priority is claimed against the wider literature on discrete orthogonal polynomials with an accumulation point, on Stieltjes-type moment problems, or on Hankel determinant asymptotics.

These are different matrices from the Li–Weil reflection pair, with Dirichlet-series or lattice nodes rather than Cayley-transformed zero ordinates. Accordingly, no novelty is claimed here for the generic presence of $N^2 \log N$ or for constrained equilibrium methods themselves. The distinction of Theorem 12.1 is the elementary two-sided endpoint-exponent principle under coarse geometric hypotheses and its application to the Li–Weil reflection measures below.

12.2. **The zeta reflection measures and the universal constant -2.** Let $\gamma_1^* < \gamma_2^* < \cdots$ denote the distinct positive ordinates and set

$$\delta_j := 1 - x_{\gamma_j^*} = \frac{2}{4(\gamma_j^*)^2 + 1}. \tag{12.40}$$

Multiplicity is deliberately omitted from this enumeration.

Lemma 12.4 (Coarse location of distinct ordinates)**.** *There are constants $c > 0$ and j_0 such that*

$$\gamma_j^* \geq c\frac{j}{\log j}, \qquad j \geq j_0. \tag{12.41}$$

This estimate requires no simplicity hypothesis.

Proof. Let $N^*(T)$ count distinct positive ordinates and let $N(T)$ count positive ordinates with multiplicity. Then $N^*(T) \leq N(T)$. The Riemann–von Mangoldt formula gives $N(T) \ll T \log T$, hence $j \leq N(\gamma_j^*) \ll \gamma_j^* \log \gamma_j^*$, which implies (12.41). Multiple zeros can only decrease $N^*(T)$ and therefore move the distinct ordinates to larger heights; the estimate is thus robust with respect to multiplicity. □

Lemma 12.5 (A deterministically separated zero subsequence)**.** *There are constants $H > 0$, $T_0 > 0$, and distinct positive ordinates $\widetilde{\gamma}_j$ such that, for all $j \geq 1$,*

$$T_0 + 2Hj \leq \widetilde{\gamma}_j \leq T_0 + 2Hj + H. \tag{12.42}$$

Consequently $\widetilde{\gamma}_j \asymp j$ and

$$\widetilde{\gamma}_b - \widetilde{\gamma}_a \geq H(b-a), \qquad 1 \leq a < b. \tag{12.43}$$

Proof. Write the Riemann–von Mangoldt formula as

$$N(T) = F(T) + R(T), \qquad F(T) := \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}, \qquad R(T) = O(\log T). \tag{12.44}$$

Thus there are absolute constants $A, T_1 > 0$ such that $|R(T)| \leq A \log T$ for $T \geq T_1$. For every fixed $H \geq 1$,

$$F(T+H) - F(T) = \frac{H}{2\pi}\log T + O_H(1), \qquad T \to \infty. \tag{12.45}$$

Choose once and for all $H > 4\pi A$. Then, for sufficiently large T,

$$N(T+H) - N(T) \geq \left(\frac{H}{2\pi} - 2A + o(1)\right)\log T > 0.$$

Hence every interval $[T, T+H]$ with $T \geq T_0$ contains a zero for some $T_0 \geq T_1$. Applying this to the alternating boxes $[T_0 + 2Hj, T_0 + 2Hj + H]$ gives (12.42); the boxes are disjoint and (12.43) follows immediately. □

Theorem 12.6 (Li–Weil determinant law)**.** *Assume* RH. *Then, for both reflection families,*

$$\boxed{\lim_{N\to\infty}\frac{\log\Delta_N^E}{N^2\log N}=\lim_{N\to\infty}\frac{\log\Delta_N^O}{N^2\log N}=-2.} \tag{12.46}$$

More quantitatively,

$$-2N^2\log N+O(N^2)\le\log\Delta_N^E\le-2N^2\log N+2N^2\log\log N+O(N^2), \tag{12.47}$$

$$-2N^2\log N+O(N^2)\le\log\Delta_N^O\le-2N^2\log N+2N^2\log\log N+O(N^2). \tag{12.48}$$

Proof. For both zero measures the support map is

$$x(t)=1-\frac{2}{4t^2+1}, \tag{12.49}$$

so

$$1-x(t)\asymp t^{-2},\qquad x'(t)\asymp t^{-3}. \tag{12.50}$$

Thus $\beta=2$. Lemma 12.4 verifies the full-sequence condition (12.4), while Lemma 12.5 verifies (12.5) with $\eta_N=0$ up to harmless constants. Along that separated subsequence, the multiplicity of every selected ordinate is at least one and $\widetilde{\gamma}_j\asymp j$. Hence each selected one-point weight in (10.12)–(10.13) is bounded below by a constant multiple of j^{-2}. Consequently its logarithmic product is bounded below by $-O(N\log N)$, which is exactly the one-sided information required in (12.6). No upper bound on zero multiplicities is used. The Chebyshev-adapted bases have leading coefficient 2^n, hence their total basis contribution is $O(N^2)$. Theorem 12.1 therefore gives the limit (12.46).

For the quantitative upper bound, let h_n be the squared norm of the monic orthogonal polynomial for either μ_E or μ_O. The competitor

$$q_n(x):=\prod_{j=1}^{n}(x-x_{\gamma_j^*}) \tag{12.51}$$

gives

$$h_n\le M\prod_{j=1}^{n}\delta_j^2,\qquad M=\mu([-1,1]). \tag{12.52}$$

Lemma 12.4 yields

$$\delta_j\ll\frac{(\log j)^2}{j^2}, \tag{12.53}$$

and therefore, after restoring the leading coefficient 2^n of U_n or P_n,

$$\log d_{n+1}^{E,O}\le-4n\log n+4n\log\log n+O(n). \tag{12.54}$$

Summing gives the upper inequalities in (12.47)–(12.48).

For the quantitative lower bound, retain the separated term $\widetilde{\gamma}_1,\dots,\widetilde{\gamma}_N$ in the nonnegative Cauchy–Binet expansion. Since $x'(t)\asymp t^{-3}$ and (12.43) holds,

$$\widetilde{x}_b-\widetilde{x}_a\gg\frac{b-a}{b^3}. \tag{12.55}$$

Hence

$$2\sum_{a<b}\log(\widetilde{x}_b-\widetilde{x}_a)\ge2\sum_{a<b}\log(b-a)-6\sum_{a<b}\log b+O(N^2) \tag{12.56}$$

$$=-2N^2\log N+O(N^2), \tag{12.57}$$

using

$$\sum_{a<b}\log(b-a)=\frac{1}{2}N^2\log N+O(N^2),\qquad \sum_{a<b}\log b=\frac{1}{2}N^2\log N+O(N^2). \tag{12.58}$$

The selected one-point weights can decrease the logarithm of this retained term by at most $O(N\log N)$, while the explicit basis prefactor contributes only $O(N^2)$; this gives the lower inequalities. □

Corollary 12.7 (Universal Cayley exponent)**.** *For every positive discrete Gram system satisfying the hypotheses of Theorem 12.1 with the Li–Cayley support map*

$$x(t)=\frac{4t^2-1}{4t^2+1}, \tag{12.59}$$

the leading determinant coefficient is -2. In particular, the value -2 is unchanged by replacing a $T\log T$ counting law by a T counting law, or by any other logarithmic refinement compatible with the theorem's node hypotheses.

Proof. The map (12.59) has endpoint exponent $\beta=2$, so the conclusion is Theorem 12.1. □

Remark 12.8 (Automorphic Li coefficients). Lagarias defines generalized Li coefficients for principal automorphic L-functions and relates them to Weil's quadratic functional [25]. Consequently, whenever a self-dual member of that class is placed under GRH and its canonical reflection compression has the same positive Cayley zero measure with the standard coarse zero-counting and separation properties, Corollary 12.7 gives the same coefficient -2. The degree and conductor can change lower-order terms but not the endpoint exponent. This is a conditional universality statement: the present paper does not identify a real even–odd reflection pair for every non-self-dual automorphic representation.

Corollary 12.9 (Condition-number obstruction)**.** *Assume* RH. *Then*

$$\boxed{\liminf_{N\to\infty}\frac{\log\kappa_2(\mathsf{E}_N)}{N\log N}\ge 2,\qquad \liminf_{N\to\infty}\frac{\log\kappa_2(\mathsf{O}_N)}{N\log N}\ge 2.} \tag{12.60}$$

More quantitatively,

$$\log\kappa_2(\mathsf{E}_N),\ \log\kappa_2(\mathsf{O}_N)\ge 2N\log N-2N\log\log N-O(N). \tag{12.61}$$

Proof. For a positive definite $N\times N$ matrix A_N,

$$\lambda_{\min}(A_N)^N\le\det A_N, \tag{12.62}$$

so

$$\log\kappa_2(A_N)\ge\log\lambda_{\max}(A_N)-\frac{1}{N}\log\det A_N. \tag{12.63}$$

For the even family $\lambda_{\max}(\mathsf{E}_N)\ge(\mathsf{E}_N)_{11}=\lambda_2>0$, while for the odd family $\lambda_{\max}(\mathsf{O}_N)\ge(\mathsf{O}_N)_{11}=\lambda_1>0$. Insert the determinant upper bounds from Theorem 12.6. This gives (12.61), and division by $N\log N$ gives (12.60). □

Corollary 12.10 (Divergence of the inverse norm under RH)**.** *Assume* RH. *Then*

$$\boxed{\log\|\mathsf{E}_N^{-1}\|_2\ge 2N\log N-2N\log\log N-O(N),} \tag{12.64}$$

and consequently $\|\mathsf{E}_N^{-1}\|_2\to\infty$ faster than C^N for every fixed $C>1$. The same statement holds for O_N^{-1}.

Proof. For a positive definite matrix, $\|A_N^{-1}\|_2 = \lambda_{\min}(A_N)^{-1}$ and $\lambda_{\min}(A_N)^N \le \det A_N$. The determinant upper bound in Theorem 12.6 therefore gives

$$\log \lambda_{\min}(\mathsf{E}_N) \le -2N \log N + 2N \log\log N + O(N),$$

which is equivalent to (12.64). The odd case is identical. □

Remark 12.11 (Congruence warning for first-difference eigenvalues). Let $\widehat{E}_N := \frac{1}{2} D_N \mathsf{E}_N D_N^T$. Then

$$\mathsf{E}_N^{-1} = \frac{1}{2} D_N^T \widehat{E}_N^{-1} D_N. \tag{12.65}$$

Hence a tiny value of $\lambda_{\min}(\widehat{E}_N)$ does force a large inverse after accounting for the congruence, but one must not write $\|\mathsf{E}_N^{-1}\|_2 = 1/\lambda_{\min}(\widehat{E}_N)$. More precisely,

$$\|\mathsf{E}_N^{-1}\|_2 \ge \frac{\sigma_{\min}(D_N)^2}{2\,\lambda_{\min}(\widehat{E}_N)}. \tag{12.66}$$

The direct identity is $\|\mathsf{E}_N^{-1}\|_2 = 1/\lambda_{\min}(\mathsf{E}_N)$.

Remark 12.12 (The condition-number bound is far from sharp). Corollary 12.9 passes through $\lambda_{\min}(A_N)^N \le \det A_N$, which replaces the smallest eigenvalue by the geometric mean of the spectrum. For these matrices the spectrum is extremely spread, so the substitution is lossy. In a direct computation with the measure carried by the first 1200 ordinates and eigenvalues evaluated at 1200 decimal digits, the observed ratios

$$\frac{\log \lambda_{\min}(\mathsf{E}_N)}{N \log N} \tag{12.67}$$

are -11.9, -8.85, -7.83, -7.31, -6.94, -6.68, -6.48, -6.32 for $N = 2, 4, \dots, 16$, decreasing slowly. The scale that (12.63) induces for the same ratio (12.67) is only -2, since the argument replaces $\log \lambda_{\min}$ by $N^{-1} \log\det$; the sign convention here is that of $\lambda_{\min}$, not of κ_2. Correspondingly, at $N = 16$ the computation gives $\log_{10} \kappa_2(\mathsf{E}_{16}) \approx 123.9$ against a proved lower bound near 38.5. The corollary is therefore correct but conservative by a factor of about three at these orders. All numbers refer to the measure truncated to the first 1200 ordinates and are reported as computation on that truncation, not as values for the exact infinite matrices, and not as an asymptotic claim.

Remark 12.13 (Why the $N^2 \log\log N$ term is unresolved). The upper term $2N^2 \log\log N$ in Theorem 12.6 is not declared to be an artifact. Positivity implies that one favorable Cauchy–Binet subset gives a lower bound, but it does not imply that the logarithm of the full infinite sum equals the logarithm of its largest term; an entropy contribution can survive below the leading scale. Moreover, the Riemann–von Mangoldt clock itself suggests that the displayed logarithmic correction is structurally plausible. If, heuristically, $t_j \asymp j/\log j$ with local spacing $t_{j+1} - t_j \asymp 1/\log j$, then for a general endpoint exponent β one has at the pairwise level

$$x(t_b) - x(t_a) \asymp \frac{(b-a)(\log b)^\beta}{b^{\beta+1}}, \tag{12.68}$$

and the corresponding Vandermonde energy has the scale

$$-\beta N^2 \log N + \beta N^2 \log\log N + O(N^2). \tag{12.69}$$

Thus for the Li–Cayley value $\beta = 2$ the coefficient $+2$ in the present upper bound is exactly what the smooth clock predicts. Proving a matching second-order law would require substantially finer information on the dense zero configuration and is deliberately separated from the universal leading theorem. Theorem 12.14 below isolates exactly what that finer information has to be.

The next statement isolates a second-order law under explicit strengthened hypotheses on the node scale, the aggregate gap geometry, and the first-N atom weights. No implication of these hypotheses from RH alone is asserted. Write

$$\Lambda_N := \sum_{1\le a<b\le N} \left[\log(t_b - t_a) - \log\frac{b-a}{\log t_b}\right]. \tag{12.70}$$

Theorem 12.14 (Conditional second-order law)**.** *Assume the hypotheses of Theorem 12.1 together with the strengthened node scale*

$$\log t_j = \log j - \log\log j + O(1), \tag{12.71}$$

the aggregate gap regularity hypothesis

$$\Lambda_N = O(N^2), \tag{12.72}$$

and the first-N weight condition

$$\sum_{j=1}^{N} \log w_j \ge -O(N^2). \tag{12.73}$$

Then

$$\log D_N[\mu] = -\beta N^2 \log N + \beta N^2 \log\log N + O(N^2). \tag{12.74}$$

For the Li–Weil reflection measures, assume RH *and, in addition, that the distinct positive ordinates satisfy*

$$\log \gamma_j^* = \log j - \log\log j + O(1) \tag{12.75}$$

and that (12.72) *holds with $t_j = \gamma_j^*$. Then*

$$\log \Delta_N^E,\ \log \Delta_N^O = -2N^2 \log N + 2N^2 \log\log N + O(N^2). \tag{12.76}$$

No claim is made that (12.75) *follows from* RH *alone.*

Proof. The upper bound is already available. Indeed $1-x(t_j) \asymp t_j^{-\beta}$ and (12.71) give $\log\big(1-x(t_j)\big) = -\beta\log j + \beta\log\log j + O(1)$, so the extremal-polynomial estimate used in Theorem 12.1 yields

$$\log h_n \le -2\beta n\log n + 2\beta n\log\log n + O(n), \tag{12.77}$$

and summation over $n = 0,\dots,N-1$, together with $\sum_{n<N} n\log n = \frac{1}{2}N^2\log N + O(N^2)$ and $\sum_{n<N} n\log\log n = \frac{1}{2}N^2\log\log N + O(N^2)$, gives the upper bound in (12.74).

For the lower bound retain in the nonnegative Cauchy–Binet expansion the single term carried by the first N atoms. By the lower derivative bound in (12.3),

$$x_b - x_a = \int_{t_a}^{t_b} x'(u)\,\mathrm{d}u \tag{12.78}$$

$$\ge c\int_{t_a}^{t_b} u^{-\beta-1}\,\mathrm{d}u \tag{12.79}$$

$$\ge c\,(t_b - t_a)t_b^{-\beta-1}, \qquad 1\le a<b\le N, \tag{12.80}$$

because $u \le t_b$ on $[t_a, t_b]$. Hence

$$2\sum_{a<b}\log(x_b - x_a) \ge 2\sum_{a<b}\log(t_b - t_a) - 2(\beta+1)\sum_{a<b}\log t_b + O(N^2). \tag{12.81}$$

By (12.72) and (12.71),

$$\sum_{a<b}\log(t_b - t_a) = \sum_{a<b}\log(b-a) - \sum_{a<b}\log\log t_b + O(N^2) = \tfrac{1}{2}N^2\log N - \tfrac{1}{2}N^2\log\log N + O(N^2), \tag{12.82}$$

while (12.71) gives

$$\sum_{a<b} \log t_b = \tfrac{1}{2}N^2 \log N - \tfrac{1}{2}N^2 \log\log N + O(N^2). \tag{12.83}$$

Combining the preceding estimates gives

$$2\sum_{a<b} \log(x_b - x_a) \geq -\beta N^2 \log N + \beta N^2 \log\log N - O(N^2). \tag{12.84}$$

Hypothesis (12.73) shows that the first-N one-point weights can lower the logarithm of the retained Cauchy–Binet term by at most $O(N^2)$. Therefore

$$\log D_N[\mu] \geq -\beta N^2 \log N + \beta N^2 \log\log N - O(N^2), \tag{12.85}$$

which matches the upper bound.

For the Li–Weil measures, (12.75) and the explicit weights in (8.15)–(8.16) give

$$\sum_{j\leq N} \log w_E(\gamma_j^*), \quad \sum_{j\leq N} \log w_O(\gamma_j^*) \geq -O(N\log N), \tag{12.86}$$

because every zero multiplicity is at least one. This one-sided lower bound is all that (12.73) requires; no upper bound on multiplicities is assumed. The Chebyshev changes of basis contribute only $O(N^2)$ to the logarithm of the two reflection determinants. Thus the same second-order formula holds for Δ_N^E and Δ_N^O under the additional hypotheses stated above. □

Remark 12.15 (Where the gap hypothesis actually bites). Hypothesis (12.72) is an upper and a lower constraint, but only the lower one has content, and only for close pairs. On one side, a counting law of the form $N(T+h) - N(T) \ll h\log T + \log T$ forces $t_b - t_a \gg (b-a)/\log t_b$ whenever $b-a \gg \log t_N$, with no further hypothesis; the error term $\log T$ is what prevents the same deduction for smaller $b-a$. Hence (12.72) constrains only the $O(N\log N)$ pairs with $b-a \ll \log t_N$, and it permits those separations to be as small as $\exp(-cN/\log N)$ on logarithmic average. It is logically independent of simplicity, because the sequence t_j contains distinct ordinates and carries no multiplicity information, and it is far weaker than any conjectured pointwise lower bound on normalized gaps. What it is not is a theorem: no unconditional lower bound of this kind for consecutive ordinates is currently available, which is precisely why (12.74) is stated conditionally.

Remark 12.16 (Size of the Cauchy–Binet defect). The gap described in Remark 12.13 can be measured. With the same finite measure on both sides, namely the first 1200 ordinates, and log det evaluated by Cholesky at 3000 decimal digits, the difference between $\log\Delta_N^E$ and the logarithm of the single Cauchy–Binet term carried by the first N atoms is as follows.

N	$\log\Delta_N^E$	single term	defect	defect/$\lvert\log\Delta_N^E\rvert$
2	−17.336	−20.421	3.085	0.178
4	−88.812	−96.619	7.807	0.088
8	−414.694	−438.761	24.067	0.058
12	−1002.853	−1048.617	45.765	0.046
16	−1869.011	−1943.919	74.908	0.040

Over the computed range the normalized defect decreases, which is consistent with a contribution of lower order relative to $N^2\log N$; a power-law fit over these orders is close to $N^{3/2}$. No asymptotic exponent is claimed, and no growth rate is asserted beyond the tabulated orders: both statements would require an independent proof, and the positivity of the Cauchy–Binet expansion gives only defect ≥ 0.

What the table does establish, without extrapolation, is that a single Cauchy–Binet term is already quantitatively inadequate at these orders. Consequently no statement below the $N^2\log N$ scale should be extracted from one term without control of $\log\sum_J$. This is the quantitative form of the caveat in Remark 12.13.

Remark 12.17 (Zero logarithmic capacity). The compact support

$$\{1\} \cup \{x_\gamma : \gamma \in \Gamma_+^*\} \tag{12.87}$$

is countable and therefore has logarithmic capacity zero. This places the two measures outside the usual positive-capacity regular regime of orthogonal-polynomial asymptotics; see Stahl–Totik [36] and Totik's survey [37]. Zero carrier capacity alone does not impose a universal regular nth-root law for the leading coefficients; the geometry and the weights of a discrete measure remain decisive. Discrete orthogonal polynomials on infinite node sets with an accumulation point form a classical subject in their own right; see [38] for a modern asymptotic treatment in a specific model. Under the change of variable $u = (1-x)^{-1}$, the endpoint law $1 - x(t) \asymp t^{-\beta}$ becomes $u \asymp t^{\beta}$, so Theorem 12.1 may also be viewed as a power-counting statement for a Stieltjes-type discrete moment problem. The comparison with Monien and Haynes–Zudilin in Remark 12.3 is therefore essential: the present claim is not that $N^2 \log N$ decay is new, but that the Li–Weil coefficient is fixed by the Cayley endpoint exponent and can be proved from the coarse hypotheses above.

13. LDL^T, LU, Cholesky, and Schur complements

Set

$$\Delta_N^E := \det \mathsf{E}_N, \qquad \Delta_N^O := \det \mathsf{O}_N, \qquad \Delta_0^E = \Delta_0^O := 1. \tag{13.1}$$

Theorem 13.1 (LDL^T decomposition)**.** *Under* RH*, there are unique unit lower-triangular matrices L_E, L_O and positive diagonal matrices D_E, D_O such that*

$$\mathsf{E}_N = L_E D_E L_E^T, \qquad \mathsf{O}_N = L_O D_O L_O^T. \tag{13.2}$$

Their diagonal pivots are

$$(D_E)_{nn} = \frac{\Delta_n^E}{\Delta_{n-1}^E}, \tag{13.3}$$

$$(D_O)_{nn} = \frac{\Delta_n^O}{\Delta_{n-1}^O}. \tag{13.4}$$

Proof. We prove the statement for a generic real symmetric positive definite matrix A_N whose leading principal blocks are denoted A_n, $1 \le n \le N$; the two reflection families then follow from Theorem 5.4. Set A_0 equal to the empty matrix and $\det A_0 := 1$.

We argue by induction on N. For $N = 1$, write $A_1 = (a_{11})$. Positive definiteness gives $a_{11} > 0$, and the factorization $A_1 = (1)(a_{11})(1)$ is unique. Assume the result for A_{N-1} and write

$$A_N = \begin{pmatrix} A_{N-1} & r \\ r^T & \alpha \end{pmatrix}. \tag{13.5}$$

Since $A_N \succ 0$, its principal block A_{N-1} is positive definite and hence invertible. By the induction hypothesis, $A_{N-1} = L_{N-1} D_{N-1} L_{N-1}^T$ with L_{N-1} unit lower triangular and D_{N-1} positive diagonal. Put

$$y := L_{N-1}^{-1} r, \qquad d_N := \alpha - r^T A_{N-1}^{-1} r, \tag{13.6}$$

and define

$$L_N := \begin{pmatrix} L_{N-1} & 0 \\ y^T D_{N-1}^{-1} & 1 \end{pmatrix}, \qquad D_N := \begin{pmatrix} D_{N-1} & 0 \\ 0 & d_N \end{pmatrix}. \tag{13.7}$$

A direct block multiplication gives $L_N D_N L_N^T = A_N$. Moreover,

$$d_N = \begin{pmatrix} -A_{N-1}^{-1} r \\ 1 \end{pmatrix}^T A_N \begin{pmatrix} -A_{N-1}^{-1} r \\ 1 \end{pmatrix} > 0, \tag{13.8}$$

so D_N is positive diagonal. Uniqueness follows recursively: in any factorization $A_N = LDL^T$ with L unit lower triangular and D diagonal, the leading $(N-1)\times(N-1)$ block fixes L_{N-1}, D_{N-1} by induction; then the last column below the diagonal is fixed by $r = L_{N-1}D_{N-1}\ell$, and the last diagonal entry is fixed by the (N,N) entry.

Finally, $\det L_n = 1$, so

$$\det A_n = \det D_n = \prod_{j=1}^{n} d_j. \tag{13.9}$$

Dividing the identities for n and $n-1$ yields $d_n = \det A_n / \det A_{n-1}$. Taking $A_n = \mathsf{E}_n$ and $A_n = \mathsf{O}_n$ proves (13.3)–(13.4). □

The first pivots are

$$d_{E,1} = \lambda_2, \tag{13.10}$$

$$d_{E,2} = \lambda_4 - \frac{(\lambda_3 - \lambda_1)^2}{\lambda_2}, \tag{13.11}$$

$$d_{O,1} = \lambda_1, \tag{13.12}$$

$$d_{O,2} = \lambda_3 - \frac{(\lambda_2 - \lambda_1)^2}{\lambda_1}. \tag{13.13}$$

The ordinary LU factorization with unit diagonal in the lower factor is

$$\mathsf{E}_N = L_E U_E, \qquad U_E = D_E L_E^T, \tag{13.14}$$

and similarly

$$\mathsf{O}_N = L_O U_O, \qquad U_O = D_O L_O^T. \tag{13.15}$$

Thus the diagonal of U_E and U_O is exactly the pivot sequence in (13.3)–(13.4).

The Cholesky factors are

$$\mathsf{E}_N = R_E R_E^T, \qquad R_E = L_E D_E^{1/2}, \tag{13.16}$$

and

$$\mathsf{O}_N = R_O R_O^T, \qquad R_O = L_O D_O^{1/2}. \tag{13.17}$$

In particular,

$$(R_E)_{nn} = \sqrt{\frac{\Delta_n^E}{\Delta_{n-1}^E}}, \tag{13.18}$$

$$(R_O)_{nn} = \sqrt{\frac{\Delta_n^O}{\Delta_{n-1}^O}}. \tag{13.19}$$

Proposition 13.2 (Schur-complement recursion)**.** *Write*

$$\mathsf{E}_N = \begin{pmatrix} \mathsf{E}_{N-1} & r_N^E \\ (r_N^E)^T & \lambda_{2N} \end{pmatrix}, \qquad \mathsf{O}_N = \begin{pmatrix} \mathsf{O}_{N-1} & r_N^O \\ (r_N^O)^T & \lambda_{2N-1} \end{pmatrix}. \tag{13.20}$$

Then under RH*,*

$$\frac{\Delta_N^E}{\Delta_{N-1}^E} = \lambda_{2N} - (r_N^E)^T \mathsf{E}_{N-1}^{-1} r_N^E > 0, \tag{13.21}$$

$$\frac{\Delta_N^O}{\Delta_{N-1}^O} = \lambda_{2N-1} - (r_N^O)^T \mathsf{O}_{N-1}^{-1} r_N^O > 0. \tag{13.22}$$

Proof. We again treat a generic block matrix

$$A_N = \begin{pmatrix} A & r \\ r^T & \alpha \end{pmatrix}, \tag{13.23}$$

where $A = A_{N-1}$ is positive definite. In particular A is invertible. Consider the two block triangular matrices

$$L := \begin{pmatrix} I & 0 \\ -r^T A^{-1} & 1 \end{pmatrix}, \qquad R := \begin{pmatrix} I & -A^{-1}r \\ 0 & 1 \end{pmatrix}. \tag{13.24}$$

Both have determinant one. Direct multiplication gives

$$LA_N R = \begin{pmatrix} A & 0 \\ 0 & \alpha - r^T A^{-1} r \end{pmatrix}. \tag{13.25}$$

Taking determinants therefore yields

$$\det A_N = \det A \, (\alpha - r^T A^{-1} r), \tag{13.26}$$

and, since $\det A > 0$,

$$\frac{\det A_N}{\det A} = \alpha - r^T A^{-1} r. \tag{13.27}$$

The scalar on the right is strictly positive because the nonzero vector

$$z := \begin{pmatrix} -A^{-1}r \\ 1 \end{pmatrix} \tag{13.28}$$

satisfies

$$z^T A_N z = \alpha - r^T A^{-1} r > 0 \tag{13.29}$$

by positive definiteness of A_N. Substituting $A = \mathsf{E}_{N-1}$, $r = r_N^E$, $\alpha = \lambda_{2N}$ and then $A = \mathsf{O}_{N-1}$, $r = r_N^O$, $\alpha = \lambda_{2N-1}$ proves both formulas and their strict positivity. □

Remark 13.3 (Geometric meaning). The quantities in (13.21) and (13.22) are the squared residual norms after orthogonally projecting the new sine mode onto the span of all preceding modes. They quantify the genuinely new Gram energy contributed by frequency N or $N - 1/2$.

14. Inverse matrices and reproducing kernels

Whenever the matrices are invertible, the adjugate formula gives

$$(\mathsf{E}_N^{-1})_{ij} = (-1)^{i+j} \frac{\det \mathsf{E}_N^{(j,i)}}{\Delta_N^E}, \tag{14.1}$$

$$(\mathsf{O}_N^{-1})_{ij} = (-1)^{i+j} \frac{\det \mathsf{O}_N^{(j,i)}}{\Delta_N^O}. \tag{14.2}$$

More structure is visible after the moment-basis transformation.

Let

$$u_N(x) := (U_0(x), \dots, U_{N-1}(x))^T, \tag{14.3}$$

$$\mathbf{p}_N(x) := (P_0(x), \dots, P_{N-1}(x))^T. \tag{14.4}$$

Define

$$\mathcal{K}_N^E(x,y) := u_N(x)^T \mathsf{E}_N^{-1} u_N(y), \tag{14.5}$$

$$\mathcal{K}_N^O(x,y) := \mathbf{p}_N(x)^T \mathsf{O}_N^{-1} \mathbf{p}_N(y). \tag{14.6}$$

Theorem 14.1 (Christoffel–Darboux interpretation)**.** *Under* RH, $\mathcal{K}_N^E$ *and* $\mathcal{K}_N^O$ *are the reproducing kernels of the polynomial spaces of degree* $< N$ *in* $L^2(\mu_E)$ *and* $L^2(\mu_O)$*, respectively. If* $\{q_j^E\}$ *and* $\{q_j^O\}$ *are the corresponding orthonormal polynomials, then*

$$\mathcal{K}_N^E(x,y) = \sum_{j=0}^{N-1} q_j^E(x) q_j^E(y), \tag{14.7}$$

$$\mathcal{K}_N^O(x,y) = \sum_{j=0}^{N-1} q_j^O(x) q_j^O(y). \tag{14.8}$$

Proof. We prove the even statement. Let

$$\mathcal{P}_{N-1} := \operatorname{span}\{U_0, \dots, U_{N-1}\} \tag{14.9}$$

inside $L^2(\mu_E)$. Under RH, $\mathsf{E}_N \succ 0$, so these basis functions are linearly independent in $L^2(\mu_E)$ and their Gram matrix is invertible. For fixed y, define

$$K_y(x) := u_N(x)^T \mathsf{E}_N^{-1} u_N(y) \in \mathcal{P}_{N-1}. \tag{14.10}$$

Take an arbitrary $f \in \mathcal{P}_{N-1}$ and write $f(x) = c^T u_N(x)$ with $c \in \mathbb{R}^N$. Then, using the moment Gram identity,

$$\langle f, K_y \rangle_{L^2(\mu_E)} = \int c^T u_N(x) u_N(x)^T \mathsf{E}_N^{-1} u_N(y) \, \mathrm{d}\mu_E(x) \tag{14.11}$$

$$= c^T \left(\int u_N(x) u_N(x)^T \, \mathrm{d}\mu_E(x) \right) \mathsf{E}_N^{-1} u_N(y) \tag{14.12}$$

$$= c^T \mathsf{E}_N \mathsf{E}_N^{-1} u_N(y) = f(y). \tag{14.13}$$

Thus K_y has the reproducing property and is the unique reproducing kernel section at y. This proves (14.5) as a reproducing kernel; the odd case is identical with p_N and O_N.

Now let $q_0^E, \dots, q_{N-1}^E$ be any orthonormal basis of $\mathcal{P}_{N-1}$. The function

$$\widetilde{K}_N^E(x,y) := \sum_{j=0}^{N-1} q_j^E(x) q_j^E(y) \tag{14.14}$$

has the same reproducing property, because for $f = \sum_j \widehat{f}_j q_j^E$ one has $\langle f, \widetilde{K}_N^E(\cdot, y) \rangle = \sum_j \widehat{f}_j q_j^E(y) = f(y)$. Uniqueness of the Riesz representer in the finite-dimensional Hilbert space therefore gives $\widetilde{K}_N^E = \mathcal{K}_N^E$. This is (14.7); the odd formula follows in the same way. □

In the monomial basis, Theorem 9.1 gives the exact inverse identities

$$\mathsf{E}_N^{-1} = (B_E^{(N)})^{-T} (H_N^E)^{-1} (B_E^{(N)})^{-1}, \tag{14.15}$$

$$\mathsf{O}_N^{-1} = (B_O^{(N)})^{-T} (H_N^O)^{-1} (B_O^{(N)})^{-1}. \tag{14.16}$$

Together with Corollary 7.4, these formulas show that the inverses simultaneously possess moment-kernel and low-displacement-rank descriptions.

Remark 14.2 (No universal sign pattern). The off-diagonal entries $\lambda_{m+n} - \lambda_{|m-n|}$ and $\lambda_{m+n-1} - \lambda_{|m-n|}$ do not have an algebraically forced sign. Consequently the matrices are not naturally Stieltjes M-matrices, and no universal checkerboard or nonpositive off-diagonal sign pattern for their inverses follows from the present theory.

15. Column recursion and the closed two-generator bordering scheme

The central matrix-first problem is whether nesting and rank-two displacement can turn the Schur complements (13.21) into a propagation law whose sign can be controlled. This section solves the recurrence part in closed form. What remains is the sign mechanism isolated later in Section 27; Remark 15.10 records the distinction.

Throughout, $\lambda_0 = 0$ and $\lambda_{-j} = \lambda_j$. Every numerical statement in this section and in Section 17 uses the Li coefficients produced by the arithmetic construction of Subsection 26.2, which is stated later only because it belongs with the archimedean–arithmetic split; it takes Stieltjes constants as input and uses no zero of ζ at any stage.

Definition 15.1 (Bordering column)**.** For $N \geq 1$ let $\mathsf{b}^{(N)}$ be the semi-infinite vector

$$\mathsf{b}^{(N)}_m := \lambda_{m+N} - \lambda_{|m-N|}, \qquad m \geq 1. \tag{15.1}$$

Its first $N-1$ entries form the off-diagonal block of E_N in (13.20), and $\mathsf{b}^{(N)}_N = \lambda_{2N}$.

Proposition 15.2 (Column wave equation)**.** *Let T denote the formal nearest-neighbour shift on semi-infinite sequences with Dirichlet condition at 0, that is $(Tv)_m = v_{m+1} + v_{m-1}$ with $v_0 := 0$. No ℓ^2 assumption is made on the bordering columns. Then for every $N \geq 2$,*

$$\boxed{\mathsf{b}^{(N+1)} + \mathsf{b}^{(N-1)} = T\,\mathsf{b}^{(N)}} \tag{15.2}$$

as an identity between semi-infinite vectors, valid unconditionally. Equivalently,

$$\mathsf{b}^{(N)} = U_{N-1}\Big(\tfrac{T}{2}\Big)\,\mathsf{b}^{(1)}, \tag{15.3}$$

where U_n is the Chebyshev polynomial of the second kind.

Proof. Fix $m \geq 1$. The left-hand side of (15.2) at index m is

$$\big(\lambda_{m+N+1} - \lambda_{|m-N-1|}\big) + \big(\lambda_{m+N-1} - \lambda_{|m-N+1|}\big), \tag{15.4}$$

and the right-hand side is

$$\big(\lambda_{m+1+N} - \lambda_{|m+1-N|}\big) + \big(\lambda_{m-1+N} - \lambda_{|m-1-N|}\big), \tag{15.5}$$

where the convention $v_0 = 0$ is automatic because $\mathsf{b}^{(N)}_0 = \lambda_N - \lambda_N = 0$. The two positive parts agree termwise. The two negative parts agree as well, since $|m-N-1| = |m-1-N|$ and $|m-N+1| = |m+1-N|$. This proves (15.2). Since $\mathsf{b}^{(0)} = 0$, the recursion $\mathsf{b}^{(N+1)} = T\mathsf{b}^{(N)} - \mathsf{b}^{(N-1)}$ with initial data $\mathsf{b}^{(0)} = 0$, $\mathsf{b}^{(1)}$ is exactly the Chebyshev recursion, giving (15.3). □

Remark 15.3 (Interpretation). Under RH the identity (15.2) is the statement $\sin((N+1)\theta) + \sin((N-1)\theta) = 2\cos\theta\,\sin(N\theta)$ transported through the Gram representation (1.9). The content of Proposition 15.2 is that it holds *unconditionally*, as an identity in the Li coefficients: the same discrete wave equation that Section 7 exhibits in the row index also governs the sequence of bordering columns. The pair $(\mathsf{b}^{(1)}, \mathsf{b}^{(2)})$ therefore generates the entire semi-infinite array, which is the sense in which the scheme below is a *two-generator* scheme.

Proposition 15.4 (Closed bordering recursion)**.** *Assume E_{N-1} is invertible and set $G := \mathsf{E}_{N-1}^{-1}$. Write $e := e_{N-1} \in \mathbb{R}^{N-1}$ for the last standard basis vector, let*

$$\omega_N := \lambda_{2N-1} - \lambda_1, \qquad r^{(N)} := \big(\mathsf{b}^{(N)}_1, \ldots, \mathsf{b}^{(N)}_{N-2}, 0\big)^T, \tag{15.6}$$

so that the off-diagonal block of E_N is

$$b_N = r^{(N)} + \omega_N\, e. \tag{15.7}$$

Put $g := Ge$, $h := Gr^{(N)}$ and $q := h + \omega_N g$. Then the next Schur pivot is

$$p_N := \frac{\Delta_N^E}{\Delta_{N-1}^E} = \lambda_{2N} - \big(r^{(N)}\big)^T h - 2\omega_N h_{N-1} - \omega_N^2 g_{N-1}. \tag{15.8}$$

If, in addition, $p_N \neq 0$—equivalently, E_N is invertible—then

$$\mathsf{E}_N^{-1} = \begin{pmatrix} G + p_N^{-1} q q^T & -p_N^{-1} q \\ -p_N^{-1} q^T & p_N^{-1} \end{pmatrix}, \qquad g^{(N)} := \mathsf{E}_N^{-1} e_N = \begin{pmatrix} -q/p_N \\ 1/p_N \end{pmatrix}. \tag{15.9}$$

In particular $g_N^{(N)} p_N = 1$ whenever the inverse exists.

Proof. Equation (15.7) is the definition of $r^{(N)}$ together with $\mathsf{b}_{N-1}^{(N)} = \lambda_{2N-1} - \lambda_1 = \omega_N$. Since G is symmetric,

$$b_N^T G b_N = \big(r^{(N)} + \omega_N e\big)^T G \big(r^{(N)} + \omega_N e\big) = \big(r^{(N)}\big)^T h + 2\omega_N\, e^T h + \omega_N^2\, e^T g, \tag{15.10}$$

and $e^T h = h_{N-1}$, $e^T g = g_{N-1}$. Substituting into the Schur complement (13.21) gives (15.8). Formula (15.9) is the standard block inverse of $\left(\begin{smallmatrix} \mathsf{E}_{N-1} & b_N \\ b_N^T & \lambda_{2N} \end{smallmatrix}\right)$ with $q = Gb_N = h + \omega_N g$, which is valid whenever $p_N \neq 0$. □

Proposition 15.5 (Closed two-vector state update)**.** *Under the hypotheses and notation of Proposition 15.4, assume in addition that $p_N \neq 0$, so that E_N^{-1} exists, and let*

$$\rho_N := \big(\mathsf{b}_1^{(N+1)}, \dots, \mathsf{b}_{N-1}^{(N+1)}\big)^T \in \mathbb{R}^{N-1}. \tag{15.11}$$

Then the column wave equation gives

$$\boxed{\rho_N = 2\mathsf{J}_E^{(N-1)} b_N - \mathsf{E}_{N-1} e + \lambda_{2N} e.} \tag{15.12}$$

If $q = h + \omega_N g = \mathsf{E}_{N-1}^{-1} b_N$ and

$$\eta_N := -\frac{q^T \rho_N}{p_N}, \tag{15.13}$$

then the second inverse generator

$$h^{(N)} := \mathsf{E}_N^{-1} r_E^{(N)}, \qquad r_E^{(N)} = (\rho_N^T, 0)^T, \tag{15.14}$$

is obtained without a new inverse column:

$$\boxed{h^{(N)} = \begin{pmatrix} 2\mathsf{J}_E^{(N-1)} q + (q_{N-1} - \eta_N) q + p_N g - e \\ \eta_N \end{pmatrix}.} \tag{15.15}$$

Together with (15.9)*, equations* (15.8) *and* (15.15) *close the inverse information needed at the next step on the pair $(g^{(N)}, h^{(N)})$ and the bordering columns.*

Proof. The first identity is the finite restriction of $\mathsf{b}^{(N+1)} + \mathsf{b}^{(N-1)} = T\mathsf{b}^{(N)}$ at the new boundary; the missing N-th coupling contributes the last two terms in (15.12). For completeness, the inverse displacement identity gives

$$\mathsf{J}_E^{(N-1)} G - G \mathsf{J}_E^{(N-1)} = \frac{1}{2}(g h^T - h g^T). \tag{15.16}$$

Apply G to (15.12), substitute $q = Gb_N = h + \omega_N g$, and use $p_N = \lambda_{2N} - b_N^T q$. The scalar terms combine to give

$$G\rho_N = 2\mathsf{J}_E^{(N-1)} q + q_{N-1} q + p_N g - e. \tag{15.17}$$

Finally multiply $r_E^{(N)} = (\rho_N^T, 0)^T$ by the block inverse (15.9). Its last component is $-q^T \rho_N / p_N = \eta_N$, and its top block is $G\rho_N - \eta_N q$. Equation (15.15) follows. □

Theorem 15.6 (Local LDL^T pivot dynamics)**.** *Let A denote either the even or the odd semi-infinite reflection array, and suppose $p_1, \dots, p_n$ are nonzero. Gaussian LDL^T elimination is then well defined through column n; in particular the leading block A_{n+1} has the factorization*

$$A_{n+1} = L_{n+1}\,\mathrm{diag}(p_1, \dots, p_{n+1})\,L_{n+1}^T, \qquad (L_{n+1})_{jj} = 1, \tag{15.18}$$

where p_{n+1} is allowed to vanish. The nth elimination column in the next leading block is also defined because its construction divides only by $p_1, \dots, p_n$. Define the first two subdiagonals

$$\ell_n := (L_{n+1})_{n+1,n}, \qquad \tau_n := (L_{n+2})_{n+2,n}, \qquad n \geq 1. \tag{15.19}$$

Set

$$\ell_0 = \sigma, \qquad \tau_0 = 0, \qquad \sigma = \begin{cases} 0, & A = \mathsf{E}, \\ 1, & A = \mathsf{O}. \end{cases} \tag{15.20}$$

Then, for every $n \geq 1$,

$$\boxed{\frac{p_{n+1}}{p_n} = 1 + (\tau_n - \tau_{n-1}) - \ell_n(\ell_n - \ell_{n-1}).} \tag{15.21}$$

Equivalently, if

$$\mathcal{H}_n := \tau_n + n - \frac{1}{2}\ell_n^2 - \frac{1}{2}\sum_{j=1}^{n}(\ell_j - \ell_{j-1})^2, \qquad \mathcal{H}_0 = -\frac{1}{2}\sigma^2, \tag{15.22}$$

then

$$\boxed{\mathcal{H}_n - \mathcal{H}_{n-1} = \frac{p_{n+1}}{p_n}.} \tag{15.23}$$

Thus, once $p_n > 0$ is known, positivity of the next pivot is equivalent to the strict increase $\mathcal{H}_n > \mathcal{H}_{n-1}$.

Proof. First suppose $p_{n+1} \neq 0$. We give the even case. On the semi-infinite reflection array the discrete wave identity is the commutation relation $\mathsf{J}_E A = A\mathsf{J}_E$. Insert $A = LDL^T$ and put $B = L^{-1}\mathsf{J}_E L$. Since J_E is tridiagonal and L is unit lower triangular, B is upper Hessenberg. The relation $BD = DB^T$ forces it to be tridiagonal. Moreover

$$2B_{n-1,n} = 1, \qquad 2B_{n,n} = \ell_n - \ell_{n-1}, \qquad 2B_{n+1,n} = \frac{p_{n+1}}{p_n}. \tag{15.24}$$

The identity $\mathsf{J}_E L = LB$, read at entry (m, n), therefore gives

$$L_{m+1,n} + L_{m-1,n} = L_{m,n-1} + (\ell_n - \ell_{n-1})L_{m,n} + \frac{p_{n+1}}{p_n}L_{m,n+1}. \tag{15.25}$$

Set $m = n + 1$. Since $L_{n,n} = L_{n+1,n+1} = 1$, $L_{n+2,n} = \tau_n$ and $L_{n+1,n-1} = \tau_{n-1}$, this is exactly (15.21). In the odd family the only change is the $-1/2$ entry at the upper-left corner of J_O; the same calculation is recovered by the boundary convention $\ell_0 = 1$. Finally,

$$\ell_n(\ell_n - \ell_{n-1}) = \frac{1}{2}(\ell_n^2 - \ell_{n-1}^2) + \frac{1}{2}(\ell_n - \ell_{n-1})^2, \tag{15.26}$$

which telescopes (15.21) into (15.23).

It remains to allow $p_{n+1} = 0$. For fixed n, the elimination coefficients ℓ_j, τ_j appearing above are rational functions of the reflection entries whose denominators involve only $p_1, \dots, p_n$. Perturb the scalar datum λ_{2n+2} in the ambient reflection sequence, without asserting that the perturbed sequence is itself a Li sequence. This leaves the previously formed pivots $p_1, \dots, p_n$ unchanged, preserves the algebraic reflection identities, and makes p_{n+1} nonzero except at one parameter value. Both sides of (15.21) are continuous as the perturbation is removed. Hence the identity extends to $p_{n+1} = 0$ as well. □

Corollary 15.7 (Defect-energy form)**.** *With σ as in Theorem 15.6, define*

$$d_n := 2n + \sigma - \ell_n, \tag{15.27}$$

$$c_n := 2n^2 + (1 + 2\sigma)n - \tau_n, \tag{15.28}$$

$$V_n := (2n + \sigma + 2)d_n - \frac{1}{2}d_n^2 - c_n, \qquad d_0 = c_0 = V_0 = 0. \tag{15.29}$$

Then

$$\boxed{\frac{p_{n+1}}{p_n} = V_n - V_{n-1} - \frac{1}{2}(d_n - d_{n-1})^2.} \tag{15.30}$$

In particular, under the inductive hypothesis $p_1, \dots, p_n > 0$,

$$p_{n+1} > 0 \quad \Longleftrightarrow \quad V_n - V_{n-1} > \frac{1}{2}(d_n - d_{n-1})^2. \tag{15.31}$$

Equivalently, the symmetric cell

$$\mathcal{C}_n = \begin{pmatrix} V_n - V_{n-1} & (d_n - d_{n-1})/\sqrt{2} \\ (d_n - d_{n-1})/\sqrt{2} & 1 \end{pmatrix} \tag{15.32}$$

satisfies

$$\det \mathcal{C}_n = \frac{p_{n+1}}{p_n}. \tag{15.33}$$

Thus the next reflection pivot is controlled by a 2×2 local cell built only from the first two subdiagonals of the LDL^T elimination factor.

Proof. Substitute the definitions (15.27) and (15.28) into (15.21), and simplify. Equation (15.33) is the Schur-complement identity for the lower-right entry 1. □

Corollary 15.8 (Cost)**.** *Suppose the inverse-producing steps under consideration have nonzero Schur pivots. Given E_{N-1}^{-1} and $\mathsf{b}^{(N-1)}, \mathsf{b}^{(N-2)}$, one step of Propositions 15.2 and 15.4 produces $\mathsf{b}^{(N)}$, p_N, $g^{(N)}$ and E_N^{-1} using one banded matrix–vector product and two dense matrix–vector products, that is $O(N^2)$ operations and no determinant evaluation. In particular the whole pivot sequence $p_1, \dots, p_N$ is obtained in $O(N^3)$ operations without ever forming Δ_N^E.*

Proof. The column-wave update is banded and costs $O(N)$. Forming the required inverse generator vectors uses dense matrix–vector products with an $(N-1) \times (N-1)$ matrix and therefore costs $O(N^2)$; the rank-one block-inverse update also costs $O(N^2)$. Thus one nonsingular bordering step is $O(N^2)$, and summing these costs through order N gives $O(N^3)$. No determinant evaluation is required, because p_n is obtained directly from the Schur formula. □

Experiment 15.9 (Numerical verification of the two-generator scheme)**.** With the arithmetic λ of Experiment 26.4 and 400 decimal working digits, the identity (15.2) holds to 6.8×10^{-400} for all $N \le 23$ and $m \le 20$, that is to working precision; it is an exact algebraic identity. The recursion (15.8)–(15.9) was run from $N = 2$ to $N = 22$, updating E_N^{-1} by (15.9) alone and never recomputing an inverse. Against direct multiprecision evaluation of $\Delta_N^E/\Delta_{N-1}^E$ and of $\mathsf{E}_N^{-1}e_N$:

N	p_N (recursion)	rel. error in p_N	rel. error in $g^{(N)}$
2	$3.360255490 \times 10^{-7}$	1.6×10^{-395}	1.6×10^{-395}
5	$6.620331547 \times 10^{-26}$	7.6×10^{-356}	7.6×10^{-356}
8	$1.511647984 \times 10^{-46}$	4.8×10^{-311}	4.8×10^{-311}
12	$1.214721178 \times 10^{-75}$	8.3×10^{-248}	8.3×10^{-248}
16	$2.446315652 \times 10^{-106}$	3.8×10^{-182}	3.8×10^{-182}
20	$4.565211501 \times 10^{-138}$	1.1×10^{-113}	1.1×10^{-113}
22	$2.854130423 \times 10^{-154}$	7.1×10^{-79}	7.1×10^{-79}

The relative error degrades by roughly fifteen digits per step, which is the expected conditioning cost of the bordering and matches the decay rate of p_N; it is not an instability of the formulas. The identity $g_N^{(N)} p_N = 1$ holds exactly at every step.

The two generators have strictly alternating signs of opposite phase for all $N \le 22$:

$$\operatorname{sign}(g_i) = (-1)^{N-1-i}, \qquad \operatorname{sign}(h_i) = (-1)^{N-i}, \qquad 1 \le i \le N-1, \tag{15.34}$$

and $q = h + \omega_N g$ inherits the phase of g. This holds with the arithmetic λ, hence independently of any zero data.

Remark 15.10 (The remaining sign problem is local). Propositions 15.4 and 15.5 close the bordering dynamics algebraically, while Theorem 15.6 removes the large inverse from the sign test altogether: once $p_n > 0$ is known, the next step is exactly the local inequality (15.31). The checkerboard pattern (15.34) is therefore evidence about the elimination geometry, not a substitute for the missing proof. The unresolved task is to derive (15.31) from structural properties of the reflection matrices under the inductive hypothesis, without replacing the local identity by a termwise bound on the ill-conditioned inverse.

16. Orthogonal polynomials and infinite Jacobi operators

Assume RH. Let π_n^E and π_n^O denote the monic orthogonal polynomials associated with μ_E and μ_O. The following elementary argument records the hypotheses behind their three-term recurrences. For either positive measure $\mu = \mu_E$ or μ_O, let $h_n = \|\pi_n\|^2_{L^2(\mu)} > 0$. Since $x\pi_n$ is monic of degree $n+1$, it has an expansion

$$x\pi_n = \pi_{n+1} + \sum_{j=0}^{n} c_{n,j}\pi_j. \tag{16.1}$$

For $j \le n-2$, symmetry of multiplication by the real variable x gives

$$\langle x\pi_n, \pi_j\rangle = \langle \pi_n, x\pi_j\rangle = 0, \tag{16.2}$$

because $x\pi_j$ has degree at most $n-1$. Thus only the $j = n$ and $j = n-1$ coefficients can remain. Moreover

$$c_{n,n-1} = \frac{\langle x\pi_n, \pi_{n-1}\rangle}{h_{n-1}} = \frac{\langle \pi_n, x\pi_{n-1}\rangle}{h_{n-1}} = \frac{h_n}{h_{n-1}} > 0, \tag{16.3}$$

because $x\pi_{n-1} = \pi_n +$ lower-degree terms. Writing $b_n := c_{n,n}$ and $(a_n)^2 := h_n/h_{n-1}$ gives

$$x\pi_n^E(x) = \pi_{n+1}^E(x) + b_n^E\pi_n^E(x) + (a_n^E)^2\pi_{n-1}^E(x), \tag{16.4}$$

$$x\pi_n^O(x) = \pi_{n+1}^O(x) + b_n^O\pi_n^O(x) + (a_n^O)^2\pi_{n-1}^O(x), \tag{16.5}$$

with $a_n^E, a_n^O > 0$ for $n \ge 1$ and the convention $\pi_{-1} = 0$. After normalizing $q_n = \pi_n/\sqrt{h_n}$, multiplication by x has the symmetric tridiagonal matrix with diagonal b_n and positive off-diagonal a_{n+1}. Because the measures are supported in $[-1,1]$, multiplication by x is a bounded self-adjoint operator of norm at most one on each L^2 space; once polynomial completeness is established below, the corresponding Jacobi matrices define bounded self-adjoint operators $\mathcal{J}_E$ and $\mathcal{J}_O$ unitarily equivalent to multiplication by x.

The first recurrence coefficients are explicit in the Li coefficients:

$$b_0^E = \frac{\lambda_3 - \lambda_1}{2\lambda_2}, \tag{16.6}$$

$$(a_1^E)^2 = \frac{\lambda_2\lambda_4 - (\lambda_3 - \lambda_1)^2}{4\lambda_2^2}, \tag{16.7}$$

$$b_0^O = \frac{\lambda_2 - 2\lambda_1}{2\lambda_1}, \tag{16.8}$$

$$(a_1^O)^2 = \frac{\lambda_1\lambda_3 - (\lambda_2 - \lambda_1)^2}{4\lambda_1^2}. \tag{16.9}$$

Proposition 16.1 (Jacobi off-diagonals from the principal determinants)**.** *Let*

$$d_n^E := \frac{\Delta_n^E}{\Delta_{n-1}^E}, \qquad d_n^O := \frac{\Delta_n^O}{\Delta_{n-1}^O}. \tag{16.10}$$

Under RH*, for every* $n \geq 1$*,*

$$(a_n^E)^2 = \frac{1}{4}\frac{d_{n+1}^E}{d_n^E} = \frac{1}{4}\frac{\Delta_{n+1}^E\Delta_{n-1}^E}{(\Delta_n^E)^2}, \tag{16.11}$$

$$(a_n^O)^2 = \frac{1}{4}\frac{d_{n+1}^O}{d_n^O} = \frac{1}{4}\frac{\Delta_{n+1}^O\Delta_{n-1}^O}{(\Delta_n^O)^2}. \tag{16.12}$$

Proof. The basis polynomial U_{n-1}, and also P_{n-1}, has leading coefficient 2^{n-1}. Therefore the squared norm h_{n-1} of the corresponding monic orthogonal polynomial is

$$h_{n-1}^E = \frac{d_n^E}{4^{n-1}}, \qquad h_{n-1}^O = \frac{d_n^O}{4^{n-1}}. \tag{16.13}$$

For monic orthogonal polynomials $(a_n)^2 = h_n/h_{n-1}$, which yields (16.11)–(16.12). □

Let Γ_+^* denote the set of *distinct* positive ordinates of nontrivial zeros and let m_γ denote multiplicity.

The measures μ_E and μ_O are convenient unnormalized representing measures. The spectral measure of the first Jacobi basis vector is instead the corresponding probability measure

$$\nu_E := \frac{\mu_E}{\lambda_2}, \qquad \nu_O := \frac{\mu_O}{\lambda_1}, \tag{16.14}$$

because $\mu_E([-1,1]) = \lambda_2$ and $\mu_O([-1,1]) = \lambda_1$. This normalization does not alter the orthogonal polynomials except for their orthonormal scaling, and it does not alter the associated Jacobi operator.

Theorem 16.2 (Exact spectrum of the infinite Jacobi operators)**.** *Assume* RH*. Then*

$$\operatorname{spec}(\mathcal{J}_E) = \operatorname{spec}(\mathcal{J}_O) = \{1\} \cup \{x_\gamma : \gamma \in \Gamma_+^*\}, \tag{16.15}$$

where

$$x_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}. \tag{16.16}$$

Each x_γ *is an isolated eigenvalue of the multiplication operator with spectral mass determined by* m_γ*, while* 1 *is the unique accumulation point of the spectrum. The inverse transformation is*

$$\gamma = \frac{1}{2}\sqrt{\frac{1+x}{1-x}} \qquad (-1 < x < 1). \tag{16.17}$$

Moreover,

$$I - \mathcal{J}_E \in \mathcal{S}_1, \qquad I - \mathcal{J}_O \in \mathcal{S}_1, \tag{16.18}$$

so both are trace-class perturbations of the identity and have essential spectrum $\{1\}$.

Proof. We treat either measure $\mu = \mu_E$ or μ_O; the support is the same in both cases. Every atom has positive mass, and $x_\gamma \to 1$ as $\gamma \to \infty$. Hence the topological support is exactly

$$K := \operatorname{supp} \mu = \{x_\gamma : \gamma \in \Gamma_+^*\} \cup \{1\}. \tag{16.19}$$

There is no atom at 1; it is present because it is the unique accumulation point of the atoms.

Because K is a compact subset of $\mathbb{R}$, the restrictions of algebraic polynomials form a unital subalgebra of $C(K)$ that separates points. The Stone–Weierstrass theorem therefore makes polynomials uniformly dense in $C(K)$. Continuous functions are dense in $L^2(\mu)$ for every finite Borel measure on a compact metric space, hence polynomials are dense in $L^2(\mu)$. Gram–Schmidt applied to $1, x, x^2, \dots$ therefore produces a complete orthonormal polynomial basis $\{p_n\}_{n\geq 0}$.

Let $\mathcal{F} : \ell^2(\mathbb{N}_0) \to L^2(\mu)$ send the standard vector e_n to p_n. Completeness makes $\mathcal{F}$ unitary. Multiplication by x is a bounded self-adjoint operator M_x with $\|M_x\| \leq 1$. Its matrix in the orthonormal polynomial basis is the Jacobi matrix $\mathcal{J}$ coming from the three-term recurrence, so

$$\mathcal{J} = \mathcal{F}^{-1} M_x \mathcal{F}. \tag{16.20}$$

Now identify $L^2(\mu)$ explicitly with a weighted sequence space. If $w_\gamma = \mu(\{x_\gamma\})$, then

$$Tf := \big(\sqrt{w_\gamma}\, f(x_\gamma)\big)_{\gamma \in \Gamma_+^*} \tag{16.21}$$

is unitary from $L^2(\mu)$ onto $\ell^2(\Gamma_+^*)$, and

$$T M_x T^{-1} = \operatorname{diag}(x_\gamma). \tag{16.22}$$

The spectrum of a bounded diagonal operator is the closure of its diagonal entries. Since the only accumulation point is 1,

$$\operatorname{spec}(\mathcal{J}) = \{x_\gamma : \gamma \in \Gamma_+^*\} \cup \{1\}. \tag{16.23}$$

Each x_γ is an isolated eigenvalue with one-dimensional eigenspace generated by the normalized atom indicator. The point 1 belongs to the spectrum as a limit point but is not an eigenvalue because $\mu(\{1\}) = 0$. Solving $x = (4\gamma^2 - 1)/(4\gamma^2 + 1)$ for $\gamma > 0$ gives (16.17).

Under the same unitary identification,

$$I - \mathcal{J} \sim \operatorname{diag}(1 - x_\gamma), \qquad 1 - x_\gamma = \frac{2}{4\gamma^2 + 1}. \tag{16.24}$$

To prove trace-class membership without appealing to an unnamed estimate, enumerate distinct ordinates increasingly as γ_j^*. Lemma 12.4 gives $\gamma_j^* \gg j/\log j$, hence

$$\sum_j (1 - x_{\gamma_j^*}) \ll \sum_{j \geq 2} \frac{(\log j)^2}{j^2} < \infty. \tag{16.25}$$

A positive diagonal operator is trace class exactly when the sum of its diagonal eigenvalues is finite. Therefore

$$\operatorname{Tr}(I - \mathcal{J}) = \sum_{\gamma \in \Gamma_+^*} \frac{2}{4\gamma^2 + 1}, \tag{16.26}$$

and the same expression holds for both E and O because their supports coincide.

For every integer $k \geq 1$, the positive operator $(I - \mathcal{J})^k$ is trace class as well, since $0 \leq 1 - x_\gamma < 1$ and $\sum (1 - x_\gamma) < \infty$. Its eigenvalues are $(1 - x_\gamma)^k$, so for both reflection families

$$\operatorname{Tr}\Big((I - \mathcal{J}_E)^k\Big) = \operatorname{Tr}\Big((I - \mathcal{J}_O)^k\Big) = \sum_{\gamma \in \Gamma_+^*} \left(\frac{2}{4\gamma^2 + 1}\right)^k. \tag{16.27}$$

Finally, trace invariance under change of orthonormal basis gives

$$\sum_{n\geq 0}(1-b_n^E)=\sum_{n\geq 0}(1-b_n^O)=\sum_{\gamma\in\Gamma_+^*}\frac{2}{4\gamma^2+1}. \tag{16.28}$$

Indeed, for either family the left side is $\sum_n\langle(I-\mathcal{J})e_n,e_n\rangle=\mathrm{Tr}(I-\mathcal{J})$; monotone convergence is automatic because $I-\mathcal{J}\geq 0$. □

Remark 16.3 (Finite truncations are not formulas for the zeros). The $N\times N$ leading Jacobi truncations have eigenvalues equal to the zeros of π_N^E or π_N^O, equivalently Gaussian-quadrature nodes for the corresponding measure. These nodes are generally not points x_γ. Exact recovery of the support in (16.15) is an infinite inverse-moment statement requiring the full coefficient sequence. Thus (16.17) converts an already recovered spectral support point into an ordinate; it is not a finite closed formula for a zeta zero.

Proposition 16.4 (Spectral masses and multiplicities)**.** *For $\gamma\in\Gamma_+^*$, the unnormalized representing measures satisfy*

$$\mu_O(\{x_\gamma\})=\frac{4m_\gamma}{4\gamma^2+1}, \tag{16.29}$$

$$\mu_E(\{x_\gamma\})=\frac{64m_\gamma\gamma^2}{(4\gamma^2+1)^2}. \tag{16.30}$$

Accordingly, the spectral measures of the first Jacobi basis vector are

$$\nu_O(\{x_\gamma\})=\frac{4m_\gamma}{\lambda_1(4\gamma^2+1)}, \tag{16.31}$$

$$\nu_E(\{x_\gamma\})=\frac{64m_\gamma\gamma^2}{\lambda_2(4\gamma^2+1)^2}. \tag{16.32}$$

Thus zero multiplicity is encoded by the distinguished cyclic spectral measure, even though the abstract multiplication operator has only one one-dimensional eigenspace for each distinct support point x_γ.

Proof. The first pair follows from (8.5), (8.6), and (8.15)–(8.16). The second pair follows by the normalization (16.14). □

Corollary 16.5 (Trace and simplicity defect)**.** *Under* RH,

$$\frac{\lambda_1}{2}-\mathrm{Tr}(I-\mathcal{J}_O)=2\sum_{\gamma\in\Gamma_+^*}\frac{m_\gamma-1}{4\gamma^2+1}\geq 0. \tag{16.33}$$

The same trace occurs for $\mathcal{J}_E$. Equality in (16.33) *holds if and only if every positive ordinate is simple.*

Proof. Equation (2.12) with $n=1$ and (2.10) give

$$\frac{\lambda_1}{2}=2\sum_{\gamma\in\Gamma_+^*}\frac{m_\gamma}{4\gamma^2+1}. \tag{16.34}$$

Subtract (16.26). Every summand in the resulting series is nonnegative, and it vanishes exactly when $m_\gamma=1$. □

Proposition 16.6 (Operator bounds and recovery of the first distinct ordinate)**.** *Assume* RH *and let γ_1 be the smallest positive zero ordinate. Then*

$$x_{\gamma_1}I\leq\mathcal{J}_E,\mathcal{J}_O\leq I, \tag{16.35}$$

and

$$\|I - \mathcal{J}_E\| = \|I - \mathcal{J}_O\| = \frac{2}{4\gamma_1^2 + 1}. \tag{16.36}$$

Consequently

$$\gamma_1 = \frac{1}{2}\sqrt{\frac{2}{\|I - \mathcal{J}_E\|} - 1} = \frac{1}{2}\sqrt{\frac{2}{\|I - \mathcal{J}_O\|} - 1}. \tag{16.37}$$

This is an identity for the complete infinite operator, not a finite formula for a zeta zero.

Proof. The spectrum in (16.15) is contained in $[x_{\gamma_1}, 1]$, which gives (16.35) by the spectral theorem. The positive compact operator $I - \mathcal{J}$ has largest eigenvalue $1 - x_{\gamma_1} = 2/(4\gamma_1^2 + 1)$, proving (16.36) and its inversion. □

Theorem 16.7 (Singular values and the Schatten threshold)**.** *Assume* RH*, and enumerate the* distinct *positive ordinates increasingly as*

$$0 < \gamma_1^* < \gamma_2^* < \cdots. \tag{16.38}$$

Then the singular values of either defect operator $K_E := I - \mathcal{J}_E$ or $K_O := I - \mathcal{J}_O$ are

$$s_k(K_E) = s_k(K_O) = \frac{2}{4(\gamma_k^*)^2 + 1}. \tag{16.39}$$

There exist positive constants c, C such that, for all sufficiently large k,

$$c\frac{(\log\log k)^2}{k^2} \le s_k(K_E) = s_k(K_O) \le C\frac{(\log k)^2}{k^2}. \tag{16.40}$$

Consequently, for every $p > 0$,

$$I - \mathcal{J}_E,\ I - \mathcal{J}_O \in \mathcal{S}_p \quad \Longleftrightarrow \quad p > \frac{1}{2}. \tag{16.41}$$

No asymptotic equivalent for (16.39) *is asserted here, because the operator counts distinct ordinates rather than zeros with multiplicity.*

Proof. Equation (16.39) follows from the diagonal multiplication representation and positivity of K_E, K_O. Let $N(T)$ count zeros with multiplicity and $N^*(T)$ count distinct positive ordinates. The Riemann–von Mangoldt formula gives $N(T) \asymp T\log T$. Under RH, Littlewood's classical bound

$$S(T) = O\left(\frac{\log T}{\log\log T}\right) \tag{16.42}$$

implies the corresponding multiplicity bound

$$m(\tfrac{1}{2} + i\gamma) = O\left(\frac{\log\gamma}{\log\log\gamma}\right); \tag{16.43}$$

see, for example, [39]. Hence

$$T\log\log T \ll N^*(T) \ll T\log T. \tag{16.44}$$

Inverting these bounds gives

$$\frac{k}{\log k} \ll \gamma_k^* \ll \frac{k}{\log\log k}, \tag{16.45}$$

which yields (16.40). The upper bound is p-summable when $2p > 1$; the lower bound forces divergence when $2p \le 1$. This proves (16.41). □

Proposition 16.8 (Endpoint exponent and critical Schatten exponent)**.** *Let a positive discrete Gram system satisfy* (12.3) *with exponent $\beta > 0$, and assume the two-sided node scale*

$$\log t_j = (1 + o(1)) \log j. \tag{16.46}$$

Let $\mathcal{J}$ denote the Jacobi operator unitarily equivalent to multiplication by x on $L^2(\mu)$, and put $K := I - \mathcal{J}$. Since the support consists of the distinct atoms $x(t_j)$ with unique accumulation point 1, the nonzero eigenvalues, hence the singular values, of the positive compact operator K are the endpoint distances $1 - x(t_j)$. Then the critical Schatten exponent of K is

$$p_c(K) := \inf\{p > 0 : \ K \in \mathcal{S}_p\} = \frac{1}{\beta}. \tag{16.47}$$

If in addition

$$\sum_j \frac{1}{t_j} = \infty, \tag{16.48}$$

then membership at the critical exponent fails, and

$$K \in \mathcal{S}_p \quad \Longleftrightarrow \quad p > \frac{1}{\beta}. \tag{16.49}$$

Proof. By (12.3),

$$c_1 t_j^{-\beta} \le 1 - x(t_j) \le c_2 t_j^{-\beta} \tag{16.50}$$

for all sufficiently large j. From (16.46), for every $\varepsilon > 0$ there is J_ε such that

$$j^{1-\varepsilon} \le t_j \le j^{1+\varepsilon}, \qquad j \ge J_\varepsilon. \tag{16.51}$$

Therefore, for fixed $p > 0$ and large j,

$$c_1^p j^{-\beta p(1+\varepsilon)} \le (1 - x(t_j))^p \le c_2^p j^{-\beta p(1-\varepsilon)}. \tag{16.52}$$

If $p > 1/\beta$, choose $\varepsilon > 0$ so small that $\beta p(1 - \varepsilon) > 1$; the upper comparison in (16.52) gives convergence. If $p < 1/\beta$, choose $\varepsilon > 0$ so small that $\beta p(1 + \varepsilon) < 1$; the lower comparison gives divergence. This proves (16.47).

Hypothesis (16.46) alone does not decide the endpoint $p = 1/\beta$. The admissible sequence $t_j = j(\log j)^2$ satisfies (16.46) while $\sum_j t_j^{-1} < \infty$, so in that case $K \in \mathcal{S}_{1/\beta}$. Under the additional hypothesis (16.48) one has

$$\sum_j (1 - x(t_j))^{1/\beta} \asymp \sum_j \frac{1}{t_j} = \infty, \tag{16.53}$$

so $K \notin \mathcal{S}_{1/\beta}$, and (16.49) follows. □

Remark 16.9 (One geometric constant, two invariants). Theorem 12.1 and Proposition 16.8 record the same quantity:

$$-\lim_{N\to\infty} \frac{\log D_N}{N^2 \log N} = \beta = \frac{1}{p_c(I - \mathcal{J})}. \tag{16.54}$$

Once their respective coarse node hypotheses are satisfied, both invariants are fixed by the same endpoint exponent. The determinant additionally uses the separated-subconfiguration and weight hypotheses of Theorem 12.1, whereas the critical Schatten exponent uses the two-sided logarithmic node scale (16.46). Neither invariant records the logarithmic Riemann–von Mangoldt refinement. The determinant is a macroscopic quantity attached to the finite compressions, the critical exponent is an ideal-theoretic invariant of the infinite defect operator, and (16.54) states that, at this coarse level, they encode the same endpoint exponent.

For the Li–Cayley map (12.59) one has $\beta = 2$, so the coefficient -2 of Theorem 12.6 and the value $1/2$ of Theorem 16.7 are one geometric constant seen twice. The divergence hypothesis (16.48)

holds in that case: by (16.45) one has $\gamma_k^* \ll k/\log\log k$, so $\sum_k (\gamma_k^*)^{-1} \gg \sum_k (\log\log k)/k = \infty$, and the strict form (16.49) is recovered.

In particular the value $1/2$ here is not the critical line. Replacing the endpoint map by one with exponent $\beta = 3$ moves the critical Schatten exponent to $1/3$ while leaving the zeros untouched, and the value is likewise unrelated to abscissas of convergence of prime series, where a coincidental $1/2$ also occurs. This reinforces Remark 12.2: the leading-order data of the pair is geometric, and arithmetic information can enter only below it.

Corollary 16.10 (Quadratic Jacobi sum rule)**.** *Under* RH*,*

$$\sum_{n\geq 0}(1-b_n^E)^2 + 2\sum_{n\geq 0}(a_{n+1}^E)^2 = \sum_{\gamma\in\Gamma_+^*}\left(\frac{2}{4\gamma^2+1}\right)^2, \tag{16.55}$$

and the identical formula holds with E replaced by O.

Proof. For $K = I - \mathcal{J}$, the Hilbert–Schmidt norm squared is the sum of the squares of all matrix entries in the Jacobi basis and also the sum of the squares of its eigenvalues. Since the off-diagonal entries occur twice, the stated identity follows. □

Proposition 16.11 (Local Jacobi inequalities and summability)**.** *Under* RH*, for either Jacobi family,*

$$a_{n+1}^2 \leq (1-b_n)(1-b_{n+1}) \qquad (n\geq 0). \tag{16.56}$$

Moreover

$$\sum_{n\geq 0}(1-b_n) < \infty, \qquad \sum_{n\geq 0} a_{n+1} < \infty, \tag{16.57}$$

and therefore

$$b_n \longrightarrow 1, \qquad a_n \longrightarrow 0. \tag{16.58}$$

Proof. The operator $I - \mathcal{J}$ is positive. Positivity of each consecutive 2×2 principal minor gives (16.56). Equation (16.28) gives the first series in (16.57). Since $1 - b_n \geq 0$,

$$a_{n+1} \leq \sqrt{(1-b_n)(1-b_{n+1})} \leq \frac{(1-b_n)+(1-b_{n+1})}{2}, \tag{16.59}$$

which gives the second series and the limits. □

Proposition 16.12 (Fredholm determinant of the Jacobi defect)**.** *Under* RH*, the trace-class operators $K_E = I - \mathcal{J}_E$ and $K_O = I - \mathcal{J}_O$ have the same Fredholm determinant*

$$D(z) := \det(I - zK_E) = \det(I - zK_O) = \prod_{\gamma\in\Gamma_+^*}\left(1 - \frac{2z}{4\gamma^2+1}\right). \tag{16.60}$$

The product converges absolutely and locally uniformly, and the zeros of D are

$$z_\gamma = \frac{4\gamma^2+1}{2}, \qquad \gamma\in\Gamma_+^*, \tag{16.61}$$

one for each distinct positive ordinate. Zeta multiplicity is not encoded as the zero multiplicity of D.

Proof. By Theorem 16.2, each defect $K = I - \mathcal{J}$ is positive and trace class, and under the multiplication representation it is diagonal on the normalized atom indicators with eigenvalues

$$\delta_\gamma := 1 - x_\gamma = \frac{2}{4\gamma^2+1} > 0, \qquad \gamma\in\Gamma_+^*. \tag{16.62}$$

The only additional spectral point is 0, coming from the accumulation $x_\gamma \to 1$; it contributes no factor to the Fredholm determinant. Since

$$\sum_{\gamma\in\Gamma_+^*} \delta_\gamma < \infty, \tag{16.63}$$

the trace-class Fredholm determinant satisfies

$$\det(I - zK) = \prod_{\gamma\in\Gamma_+^*} (1 - z\delta_\gamma). \tag{16.64}$$

For completeness, local uniform convergence follows directly: on $|z| \le R$, choose a tail for which $R\delta_\gamma \le 1/2$; then $|\log(1 - z\delta_\gamma)| \le 2R\delta_\gamma$, and the Weierstrass test applies. Thus the product defines an entire function and can vanish only when one factor vanishes, namely at

$$z = \delta_\gamma^{-1} = \frac{4\gamma^2 + 1}{2}. \tag{16.65}$$

Each distinct support point supplies one one-dimensional eigenspace of the multiplication operator, so these zeros are indexed by distinct ordinates rather than by zeta multiplicity. The two reflection families have the same support and hence the same eigenvalue list $\{\delta_\gamma\}$, proving equality of their Fredholm determinants. □

Remark 16.13 (What the Jacobi operator does and does not retain). The abstract operator $\mathcal{J}$ determines the set of distinct transformed ordinates through its spectrum, but not the zeta multiplicities. The normalized cyclic pair $(\mathcal{J}, e_0)$ determines the probability spectral measure and therefore the relative masses. To recover the integer multiplicities from the odd realization one must also retain its scale λ_1, equivalently the unnormalized measure μ_O. Thus

$$(\mathcal{J}_O, e_0, \lambda_1) \quad \text{determines} \quad \{(\gamma, m_\gamma) : \gamma \in \Gamma_+^*\}. \tag{16.66}$$

The original semi-infinite matrix O already retains this scale because $o_{11} = \lambda_1$.

Proposition 16.14 (Unitary equivalence of the two Jacobi operators)**.** *Under* RH*, $\mathcal{J}_E$ and $\mathcal{J}_O$ are unitarily equivalent.*

Proof. From (8.9),

$$\mathrm{d}\mu_E(x) = 2(1 + x)\ \mathrm{d}\mu_O(x). \tag{16.67}$$

The actual support is contained in $[x_{\gamma_1}, 1)$ with $x_{\gamma_1} > -1$, so $2(1+x)$ is bounded above and bounded away from zero on the support. Define

$$(Uf)(x) := \sqrt{2(1 + x)}\, f(x). \tag{16.68}$$

Then for every $f \in L^2(\mu_E)$,

$$\|Uf\|^2_{L^2(\mu_O)} = \int 2(1 + x)|f(x)|^2\ \mathrm{d}\mu_O(x) = \int |f(x)|^2\ \mathrm{d}\mu_E(x), \tag{16.69}$$

so U is an isometry. Because the multiplier $[2(1 + x)]^{-1/2}$ is bounded on the support, every $g \in L^2(\mu_O)$ has the preimage $f = g/[2(1 + x)]^{1/2} \in L^2(\mu_E)$; hence U is onto and therefore unitary.

Let M_x denote multiplication by x. Since both U and M_x are multiplication operators, $UM_x = M_xU$ on all of $L^2(\mu_E)$. Let $\mathcal{F}_E : \ell^2(\mathbb{N}_0) \to L^2(\mu_E)$ and $\mathcal{F}_O : \ell^2(\mathbb{N}_0) \to L^2(\mu_O)$ be the unitary maps sending the standard basis to the corresponding orthonormal polynomial bases. By definition

$$\mathcal{J}_E = \mathcal{F}_E^{-1} M_x \mathcal{F}_E, \qquad \mathcal{J}_O = \mathcal{F}_O^{-1} M_x \mathcal{F}_O. \tag{16.70}$$

Thus $V := \mathcal{F}_O^{-1} U \mathcal{F}_E$ is unitary and

$$V\mathcal{J}_E = \mathcal{F}_O^{-1} U M_x \mathcal{F}_E = \mathcal{F}_O^{-1} M_x U \mathcal{F}_E = \mathcal{J}_O V, \tag{16.71}$$

which proves unitary equivalence. □

Theorem 16.15 (Darboux factorization of the Jacobi pair)**.** *Assume* RH. *Since* -1 *lies strictly below the common spectral support,* $\mathcal{J}_O + I$ *is strictly positive. Let*

$$\mathcal{J}_O + I = LL^* \tag{16.72}$$

be its positive lower-bidiagonal Cholesky factorization. Then

$$\mathcal{J}_E + I = L^*L. \tag{16.73}$$

Writing

$$L = \begin{pmatrix} r_0 & & & \\ s_0 & r_1 & & \\ & s_1 & r_2 & \\ & & \ddots & \ddots \end{pmatrix}, \qquad r_n > 0, \tag{16.74}$$

one obtains

$$1 + b_n^O = r_n^2 + s_{n-1}^2, \qquad a_{n+1}^O = r_n s_n, \tag{16.75}$$

$$1 + b_n^E = r_n^2 + s_n^2, \qquad a_{n+1}^E = s_n r_{n+1}, \tag{16.76}$$

with $s_{-1} := 0$.

Proof. Let $\{p_n^E\}_{n\geq 0}$ and $\{p_n^O\}_{n\geq 0}$ be the orthonormal polynomial bases in $L^2(\mu_E)$ and $L^2(\mu_O)$, chosen with positive leading coefficients. Consider the bounded operator

$$T : L^2(\mu_E) \longrightarrow L^2(\mu_O), \qquad (Tf)(x) := (x+1)f(x). \tag{16.77}$$

It is bounded because, using $\mathrm{d}\mu_E = 2(1+x)\,\mathrm{d}\mu_O$ and $0 < 1 + x \leq 2$ on the support,

$$\|Tf\|^2_{L^2(\mu_O)} = \int (1+x)^2 |f(x)|^2 \,\mathrm{d}\mu_O(x) \tag{16.78}$$

$$= \frac{1}{2}\int (1+x)|f(x)|^2 \,\mathrm{d}\mu_E(x) \leq \|f\|^2_{L^2(\mu_E)}. \tag{16.79}$$

For $f \in L^2(\mu_E)$ and $g \in L^2(\mu_O)$,

$$\langle Tf, g\rangle_O = \int (x+1) f\overline{g} \,\mathrm{d}\mu_O = \frac{1}{2}\int f\overline{g} \,\mathrm{d}\mu_E. \tag{16.80}$$

The identity embedding $g \mapsto g$ from $L^2(\mu_O)$ to $L^2(\mu_E)$ is bounded, since

$$\|g\|^2_{L^2(\mu_E)} = 2\int (1+x)|g|^2 \,\mathrm{d}\mu_O \leq 4\|g\|^2_{L^2(\mu_O)}. \tag{16.81}$$

Hence the adjoint is the everywhere-defined bounded operator

$$T^* g = \frac{1}{2} g. \tag{16.82}$$

Consequently

$$T^*Tf = \frac{1}{2}(x+1)f, \qquad TT^*g = \frac{1}{2}(x+1)g. \tag{16.83}$$

Let R be the matrix of T from the E orthonormal basis to the O orthonormal basis,

$$R_{mn} := \langle p_m^O, Tp_n^E\rangle_O. \tag{16.84}$$

The polynomial $(x+1)p_n^E$ has degree $n+1$ and, by the same orthogonality calculation used in Proposition 16.17, is orthogonal in $L^2(\mu_O)$ to every polynomial of degree $< n$. Therefore

$$Tp_n^E = \alpha_n p_n^O + \beta_n p_{n+1}^O, \tag{16.85}$$

with no other orthonormal-polynomial components. Comparison of positive leading coefficients gives $\beta_n > 0$. The diagonal coefficient is also strictly positive:

$$\alpha_n = \langle p_n^O, (x+1)p_n^E\rangle_O = \frac{1}{2}\langle p_n^O, p_n^E\rangle_E.$$

Writing $p_n^O = (\kappa_n^O/\kappa_n^E)p_n^E +$ lower-degree terms, where $\kappa_n^{E,O} > 0$ are the leading coefficients, and using the E-orthogonality of p_n^E to lower-degree polynomials gives

$$\alpha_n = \frac{1}{2}\,\frac{\kappa_n^O}{\kappa_n^E} > 0.$$

Thus R is lower bidiagonal with positive diagonal and positive subdiagonal. In coefficient space, (16.83) becomes

$$R^*R = \frac{1}{2}(\mathcal{J}_E + I), \qquad RR^* = \frac{1}{2}(\mathcal{J}_O + I). \tag{16.86}$$

Set

$$L := \sqrt{2}\,R. \tag{16.87}$$

Then L is lower bidiagonal and

$$\mathcal{J}_O + I = LL^*, \qquad \mathcal{J}_E + I = L^*L. \tag{16.88}$$

Since the common spectrum is contained in $[x_{\gamma_1}, 1]$ with $x_{\gamma_1} > -1$, both operators $\mathcal{J}_{E,O} + I$ are bounded below by $(1 + x_{\gamma_1})I > 0$. Hence the bidiagonal factor has no zero diagonal entry; with the positive-leading-coefficient convention we take $r_n > 0$. This establishes the asserted Cholesky/Darboux factorization without an appeal to a formal infinite Gaussian elimination.

Finally, write $L_{nn} = r_n$ and $L_{n+1,n} = s_n$, with $s_{-1} = 0$. Direct multiplication gives

$$(LL^*)_{nn} = r_n^2 + s_{n-1}^2, \qquad (LL^*)_{n,n+1} = r_n s_n, \tag{16.89}$$

$$(L^*L)_{nn} = r_n^2 + s_n^2, \qquad (L^*L)_{n,n+1} = s_n r_{n+1}. \tag{16.90}$$

Comparison with the Jacobi entries proves (16.75)–(16.76). □

Proposition 16.16 (Exact relation between the Weyl functions)**.** *Define, for z off the common spectrum,*

$$m_E(z) := \int \frac{\mathrm{d}\nu_E(x)}{x - z}, \qquad m_O(z) := \int \frac{\mathrm{d}\nu_O(x)}{x - z}. \tag{16.91}$$

Then

$$m_E(z) = \frac{2\lambda_1}{\lambda_2}\left[1 + (z+1)m_O(z)\right], \tag{16.92}$$

and, for $z \neq -1$, equivalently

$$m_O(z) = \frac{\dfrac{\lambda_2}{2\lambda_1}m_E(z) - 1}{z+1}. \tag{16.93}$$

Proof. From (8.9) and (16.14),

$$\mathrm{d}\nu_E(x) = \frac{2\lambda_1}{\lambda_2}(1+x)\,\mathrm{d}\nu_O(x). \tag{16.94}$$

Use

$$\frac{1+x}{x-z} = 1 + \frac{z+1}{x-z} \tag{16.95}$$

and $\nu_O([-1,1]) = 1$. □

Proposition 16.17 (Christoffel relation between orthogonal systems)**.** *Under* RH*,*

$$\pi_n^E(x) = \frac{\pi_{n+1}^O(x) - \dfrac{\pi_{n+1}^O(-1)}{\pi_n^O(-1)}\pi_n^O(x)}{x+1}. \tag{16.96}$$

The denominator $\pi_n^O(-1)$ *never vanishes.*

Proof. The scalar factor 2 in $\mathrm{d}\mu_E = 2(1+x)\,\mathrm{d}\mu_O$ does not affect orthogonality. Let q be any polynomial with $\deg q < n$. Since π_n^E is orthogonal to every such q in $L^2(\mu_E)$,

$$0 = \int \pi_n^E(x) q(x)\ \mathrm{d}\mu_E(x) = 2\int (x+1)\pi_n^E(x) q(x)\ \mathrm{d}\mu_O(x). \tag{16.97}$$

Thus the monic polynomial $(x+1)\pi_n^E(x)$ has degree $n+1$ and is orthogonal, with respect to μ_O, to all polynomials of degree at most $n-1$.

Because the monic orthogonal polynomials $\pi_0^O, \dots, \pi_{n+1}^O$ form a basis of the polynomials of degree at most $n+1$, write

$$(x+1)\pi_n^E(x) = \sum_{j=0}^{n+1} c_j \pi_j^O(x). \tag{16.98}$$

Taking the $L^2(\mu_O)$ inner product with π_j^O for $j < n$ shows $c_j = 0$. Comparing leading coefficients gives $c_{n+1} = 1$, so

$$(x+1)\pi_n^E(x) = \pi_{n+1}^O(x) + c_n \pi_n^O(x). \tag{16.99}$$

Every zero of π_n^O lies in the open convex hull of the support of the positive measure μ_O; the support is contained in $(-1,1)$ and has left endpoint $x_{\gamma_1} > -1$. Hence $\pi_n^O(-1) \neq 0$. Evaluating the preceding identity at $x = -1$ gives $c_n = -\pi_{n+1}^O(-1)/\pi_n^O(-1)$, which is exactly (16.96) after division by $x+1$. The numerator vanishes at $x = -1$, so the quotient is indeed a polynomial of degree n. □

Corollary 16.18 (Determinant bridge)**.** *Under* RH*,*

$$\det \mathsf{E}_N = 2^N (-1)^N \pi_N^O(-1) \det \mathsf{O}_N. \tag{16.100}$$

Proof. We first derive the required Christoffel determinant identity from the moment determinants. Let $\mu_k := \int x^k\ \mathrm{d}\mu_O(x)$ and

$$D_N[\mu_O] := \det(\mu_{i+j})_{i,j=0}^{N-1} > 0. \tag{16.101}$$

The monic orthogonal polynomial of degree N has the determinantal representation

$$\pi_N^O(c) = \frac{1}{D_N[\mu_O]} \det \begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_N \\ \mu_1 & \mu_2 & \cdots & \mu_{N+1} \\ \vdots & \vdots & & \vdots \\ \mu_{N-1} & \mu_N & \cdots & \mu_{2N-1} \\ 1 & c & \cdots & c^N \end{pmatrix}. \tag{16.102}$$

Indeed, expansion along the last row shows that the right side is monic of degree N; integrating it against $c^k\,\mathrm{d}\mu_O(c)$ for $0 \le k < N$ replaces the last row by the already present moment row $(\mu_k, \dots, \mu_{k+N})$, so the determinant vanishes. Thus it is exactly the monic orthogonal polynomial.

Starting from the determinant in (16.102), perform for $j = N-1, N-2, \dots, 0$ the column operation

$$C_{j+1} \leftarrow C_{j+1} - cC_j. \tag{16.103}$$

The determinant is unchanged. The last row becomes $(1,0,\dots,0)$, while after deleting its first column the remaining $N\times N$ minor has entries $\mu_{i+j+1}-c\mu_{i+j}$, $0\le i,j<N$. Expansion along the last row therefore gives

$$D_N[\mu_O]\,\pi_N^O(c)=(-1)^N\det(\mu_{i+j+1}-c\mu_{i+j})_{i,j=0}^{N-1}. \tag{16.104}$$

But the matrix on the right is precisely the moment matrix of the measure $(x-c)\,\mathrm{d}\mu_O(x)$. Hence

$$D_N[(x-c)\mu_O]=(-1)^N\pi_N^O(c)D_N[\mu_O]. \tag{16.105}$$

Taking $c=-1$ proves

$$D_N[(x+1)\mu_O]=(-1)^N\pi_N^O(-1)D_N[\mu_O]. \tag{16.106}$$

Now $\mu_E=2(x+1)\mu_O$. Multiplying a measure by the scalar 2 multiplies every entry of its $N\times N$ moment matrix by 2, and therefore its determinant by 2^N. Finally, Theorem 9.1 gives

$$\det \mathsf{E}_N=2^{N(N-1)}D_N[\mu_E],\qquad \det \mathsf{O}_N=2^{N(N-1)}D_N[\mu_O]. \tag{16.107}$$

The identical basis factors cancel in the ratio. Combining these identities with (16.106) yields

$$\det \mathsf{E}_N=2^N(-1)^N\pi_N^O(-1)\det \mathsf{O}_N, \tag{16.108}$$

as claimed. □

Corollary 16.19 (Bridge between the elimination pivots)**.** *Under* RH*,*

$$d_n^E=-2\frac{\pi_n^O(-1)}{\pi_{n-1}^O(-1)}d_n^O\qquad(n\ge 1), \tag{16.109}$$

where d_n^E,d_n^O are defined in (16.10)*.*

Proof. Divide (16.100) for $N=n$ by the same identity for $N=n-1$. □

17. The Christoffel margin: exact factorisation of the pivot ratio and a two-sided band

Corollary 16.18 links the two determinant sequences through the single number $\pi_N^O(-1)$. This section identifies that number as a truncated Fredholm determinant of the Jacobi defect, deduces an exact factorisation of the ratio of the two pivot sequences into a universal constant times an arithmetic factor, and turns the resulting monotonicity into a two-sided band which is a necessary condition for RH requiring no arithmetic input at all.

Definition 17.1 (Christoffel factor)**.** For $N\ge 1$ set

$$F_N:=\frac{(-1)^N\pi_N^O(-1)}{2^N},\qquad F_0:=1. \tag{17.1}$$

Corollary 16.18 then reads

$$\boxed{\Delta_N^E=4^N F_N\,\Delta_N^O} \tag{17.2}$$

Proposition 17.2 (F_N is a truncated Fredholm determinant of the Jacobi defect)**.** *Under* RH*, let $\mathcal{J}_O^{(N)}$ be the leading $N\times N$ truncation of the odd Jacobi operator and let $\mathcal{D}_N:=I_N-\mathcal{J}_O^{(N)}$ be its defect (the symbol $\mathcal{K}$ is reserved for the Christoffel–Darboux kernels of Section 14). Then*

$$\boxed{F_N=\det\left(\frac{I_N+\mathcal{J}_O^{(N)}}{2}\right)=\det\Big(I_N-\tfrac12\mathcal{D}_N\Big)=\sum_{k=0}^{N}\frac{(-1)^k}{2^k}\,e_k(\mathcal{D}_N),} \tag{17.3}$$

where e_k denotes the k-th elementary symmetric function of the eigenvalues.

Proof. The monic orthogonal polynomial of degree N is the characteristic polynomial of the leading Jacobi truncation, $\pi_N^O(x) = \det(xI_N - \mathcal{J}_O^{(N)})$; this is the statement recalled in Remark 16.3. Evaluating at $x = -1$,

$$\pi_N^O(-1) = \det(-I_N - \mathcal{J}_O^{(N)}) = (-1)^N \det(I_N + \mathcal{J}_O^{(N)}), \tag{17.4}$$

so $(-1)^N \pi_N^O(-1) = \det(I_N + \mathcal{J}_O^{(N)})$ and (17.1) gives the first equality in (17.3). The second is $I + \mathsf{J} = 2I - (I - \mathsf{J})$ divided by 2, and the third is the expansion of $\det(I - zK)$ in elementary symmetric functions at $z = 1/2$. □

Theorem 17.3 (Margin factorisation)**.** *Let $p_N^E := \Delta_N^E/\Delta_{N-1}^E$ and $p_N^O := \Delta_N^O/\Delta_{N-1}^O$ be the Cholesky pivots* (13.3)–(13.4)*, and define the* margin

$$\kappa_N := \frac{p_N^E}{p_N^O}. \tag{17.5}$$

Then, under RH*,*

$$\boxed{\kappa_N = 4\,\frac{F_N}{F_{N-1}}} \qquad (N \geq 1). \tag{17.6}$$

Within any positive even–odd pair with the same Christoffel relation $\mathrm{d}\mu_E = 2(1+x)\,\mathrm{d}\mu_O$ and the same polynomial normalizations used here, the constant 4 is independent of the particular measure. The measure-dependent information in this factorization sits in F_N/F_{N-1}.

Proof. Divide (17.2) at N by the same identity at $N-1$. □

Theorem 17.4 (Monotonicity, band, and Fredholm limit)**.** *Let $\mathcal{D} = I - \mathcal{J}_O$ be the Jacobi defect of Proposition 16.11, a positive trace-class operator satisfying $\operatorname{Tr}\mathcal{D} \leq \lambda_1/2$, with equality if and only if every positive zero ordinate is simple. Under* RH*:*

(i) F_N is nonincreasing in N and $0 < F_N \leq 1$;

(ii) $\det(I - \frac{1}{2}\mathcal{D}) \leq F_N \leq 1$ for every N, and $F_N \to \det(I - \frac{1}{2}\mathcal{D})$ as $N \to \infty$, where

$$\det\Big(I - \tfrac{1}{2}\mathcal{D}\Big) = \prod_{\gamma \in \Gamma_+^*} \frac{4\gamma^2}{4\gamma^2 + 1}; \tag{17.7}$$

(iii) consequently

$$\boxed{4\det\Big(I - \tfrac{1}{2}\mathcal{D}\Big) \;\leq\; \kappa_N \;\leq\; 4 \qquad (N \geq 1),} \tag{17.8}$$

and the weaker but entirely explicit form

$$4 - \lambda_1 \;\leq\; \kappa_N \;\leq\; 4. \tag{17.9}$$

Proof. Write $A := \frac{1}{2}(I + \mathcal{J}_O) = I - \frac{1}{2}\mathcal{D}$ and A_N for its leading $N \times N$ truncation, so that $F_N = \det A_N$ by Proposition 17.2. By Theorem 16.2 the spectrum of $\mathcal{J}_O$ lies in $[x_{\gamma_1}, 1]$, hence $0 \prec A \preceq I$ and every diagonal entry of A in the Jacobi basis satisfies $0 < A_{nn} \leq 1$.

(i) Positivity of F_N is positivity of the principal minors of the positive definite operator A. For monotonicity, bordering gives

$$\det A_N = \det A_{N-1}(A_{NN} - v^T A_{N-1}^{-1} v) \leq \det A_{N-1}\, A_{NN} \leq \det A_{N-1}, \tag{17.10}$$

where v is the last off-diagonal block and $A_{N-1}^{-1} \succ 0$. Also $F_1 = A_{11} \leq 1$.

(ii) Let $\mu_1 \geq \mu_2 \geq \cdots$ be the eigenvalues of $\mathcal{D}$ and $\mu_1^{(N)} \geq \cdots \geq \mu_N^{(N)}$ those of the compression $\mathcal{D}_N = P_N \mathcal{D} P_N$. Cauchy interlacing for compressions gives $\mu_j^{(N)} \leq \mu_j$ for $1 \leq j \leq N$, and $0 \leq \mu_j < 1$

for all j because $\|\mathcal{D}\| = 1 - x_{\gamma_1} < 1$ by (16.36). Hence

$$F_N = \prod_{j=1}^{N}\Big(1 - \tfrac{1}{2}\mu_j^{(N)}\Big) \geq \prod_{j=1}^{N}\Big(1 - \tfrac{1}{2}\mu_j\Big) \geq \prod_{j\geq 1}\Big(1 - \tfrac{1}{2}\mu_j\Big) = \det\Big(I - \tfrac{1}{2}\mathcal{D}\Big), \tag{17.11}$$

each factor being in $(0, 1]$. Since $\mathcal{D}$ is trace class, $\mathcal{D}_N \to \mathcal{D}$ in trace norm and the Fredholm determinant is trace-norm continuous, so $F_N \to \det(I - \mathcal{D}/2)$. Formula (17.7) is $D(1/2)$ in the notation of Proposition 16.12, since $1 - \frac{1}{4\gamma^2+1} = \frac{4\gamma^2}{4\gamma^2+1}$.

(iii) Combining (i), (ii) with (17.6) gives $\kappa_N = 4F_N/F_{N-1} \leq 4$ and $\kappa_N \geq 4\det(I-\mathcal{D}/2)$, because $F_N \geq \det(I - \mathcal{D}/2)$ and $F_{N-1} \leq 1$. Finally, for a positive operator with eigenvalues in $[0, 1)$ one has $\prod(1-\mu_j/2) \geq 1 - \frac{1}{2}\sum\mu_j$, and $\operatorname{Tr}\mathcal{D} \leq \lambda_1/2$ by Corollary 16.5, whence $\det(I-\mathcal{D}/2) \geq 1 - \lambda_1/4$ and $4\det(I - \mathcal{D}/2) \geq 4 - \lambda_1$. □

Corollary 17.5 (An arithmetic-free necessary condition)**.** *If* RH *holds, then for every* $N \geq 1$*,*

$$\Delta_N^E\,\Delta_{N-1}^O \leq 4\,\Delta_{N-1}^E\,\Delta_N^O.$$

Equivalently $F_N \leq F_{N-1}$*. No zero, prime, or Li coefficient beyond those defining the two determinants enters this inequality.*

Proof. This is the upper half of (17.8), whose only input is $\mathcal{J}_O \preceq I$, that is, that the odd reflection measure is supported in $(-\infty, 1]$. Under RH the support is $\{x_\gamma\} \cup \{1\} \subset (0, 1]$. □

Remark 17.6 (What is universal and what is arithmetic)*.* Identity (17.6) is an exact algebraic statement about *any* positive sine Gram pair, not a small-angle approximation: it was verified to working precision for angles drawn uniformly from $(0, 0.08)$, where the first tested values are

$$\kappa_N = 3.9959\ldots,\ 3.9967\ldots,\ 3.9970\ldots.$$

For angles drawn uniformly from $(0, \pi)$, the same identity holds exactly; for $N = 1, \ldots, 6$ the tested values are

$$1.3484,\ 1.3844,\ 1.1409,\ 1.4586,\ 1.2812,\ 1.4200,$$

and F_N falls to 1.4×10^{-3}. What requires the zeta geometry is not the factorisation but the *value*. In the finite zeta computations reported below, κ_N is close to 4, consistently with the small Cayley angles and with F_N/F_{N-1} being close to 1. The universal object is the constant 4 together with the exact factorisation, while the arithmetic object is F_N.

Remark 17.7 (Trace normalization)*.* Writing $q_n := 2\operatorname{Tr}\mathcal{J}_O^{(n)}$, the identity $2n - q_n = 2\operatorname{Tr}\mathcal{D}_n$ is immediate from $\mathcal{D}_n = I_n - \mathcal{J}_O^{(n)}$, and by (16.28) the increasing sequence $\operatorname{Tr}\mathcal{D}_n = \sum_{j<n}(1 - b_j^O)$ converges to $\operatorname{Tr}\mathcal{D} \leq \lambda_1/2$, with equality exactly in the simple-zero case. This is the first-order shadow of Theorem 17.4(ii): $\log F_n$ decreases to $\log\det(I - \mathcal{D}/2)$, whose linearization is $-\frac{1}{2}\operatorname{Tr}\mathcal{D} = -\lambda_1/4$.

Experiment 17.8 (The Christoffel margin for the arithmetic λ)**.** All entries below use the arithmetic λ of Experiment 26.4 at 250 working digits. The columns F_N and $\det\big((I + \mathcal{J}_O^{(N)})/2\big)$ were computed by two independent routes: the first from $\Delta_N^E/(4^N\Delta_N^O)$, the second from the Jacobi matrix reconstructed from the moments of μ_O by Cholesky.

N	F_N	$\kappa_N = p_N^E/p_N^O$	$\lvert\kappa_N - 4F_N/F_{N-1}\rvert$	$\operatorname{Tr}\mathcal{D}_N$
1	0.99959840336042631	3.9983936134417052	0	0.000803193
2	0.99872447452420145	3.9965028802235502	7.0×10^{-251}	0.002551298
3	0.99827559875164635	3.9982022037748942	0	0.003450031
4	0.99797192293554315	3.9987831984815297	3.5×10^{-251}	0.004058371
5	0.99770423468019879	3.9989270709959176	3.5×10^{-251}	0.004594797
6	0.99748822571541767	3.9991339759529021	0	0.005027778
7	0.99733892798897730	3.9994013053083076	7.0×10^{-251}	0.005327110
8	0.99720226204496688	3.9994518776308632	0	0.005601162
9	0.99706026967019839	3.9994304370129394	3.5×10^{-251}	0.005885933
10	0.99694804248538865	3.9995497676992109	3.5×10^{-251}	0.006111041

The two routes to F_N agree to 1.1×10^{-182} at $N=10$, and (17.6) holds to the last working digit; the entries in the fourth column are pure rounding. The band (17.8) is satisfied with $4\det(I-\mathcal{D}/2) = 3.976966$ from the first 1500 ordinates with the Riemann–von Mangoldt tail, against the explicit bound $4-\lambda_1 = 3.976904$. The Fredholm value itself is

$$\det\Big(I-\tfrac{1}{2}\mathcal{D}\Big) = \prod_{\gamma>0}\frac{4\gamma^2}{4\gamma^2+1} = 0.9942415\ldots, \tag{17.12}$$

whose truncation to the first 1200 ordinates is 0.9943992 and to the first 1500 ordinates 0.9943764; the difference is the tail $\sum_{\gamma>T}(4\gamma^2+1)^{-1} = 1.36\times10^{-4}$. Independently, a first-1500-ordinate evaluation supplemented by the same Riemann–von Mangoldt tail approximation gives $\sum 2/(4\gamma^2+1) = 0.0115479184$, within 6.4×10^{-8} of the rigorous arithmetic upper bound $\lambda_1/2 = 0.0115478545$. Since Corollary 16.5 proves $\operatorname{Tr}\mathcal{D}\le\lambda_1/2$, the small numerical overshoot measures the tail approximation error and is not interpreted as an excess of the exact trace. Convergence of F_N to (17.12) is slow: at $N=10$ only about a fifth of the estimated total defect has been accumulated.

Experiment 17.9 (The binding component of the local Jacobi inequalities)**.** Proposition 16.11 gives $a_{n+1}^2 \le (1-b_n)(1-b_{n+1})$ for every $n\ge0$. Define the normalized margin

$$\varrho_n := \frac{a_{n+1}^2}{(1-b_n)(1-b_{n+1})} \in [0,1], \tag{17.13}$$

so that ϱ_n close to 1 means the corresponding 2×2 minor is close to degenerate. With the arithmetic λ one finds, for both families,

n	0	1	2	3	4	5	6	7
ϱ_n (even)	0.6482	0.2100	0.2222	0.2353	0.2665	0.2401	0.2258	0.2633
ϱ_n (odd)	0.6482	0.2099	0.2222	0.2353	0.2665	0.2401	0.2258	0.2633

The binding index is $n=0$ in every positive definite case tested, with $\varrho_0 = 0.648160$ (even) and 0.648207 (odd), a factor of about three above all later values. This is the precise sense in which "the first component binds". The two families agree to four digits because they differ by a Christoffel transform at a point outside the support, which perturbs the low-index recurrence coefficients only at order $1-x_{\gamma_1}$.

18. Hermitian congruence reduction and inertia

For the real symmetric matrices considered here, the intrinsic analogue of a reduction of the quadratic form is congruence diagonalization. From (13.2),

$$L_E^{-1}\mathsf{E}_N L_E^{-T} = D_E = \operatorname{diag}\left(\frac{\Delta_1^E}{\Delta_0^E}, \frac{\Delta_2^E}{\Delta_1^E}, \dots, \frac{\Delta_N^E}{\Delta_{N-1}^E}\right), \tag{18.1}$$

$$L_O^{-1}\mathsf{O}_N L_O^{-T} = D_O = \operatorname{diag}\left(\frac{\Delta_1^O}{\Delta_0^O}, \frac{\Delta_2^O}{\Delta_1^O}, \dots, \frac{\Delta_N^O}{\Delta_{N-1}^O}\right). \tag{18.2}$$

Thus the signature is read from the signs of the pivot quotients whenever no principal determinant vanishes.

Remark 18.1 (Hermite normal form). The integer Hermite normal form is not intrinsic to E_N or O_N: the entries are real numbers built from Li coefficients, and no natural embedding of these matrices into a specified principal ideal domain is part of the construction. No transcendence property of the Li coefficients is assumed or known here. Hermite normal form is an arithmetic reduction after a coefficient ring such as $\mathbb{Z}$ has been fixed, as in [21]. For the present real symmetric forms, the intrinsic reduction is the congruence diagonalization in (18.1)–(18.2).

19. Geometric transformations

Under RH, each matrix defines an inner product on $\mathbb{R}^N$:

$$\langle c, d\rangle_E := c^T \mathsf{E}_N d, \qquad \langle c, d\rangle_O := c^T \mathsf{O}_N d. \tag{19.1}$$

The associated synthesis maps are

$$(\mathcal{S}_E c)_\gamma := 2\sum_{n=1}^{N} c_n \sin(n\theta_\gamma), \tag{19.2}$$

$$(\mathcal{S}_O c)_\gamma := 2\sum_{n=1}^{N} c_n \sin((n-\tfrac{1}{2})\theta_\gamma), \tag{19.3}$$

so that

$$\mathsf{E}_N = \mathcal{S}_E^*\mathcal{S}_E, \qquad \mathsf{O}_N = \mathcal{S}_O^*\mathcal{S}_O. \tag{19.4}$$

Hence the matrices are pullback metrics from the zero-side ℓ^2 geometry.

If $\sigma_1 \ge \cdots \ge \sigma_N > 0$ are the eigenvalues of E_N, then

$$\sigma_1 = 4\max_{\|c\|_2=1} \sum_{\gamma>0} \left|\sum_{n=1}^{N} c_n \sin(n\theta_\gamma)\right|^2, \tag{19.5}$$

$$\sigma_N = 4\min_{\|c\|_2=1} \sum_{\gamma>0} \left|\sum_{n=1}^{N} c_n \sin(n\theta_\gamma)\right|^2. \tag{19.6}$$

Thus σ_N measures the least zero-side energy of a normalized sine polynomial of degree N. Analogous formulas hold for O_N with half-integer modes.

The positive square root $\mathsf{E}_N^{1/2}$ maps the Euclidean unit ball to the ellipsoid whose principal semiaxes have lengths $\sqrt{\sigma_j}$. Cholesky gives a triangular coordinate realization of the same geometry, while L_E^{-1} gives the Gram–Schmidt coordinates in which the form becomes diagonal. The Chebyshev changes $B_E^{(N)}$ and $B_O^{(N)}$ convert from reflection-adapted harmonic coordinates to monomial moment coordinates. Passing to the infinite orthonormal-polynomial bases yields the Jacobi operators $\mathcal{J}_E$ and $\mathcal{J}_O$, in which multiplication by x is tridiagonal.

Corollary 19.1 (Eigenvalue interlacing)**.** *Under* RH*, the eigenvalues of* E_N *interlace those of* E_{N+1}*, and the same holds for* O_N *and* O_{N+1}*.*

Proof. Each $N \times N$ matrix is a leading principal submatrix of the next truncation, so Cauchy's interlacing theorem applies. □

20. Generating functions

The complete infinite arrays have compact bivariate generating functions. Let $\Lambda(z)$ be given by (2.8).

Theorem 20.1 (Bivariate generating functions)**.** *As identities of formal power series, one has*

$$\sum_{m,n\geq 1} e_{mn} x^m y^n = \frac{y\Lambda(x) - x\Lambda(y)}{x-y} - \frac{xy}{1-xy}\big(\Lambda(x)+\Lambda(y)\big), \tag{20.1}$$

$$\sum_{m,n\geq 1} o_{mn} x^m y^n = \frac{xy(\Lambda(x) - \Lambda(y))}{x-y} - \frac{xy}{1-xy}\big(\Lambda(x)+\Lambda(y)\big). \tag{20.2}$$

Analytically, these formulas hold whenever $|x|, |y| < R_\xi$*; under* RH *this is the full bidisk* $|x|, |y| < 1$*.*

Proof. For the Hankel part,

$$\sum_{m,n\geq 1} \lambda_{m+n} x^m y^n = \frac{y\Lambda(x) - x\Lambda(y)}{x-y}, \tag{20.3}$$

and

$$\sum_{m,n\geq 1} \lambda_{m+n-1} x^m y^n = \frac{xy(\Lambda(x) - \Lambda(y))}{x-y}. \tag{20.4}$$

For the Toeplitz term,

$$\sum_{m,n\geq 1} \lambda_{|m-n|} x^m y^n = \frac{xy}{1-xy}\big(\Lambda(x)+\Lambda(y)\big), \tag{20.5}$$

because $\lambda_0 = 0$. Subtraction yields the result. □

Substituting (2.8) gives bivariate expressions directly in ξ'/ξ evaluated at the Cayley variable $1/(1-z)$.

21. Finite-dimensional inequalities

Positive semidefiniteness generates a hierarchy of inequalities among Li coefficients.

Corollary 21.1 (Two-by-two inequalities)**.** *Under* RH*, for all* $m, n \geq 1$*,*

$$\big(\lambda_{m+n} - \lambda_{|m-n|}\big)^2 \leq \lambda_{2m}\lambda_{2n}, \tag{21.1}$$

$$\big(\lambda_{m+n-1} - \lambda_{|m-n|}\big)^2 \leq \lambda_{2m-1}\lambda_{2n-1}. \tag{21.2}$$

Proof. Apply nonnegativity of every 2×2 principal determinant. □

Special cases include

$$(\lambda_3 - \lambda_1)^2 \leq \lambda_2\lambda_4, \tag{21.3}$$

$$(\lambda_2 - \lambda_1)^2 \leq \lambda_1\lambda_3, \tag{21.4}$$

$$(\lambda_{n+1} - \lambda_{n-1})^2 \leq \lambda_2\lambda_{2n}, \tag{21.5}$$

$$(\lambda_n - \lambda_{n-1})^2 \leq \lambda_1\lambda_{2n-1}. \tag{21.6}$$

For three indices a, b, c, set

$$x^E_{ab} := \lambda_{a+b} - \lambda_{|a-b|}. \tag{21.7}$$

Then the 3×3 principal determinant gives

$$\begin{aligned} \lambda_{2a}\lambda_{2b}\lambda_{2c} + 2x^E_{ab}x^E_{ac}x^E_{bc} - \lambda_{2a}(x^E_{bc})^2 \\ - \lambda_{2b}(x^E_{ac})^2 - \lambda_{2c}(x^E_{ab})^2 \geq 0. \end{aligned} \tag{21.8}$$

The odd analogue follows after replacing x^E_{ab} by

$$x^O_{ab} := \lambda_{a+b-1} - \lambda_{|a-b|} \tag{21.9}$$

and the diagonal factors λ_{2a} by λ_{2a-1}.

22. Trace, norm, and conditioning data

The elementary spectral invariants have direct Li expressions:

$$\operatorname{tr} \mathsf{E}_N = \sum_{n=1}^{N} \lambda_{2n}, \tag{22.1}$$

$$\operatorname{tr} \mathsf{O}_N = \sum_{n=1}^{N} \lambda_{2n-1}, \tag{22.2}$$

$$\operatorname{tr} \mathsf{E}_N + \operatorname{tr} \mathsf{O}_N = \sum_{k=1}^{2N} \lambda_k. \tag{22.3}$$

The Frobenius norms are

$$\|\mathsf{E}_N\|_F^2 = \sum_{m,n=1}^{N} \big(\lambda_{m+n} - \lambda_{|m-n|}\big)^2, \tag{22.4}$$

$$\|\mathsf{O}_N\|_F^2 = \sum_{m,n=1}^{N} \big(\lambda_{m+n-1} - \lambda_{|m-n|}\big)^2. \tag{22.5}$$

Under RH, the condition numbers

$$\kappa_2(\mathsf{E}_N) = \frac{\sigma_{\max}(\mathsf{E}_N)}{\sigma_{\min}(\mathsf{E}_N)}, \qquad \kappa_2(\mathsf{O}_N) = \frac{\sigma_{\max}(\mathsf{O}_N)}{\sigma_{\min}(\mathsf{O}_N)} \tag{22.6}$$

measure the anisotropy of the integer and half-integer zero-side sampling geometries. Corollary 12.9 gives the nonperturbative lower bound

$$\log \kappa_2(\mathsf{E}_N),\ \log \kappa_2(\mathsf{O}_N) \geq 2N \log N - 2N \log\log N - O(N), \tag{22.7}$$

so ill-conditioning is forced asymptotically by the determinant law and is not merely an implementation issue. The smallest eigenvalue is particularly informative because it is the minimum energy in (19.6) and its odd analogue.

23. Literature position and comparison with other matrix criteria

For avoidance of any ambiguity about origin, the input used to manufacture the reflection matrices is copied algebraically from the published Li–Weil framework: Li supplies the coefficients and criterion [26]; Bombieri–Lagarias supply the general multiset criterion, explicit-formula interpretation and Li test functions in inverse-Mellin coordinates [3]; and Lagarias supplies the exact bilateral Gram identity used here [25, Theorem 3.1]:

$$\langle G_n, G_m \rangle_W = \lambda_n + \lambda_{-m} - \lambda_{n-m}. \tag{23.1}$$

Accordingly, neither the Li positivity mechanism nor the full Li–Weil Gram kernel is new here. The specific origin of E_N is the reflection map $G_n \mapsto (G_n - G_{-n})/\sqrt{2}$ applied to Lagarias's known bilateral kernel; the odd family is obtained from $G_n \mapsto (G_n - G_{1-n})/\sqrt{2}$. This is the precise sense in which the reflection matrices are transformations of known Li–Weil data rather than independent replacements for it. Bombieri studied other finite-dimensional truncations of Weil's quadratic functional [4]. Matiyasevich gave matrix/eigenvalue reformulations based on Taylor coefficients and Toeplitz/Hankel rearrangements [42, 43]. Suzuki developed norm, screw-function, moment, and Hilbert-space realizations of the same general positivity framework [32, 33, 35].

The localisation of Li-type criteria in height, and the extraction of zero-free regions from finitely many coefficients, are also established subjects. Freitas constructed a family of real functions interpolating the Li coefficients and characterising zero-free strips inside the critical strip [7]. Droll developed variations of Li's criterion for an extension of the Selberg class [6], and Bucur, Ernvall-Hytönen, Odžak and Smajlović treated Li-type criteria for zero-free regions of Dirichlet series with real coefficients [5]. Palojärvi proved τ-Li-type criteria with explicit indices N_1, N_2 such that nonnegativity of the coefficients between them forces a zero-free region, and a converse producing a zero from a single negative coefficient [10]. Sekatskii generalized the Bombieri–Lagarias theorem and Li's criterion and gave an arithmetic interpretation [11]. For Hecke L-functions, Omar, Ouni and Mazhouda computed the associated Li coefficients and conjectured the relevant monotonicity in 2014 [8]; the equivalence with the Riemann hypothesis was proved subsequently by Omar and Ouni [9]. Accordingly, the detection experiments in Subsection 25.1 are presented as calibration of the sensitivity of the reflection pivots against the scalar criterion, and no height-localised criterion is claimed as new here.

The scalar Laguerre direction also has substantial precedent. Coffey reformulated the Li/Keiper constants using theta–Laguerre calculus and integral representations, and decomposed the associated scalar quantities into superdominant and subdominant components whose sufficiently strong estimation would imply the Riemann hypothesis [13]. Related work develops the polygamma–Stieltjes-constant decomposition and further sums over the nontrivial zeros [14, 15]. Accordingly, no novelty is claimed here for the scalar appearance of associated Laguerre polynomials, for scalar archimedean–arithmetic decomposition as such, or for the general strategy of estimating a subdominant Keiper–Li component. The distinction retained in Section 26 is the reflection-matrix cancellation: the entries are differences of scalar coefficients, and the corresponding pole counterterms disappear identically before estimation.

The orthogonal-polynomial machinery used after the reflection reduction is also standard. Hankel moments and the Cauchy–Binet, Vandermonde, Christoffel, Christoffel–Darboux, Jacobi, and Darboux constructions are classical tools of moment and orthogonal-polynomial theory; see [12, 36]. Infinite discrete node sets with an accumulation point have an established asymptotic theory in model situations [38]. Section 12 therefore applies these standard tools rather than presenting them as new techniques.

It is especially important to separate the determinant result here from earlier arithmetic Hankel asymptotics. Monien's determinants are

$$H_n^{(r)}[\zeta] = \det\big(\zeta(i+j+r)\big)_{1\le i,j\le n}, \tag{23.2}$$

and their discrete Vandermonde representation is supported on reciprocal integers; Monien obtains the leading $-n^2 \log n$ scale for the zeta model [27]. Haynes–Zudilin study the different arithmetic-progression matrices $\det(\zeta(a(i+j)+b))$ and obtain general $N^2 \log N$ upper decay estimates, while recording the expected coefficient $-a$ for the full leading law [23]. Thus the generic presence of $N^2 \log N$ in discrete Hankel determinants is explicitly *not* a novelty claim of this paper. The Li–Weil matrices here have entries formed from reflection differences of Li coefficients, and their moment nodes are the Cayley transforms of zero ordinates rather than Dirichlet-series nodes. Theorem 12.1 identifies the common geometric mechanism in a general sufficient-condition form: the coefficient

is the endpoint exponent $-\beta$. The zeta reflection matrices then have $\beta = 2$. In particular, the leading -2 does not encode the $T \log T$ refinement of Riemann–von Mangoldt; that refinement can first affect lower orders.

Within that background, the paper singles out the pair

$$\mathsf{E}_N = (\lambda_{m+n} - \lambda_{|m-n|}), \qquad \mathsf{O}_N = (\lambda_{m+n-1} - \lambda_{|m-n|}), \tag{23.3}$$

as the two reflection-antisymmetric compressions associated with the involutions $k \mapsto -k$ and $k \mapsto 1 - k$. The pair-specific results are the exact algebraic reconstruction and information kernel of Section 4, the universal rank-at-most-two displacement identities of Section 7, the two explicit zero measures and their relation

$$\mathrm{d}\mu_E = 2(1 + x)\, \mathrm{d}\mu_O, \tag{23.4}$$

the resulting determinant bridge, Weyl-function relation and Darboux partner formulas, and the infinite Jacobi realization whose spectral support is the Cayley image of the distinct positive zero ordinates. Theorem 12.6 is likewise a statement about this pair, while Theorem 12.1 shows that its leading constant is geometric: -2 is the endpoint exponent of the Li–Cayley map. The precise Riemann–von Mangoldt density, including its logarithmic factor, is not recorded at this order.

The matrices are different from Suzuki's Hankel moment matrices in their native coordinates. Their conversion to Hankel form by an explicit Chebyshev change of basis is a derived representation of the reflection coordinates, not an independent RH criterion. Likewise, Theorem 5.4 is an exact reformulation of Li/Weil positivity for this pair and is not claimed as new arithmetic information.

Separate matrix approaches to the Riemann hypothesis include divisibility, Mertens-type, Redheffer, and multiplicative Toeplitz constructions; examples are [29, 30, 40, 19, 20]. Those encode different arithmetic data. The scope of the present paper is narrower: a complete structural characterization of the two reflection compressions and the consequences that are specific to them.

24. A de Bruijn–Newman falsification filter for matrix proof mechanisms

The de Bruijn–Newman deformation provides a way to separate identities that belong to the reflection/functional-equation geometry from input that is genuinely specific to the Euler product. In the normalization of de Bruijn, Newman, and Rodgers–Tao, set

$$H_t(z) := \int_0^\infty \mathrm{e}^{tu^2} \Phi(u) \cos(zu)\, \mathrm{d}u, \qquad t \in \mathbb{R}, \tag{24.1}$$

where, in the normalization of Rodgers–Tao,

$$\Phi(u) = \sum_{n=1}^{\infty} \left(2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}\right) \exp\left(-\pi n^2 \mathrm{e}^{4u}\right). \tag{24.2}$$

The kernel is even by the theta transformation and is strictly positive. For $u \geq 0$, strict positivity is already termwise in (24.2), since

$$2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u} = \pi n^2 \mathrm{e}^{5u}\big(2\pi n^2 \mathrm{e}^{4u} - 3\big) > 0.$$

At $t = 0$, H_0 is a fixed rescaling of the Riemann Ξ-function. De Bruijn proved forward preservation of real zeros under the heat flow, Newman proved the existence of a finite constant Λ such that all zeros of H_t are real exactly for $t \geq \Lambda$, and Rodgers–Tao proved

$$\boxed{\Lambda \geq 0.} \tag{24.3}$$

See [41, 44, 45].

For each real t, introduce the centered entire function

$$\Xi_t(s) := \frac{H_t\left(-2i(s - \frac{1}{2})\right)}{H_t(-i)}. \tag{24.4}$$

The denominator is strictly positive. Indeed, because $\Phi(u) > 0$,

$$H_t(iy) = \int_0^\infty \mathrm{e}^{tu^2}\Phi(u)\cosh(yu)\,\mathrm{d}u > 0 \qquad (y \in \mathbb{R}). \tag{24.5}$$

Consequently $\Xi_t(x) > 0$ for every real x. Evenness and reality of H_t give

$$\Xi_t(s) = \Xi_t(1-s) = \overline{\Xi_t(\overline{s})}. \tag{24.6}$$

At $t = 0$, Ξ_0 is the Riemann ξ-function divided by its value at 1, so the following coefficients agree with the usual Li coefficients. Define $\lambda_n^{(t)}$ by the local generating identity

$$\log \Xi_t\Big(\frac{1}{1-z}\Big) = \sum_{n\geq 1} \frac{\lambda_n^{(t)}}{n} z^n \qquad (|z| \text{ sufficiently small}), \tag{24.7}$$

and form

$$\mathsf{E}_N^{(t)} := \big(\lambda_{m+n}^{(t)} - \lambda_{|m-n|}^{(t)}\big)_{m,n=1}^{N}. \tag{24.8}$$

This definition avoids imposing a zero-summation convention at negative heat time.

Lemma 24.1 (Even Li positivity detects the heat-flow critical line)**.** *Fix $t \in \mathbb{R}$. If*

$$\lambda_{2n}^{(t)} \geq 0 \qquad (n \geq 1), \tag{24.9}$$

then every zero of H_t is real.

Proof. Put

$$L_t(z) := \log \Xi_t\Big(\frac{1}{1-z}\Big)$$

for the branch determined near $z = 0$, and define the even coefficient series

$$F_t(w) := \sum_{n\geq 1} \frac{\lambda_{2n}^{(t)}}{2n} w^n. \tag{24.10}$$

Near the origin one has the exact identity

$$F_t(z^2) = \frac{1}{2}\big(L_t(z) + L_t(-z)\big). \tag{24.11}$$

Suppose that H_t has a nonreal zero. Under the map $\rho = \frac{1}{2} + \frac{i}{2}z$, this gives a zero ρ of Ξ_t with $\Re\rho \neq \frac{1}{2}$. Set $q = 1 - 1/\rho$. The symmetry $\rho \mapsto 1-\rho$ sends q to q^{-1}; hence one member of the reflected pair has $|q| < 1$. Therefore $L_t(z)$ has a logarithmic singularity at some point of modulus strictly less than one. Passing to the even part cannot cancel such a singularity: Ξ_t is entire, so all logarithmic singularities here come from zeros and carry nonnegative integer multiplicity; if both terms in (24.11) are singular at the same point, their multiplicities add. Thus the radius of convergence R_t of F_t satisfies $R_t < 1$.

Under (24.9), Pringsheim's theorem forces the positive real point $w = R_t$ to be a singularity of F_t. But for every $0 < w < 1$, with $z = \sqrt{w}$, the two arguments

$$\frac{1}{1-z} > 1, \qquad \frac{1}{1+z} \in \Big(\frac{1}{2}, 1\Big)$$

are real. Equation (24.5) therefore shows that both terms on the right of (24.11) are analytic there. Hence F_t is analytic on $(0, 1)$, a contradiction. All zeros of H_t are therefore real. □

Proposition 24.2 (Heat-flow sign test)**.** *For every $t < \Lambda$ there exists at least one N for which $\mathsf{E}_N^{(t)} \not\succeq 0$. Consequently, since $\Lambda \geq 0$, for every $\varepsilon > 0$ there exists $N = N(\varepsilon)$ such that*

$$\boxed{\mathsf{E}_N^{(-\varepsilon)} \not\succeq 0.} \tag{24.12}$$

For $t \geq \Lambda$, the zero set is real by definition of the de Bruijn–Newman constant. The negative-time implication above, which is the part used as a falsification test, requires only Lemma 24.1 and no Bombieri–Lagarias summability assumption for the deformed zero set.

Proof. Let $t < \Lambda$. By the definition of Λ, H_t has a nonreal zero. If $\mathsf{E}_N^{(t)} \succeq 0$ for every N, then its diagonal would give

$$\lambda_{2N}^{(t)} = (\mathsf{E}_N^{(t)})_{NN} \geq 0 \qquad (N \geq 1),$$

contradicting Lemma 24.1. Hence some finite even reflection block fails to be positive semidefinite. Taking $t = -\varepsilon$ and using $\Lambda \geq 0$ proves (24.12). □

Corollary 24.3 (Falsification principle)**.** *No proof of $\mathsf{E}_N \succeq 0$ for all N can be based solely on identities and inequalities that remain valid, with the same sign content, for every negative-time family $H_{-\varepsilon}$. Any successful proof must use an input that distinguishes the zeta function from those heat deformations.*

Proof. If such a proof used only statements that remained valid with the same sign content for every $H_{-\varepsilon}$, it would prove $\mathsf{E}_N^{(-\varepsilon)} \succeq 0$ for every N. This contradicts Proposition 24.2. Hence at least one zeta-specific input that fails under negative heat deformation is necessary. □

Remark 24.4 (What survives and what does not). The algebraic reflection form, the discrete wave equation, rank-two displacement, the Schur bordering identities, the inverse displacement identity, and the finite-rank update formulas depend only on the coefficient/reflection algebra and therefore survive the heat deformation. The positive-measure consequences do *not* survive once the zeros leave the line, because their positivity is exactly the property being tested.

There is also a useful distinction on the archimedean–arithmetic split. The sequence A_n is *fixed by decree as a reference sequence*; it is not inherited from H_t. After making that choice one may write $\lambda_n^{(t)} = A_n + Z_n^{(t)}$, and then the explicit matrix S_N and the scalar c_∞ are identical by construction. What fails for H_t is precisely the zeta-specific von Mangoldt–Laguerre representation of Z_n, equivalently the representation of $\widehat{\mathsf{Z}}_N^E$ through $\Lambda(n)$, $\psi(x) - x$, and the Euler product. Thus, within the machinery actually developed in this paper, the prime–continuum representation is the only identified zeta-specific door through which a sign proof can distinguish ξ from $H_{-\varepsilon}$. This statement is internal to the present framework; it does not claim that every conceivable proof of RH must literally use formula (26.53).

Remark 24.5 (Criticality). Rodgers–Tao gives $\Lambda \geq 0$, while RH is equivalent to $\Lambda \leq 0$. Hence, *conditional on RH*, one has $\Lambda = 0$: the Riemann Ξ-function then lies exactly on the heat-flow boundary of the all-real-zero region. The very small reflection pivots, near-unit low-zero leverage, and other finite margins reported below are consistent with such criticality, but they do not establish $\Lambda = 0$, nor does the heat-flow theorem by itself identify a quantitative rate for any one of those finite margins.

25. Computational realization

For a supplied list $\lambda_1, \ldots, \lambda_{2N}$, both matrices are formed in $O(N^2)$ scalar operations. With the zero final component convention in (7.4)–(7.5), the displacement checks also require no coefficient beyond λ_{2N}. The basic numerical diagnostics are:

(i) the smallest eigenvalues of E_N and O_N;
(ii) the determinant ratios $\Delta_n^E/\Delta_{n-1}^E$ and $\Delta_n^O/\Delta_{n-1}^O$;
(iii) the Cholesky factors and Schur residuals in (13.21)–(13.22);
(iv) the initial Jacobi recurrence coefficients determined by the available moments, together with eigenvalues of the corresponding finite Jacobi truncations as quadrature nodes;
(v) the displacement residuals in (7.6)–(7.7), which provide exact algebraic consistency checks;

(vi) the Christoffel identity (8.9) and determinant bridge (16.100).

Experiment 25.1 (Single-zero leverage and transport at $N = 10$)**.** The exact formulas above were evaluated for E_{10}. The Li coefficients $\lambda_1, \dots, \lambda_{20}$ were computed at 160 decimal digits from the Taylor expansion of $\log \xi(1+z)$, and the required ordinates were evaluated at high precision. The determinant responses were computed from the 2×2 formula (11.43), not by subtracting two ill-conditioned determinants. As an independent audit, for $j = 1, 10, 100$ with a 10% local-gap motion, direct multiprecision recomputation of the perturbed 10×10 determinant agrees with (11.43) to more than 80 decimal digits.

For a zero γ_j, let

$$h_j := \gamma_{j+1} - \gamma_j, \qquad \gamma_j(c) := \gamma_j + c h_j. \tag{25.1}$$

The table gives the even leverage ℓ^E_{10,γ_j}, the exact determinant ratio after deletion through $1-\ell^E_{10,\gamma_j}$, and the logarithmic determinant response when the zero is moved by 10% and 50% of its next local gap.

j	γ_j	ℓ^E_{10,γ_j}	$1-\ell^E_{10,\gamma_j}$	$\log(\Delta'/\Delta)_{c=0.1}$	$\log(\Delta'/\Delta)_{c=0.5}$
1	14.1347	$1-6.90\times10^{-19}$	6.90×10^{-19}	-2.2354	-10.9973
2	21.0220	0.9999999998	1.58×10^{-10}	-0.9565	-5.1352
3	25.0109	0.9999995136	4.86×10^{-7}	-0.8360	-4.8068
4	30.4249	0.9975186583	2.48×10^{-3}	-0.3917	-2.2879
5	32.9351	0.9644640587	3.55×10^{-2}	-0.1175	-1.1180
10	49.7738	0.2310567145	0.76894329	-6.20×10^{-3}	-3.24×10^{-2}
20	77.1448	0.0656564393	0.93434356	-7.04×10^{-4}	-3.44×10^{-3}
50	143.1118	0.0160389932	0.98396101	-9.89×10^{-5}	-4.89×10^{-4}
100	236.5242	0.0034507536	0.99654925	-3.80×10^{-6}	-1.89×10^{-5}
200	396.3819	0.0016652308	0.99833477	-8.26×10^{-7}	-4.12×10^{-6}

The total leverage is exactly 10. Numerically, the cumulative even leverage carried by the first M positive zeros is

M	1	2	3	4	5	10	20	50	100	200
$\sum_{j\le M}\ell^E_{10,\gamma_j}$	1.0000	2.0000	3.0000	3.9975	4.9620	6.7290	7.7976	8.6834	9.0382	9.2658

Thus the first five zeros already carry about 49.6% of the total leverage, the first ten about 67.3%, and the first two hundred about 92.7%. This is a property of the *full matrix determinant*: the leverage is obtained from E_{10}^{-1} and is the exact marginal of the complete Cauchy–Binet sum.

For illustration of the Li-side representation, moving the tenth zero by $0.1h_{10} = 0.3196489000\dots$ changes the contribution of that conjugate pair by

$$\delta\lambda_1 = -5.13\times10^{-6}, \quad \delta\lambda_5 = -1.28\times10^{-4}, \quad \delta\lambda_{10} = -5.10\times10^{-4}, \quad \delta\lambda_{20} = -1.999\times10^{-3}, \tag{25.2}$$

while the exact determinant response is

$$\log\frac{\det \mathsf{E}'_{10}}{\det \mathsf{E}_{10}} = -0.00620213676254\dots, \qquad \frac{\det \mathsf{E}'_{10}}{\det \mathsf{E}_{10}} = 0.9938170567868\dots. \tag{25.3}$$

For comparison, the same 10% local-gap motion gives

$$\frac{\det \mathsf{E}'_{10}}{\det \mathsf{E}_{10}} = 0.1069489018\dots \quad (j=1), \qquad \frac{\det \mathsf{E}'_{10}}{\det \mathsf{E}_{10}} = 0.9999962023\dots \quad (j=100). \tag{25.4}$$

This strong localization is consistent with Theorem 11.8.

Experiment 25.2 (Conditioning of the reflection matrices)**.** Direct determinant evaluation is unstable at very small orders. In the computation used here, Li coefficients carried to 40 significant decimal digits and passed to a direct LU determinant produced a negative value for Δ^E_7. This sign

is numerical: under RH, Theorem 5.4 gives $\Delta_7^E > 0$, and the zero-side Cauchy–Binet representation is a sum of nonnegative terms.

As a diagnostic, retaining only one Cauchy–Binet term, formed from tabulated low ordinates, gives positive one-term estimates whose base-10 logarithms for $N = 1, 2, 4, 7, 10, 14, 20$ are approximately

$$-1.7,\ -8.9,\ -42.0,\ -142.5,\ -308.0,\ -634.6,\ -1361.2. \tag{25.5}$$

These decimal values are numerical estimates rather than interval-certified bounds. Their role is only to illustrate the scale proved analytically in Theorem 12.6.

Experiment 25.3 (Critical sine-polynomial variational identity and finite profile data)**.** Consider the even prime–archimedean normal form introduced in Section 26, and let $v^{(N)}$ denote a unit critical direction obtained numerically from the smallest-eigenvalue problem, with a consistent global sign convention. A recheck using a 1500-ordinate model, inverse iteration and 250 decimal working digits gives peak indices $m = 3, 3, 3, 5$ for $N = 12, 20, 32, 48$. Representative normalized amplitudes are

N	$m/N = 0.05$	$m/N = 0.30$	$m/N = 0.50$
12	0.636	0.750	0.206
20	0.493	0.523	0.032
32	0.893	0.308	0.011
48	0.570	0.188	0.056

Over this finite range the mass remains concentrated at low indices and the data do not support a stable nonzero profile in the variable m/N. They do not establish the absence of such a profile as $N \to \infty$.

The correct exact interpretation is instead variational. Under RH, for every real vector $v = (v_1, \dots, v_N)^T$ define

$$P_v(\theta) := \sum_{m=1}^{N} v_m \sin(m\theta). \tag{25.6}$$

Then the sine Gram representation gives

$$\boxed{v^T \mathsf{E}_N v = 4 \sum_{\gamma > 0} \left| P_v(\theta_\gamma) \right|^2.} \tag{25.7}$$

Hence the smallest-eigenvalue direction is exactly the unit coefficient vector whose sine polynomial minimizes the discrete squared energy on the Cayley nodes $\{\theta_\gamma\}$. Since $\theta_\gamma \downarrow 0$, suppression of this energy naturally favors high-order flatness at the origin. Taylor expansion gives

$$P_v(\theta) = \sum_{j \geq 0} \frac{(-1)^j \theta^{2j+1}}{(2j+1)!} \left(\sum_{m=1}^{N} m^{2j+1} v_m \right), \tag{25.8}$$

so the exact conditions

$$\sum_{m=1}^{N} m^{2j+1} v_m = 0, \qquad 0 \leq j < r, \tag{25.9}$$

are equivalent to $P_v(\theta) = O(\theta^{2r+1})$ at the origin. The observed strictly alternating signs are consistent with these successive odd-moment cancellations, although alternation alone does not prove maximal flatness.

Numerically, the visible core at the largest tested orders is concentrated on low indices of size comparable with

$$\theta_1^{-1} = \left(2\arctan\frac{1}{2\gamma_1}\right)^{-1} \approx 14.1. \tag{25.10}$$

This suggests that the first Cayley node sets the scale of the dominant visible core. It does not imply finite support: if the minimum energy tends to zero, an exactly fixed finite-support limiting polynomial would vanish at infinitely many nodes accumulating at 0 and hence would be identically zero. The numerically tiny high-index tail may therefore be essential for purchasing additional orders of flatness as N grows.

Remark 25.4 (Matrix-specific flatness problem). The finite computations do not exhibit a stable m/N continuum profile over the tested orders. The matrix-specific object is instead the sequence of odd moments

$$M_{2j+1}^{(N)} := \sum_{m=1}^{N} m^{2j+1} v_m^{(N)}. \tag{25.11}$$

A direct high-precision diagnostic is to determine how many consecutive moments $M_1^{(N)}, M_3^{(N)}, \dots$ are suppressed to the numerical scale of the smallest-eigenvalue computation. Equivalently, if

$$V_{r,N} := \left(m^{2j+1}\right)_{\substack{0\le j<r\\ 1\le m\le N}}, \tag{25.12}$$

then one should measure the distance of $v^{(N)}$ to $\ker V_{r,N}$ as r grows. A growing flatness order would explain how additional matrix dimensions can reduce the sine energy without producing a macroscopic m/N profile. This phenomenon has no scalar Keiper–Li analogue because it is a statement about the geometry of a smallest-eigenvalue direction in the reflection Gram matrix.

Remark 25.5 (Numerical evaluation). Theorem 12.6 shows that

$$\Delta_N^{E,O} = \exp\bigl(-(2+o(1))N^2\log N\bigr), \tag{25.13}$$

so forming a determinant directly in fixed precision rapidly loses meaning. The theorem by itself does not prescribe a sharp number of working digits for a stable factorization, because that also depends on the smallest pivots and on entrywise cancellation. Numerically one should work with multiprecision LDL^T or Cholesky pivots and accumulate $\log\Delta_N^{E,O} = \sum_n \log d_n^{E,O}$; when certified positivity is required, interval arithmetic or certified zero-side estimates are preferable. The Cauchy–Binet formulas are especially useful for sign certification under RH because their individual terms are nonnegative.

The rank-at-most-two displacement structure is particularly useful computationally: structured Toeplitz–Hankel algorithms can exploit low displacement rank, and Corollary 7.4 shows that this feature survives inversion. The matrices therefore offer a natural setting for numerical experiments on Li coefficients without destroying the underlying reflection geometry.

25.1. **Off-line detection: how early a reflection pivot sees a zero off the line.** The experiments of this subsection use a deliberately constructed *model* measure and therefore do not bear on ζ itself; their purpose is to calibrate the sensitivity of the reflection pivots against the scalar Li criterion.

Definition 25.6 (Count- and symmetry-preserving deformation)**.** Fix an ordinate γ and $\beta \in (0,1)$. The model $M(\beta,\gamma)$ is obtained from the on-line configuration by replacing the conjugate pair

$\{\frac{1}{2} \pm i\gamma\}$ by the four points $\{\beta \pm i\gamma,\ 1-\beta \pm i\gamma\}$, each carrying weight $\frac{1}{2}$; all other ordinates remain on the critical line. The resulting coefficients

$$\lambda_k^M = \sum_\rho w_\rho \Big[1 - \big(1 - \rho^{-1}\big)^k\Big] \tag{25.14}$$

are real, the total zero mass is unchanged, the set is invariant under both $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1-\rho$, and $M(\frac{1}{2}, \gamma)$ coincides identically with the on-line configuration. The last property is what makes $\beta \mapsto \lambda_k^M$ a genuine deformation with vanishing signal at $\beta = 1/2$; models that replace a pair by an unweighted quadruple fail it and produce a spurious β-independent signal.

Experiment 25.7 (Detection index of the reflection pivot)**.** Let $N^*(\beta, \gamma)$ be the smallest N for which the Cholesky pivot p_N^E of the model $M(\beta, \gamma)$ fails to be positive. Computations use $K = 300$ background ordinates, $N \le 34$ and 900 decimal working digits. As a control, the unperturbed on-line configuration has all pivots strictly positive up to $N = 34$, with $\log_{10} p_{34} = -255.2$, so the working precision exceeds the pivot scale by a wide margin and every reported sign change is genuine.

γ	$\beta = 0.45$	$\beta = 0.49$	$\beta = 0.499$	$\beta = 0.4999$	$\gamma/2$
14.1347	5	6	7	8	7.07
21.0220	8	9	10	11	10.51
25.0109	10	11	12	14	12.51
30.4249	13	14	16	17	15.21
32.9351	14	16	17	18	16.47
40.9187	19	20	22	23	20.46
49.7738	24	26	28	29	24.89
65.1125	34	> 34	> 34	> 34	32.56
77.1448	> 34	> 34	> 34	> 34	38.57

The detection index remained unchanged across the tested background truncations: $K = 300$, 600 and 1200 give identical values, for instance $13, 14, 16$ at $\gamma = 30.4249$ for the three smallest β. This is a numerical stability statement for the tested truncations only; no background-independence theorem is inferred from it.

The natural variable is not γ but the Cayley angle. With $\theta_\gamma = 2\arctan(1/2\gamma)$ and $\beta = 0.499$,

γ	14.1347	21.0220	25.0109	30.4249	40.9187	49.7738
$N^*\theta_\gamma$	0.495	0.476	0.480	0.526	0.538	0.563

so that

$$N^*\,\theta_\gamma \approx \frac{1}{2}. \tag{25.15}$$

The observed values suggest a resolution scale in the Cayley angle. The top available sine mode is $\sin(N\theta)$, so departure from the small-angle linear regime is a natural heuristic for the numerical scale, but the data do not prove that this mechanism determines the threshold. Since $\theta_\gamma \sim 1/\gamma$, the measured values in (25.15) are consistent with $N^* \approx \gamma/2$ over the tested range. The dependence on β is also empirical: in the table N^* increases as $\beta \uparrow 1/2$ toward the observed near-critical scale.

Experiment 25.8 (Comparison with the scalar Li criterion)**.** At fixed low height the numerical separation from the scalar Li crossing estimate is large. For an off-line pair the Cayley image $z = (\rho - 1)/\rho$ of the zero with $\Re\rho < 1/2$ has $|z| > 1$, and the scalar coefficient λ_n first becomes negative near the crossing of $|z|^n$ with the on-line growth $\frac{n}{2}\log n$. Solving $n \log|z| = \log(\frac{n}{2}\log n)$ at $\gamma = 14.1347$:

β	$\log\lvert z\rvert$	scalar n	matrix N^*
0.45	2.50×10^{-4}	5.0×10^{4}	5
0.49	5.00×10^{-5}	2.9×10^{5}	6
0.499	5.00×10^{-6}	3.4×10^{6}	7
0.4999	5.00×10^{-7}	3.9×10^{7}	8

The reflection pivot at $N^*=8$ uses $\lambda_1,\dots,\lambda_{16}$. These figures show a large finite-height separation between the model pivot detector and the scalar crossing estimate. The same data also show that the required matrix dimension grows with height, so the comparison is not a uniform complexity theorem and should not be read as a general improvement of Li's criterion.

Experiment 25.9 (Relative signal across raw and processed observables)**.** Define the relative signal of an observable X as $|X(M(\beta,\gamma))-X(M(\frac12,\gamma))|/|X(M(\frac12,\gamma))|$, maximized over $n\le30$ for $X=\lambda_n$ and over $N\le10$ for $X=\kappa_N$ and $X=F_N$. With $K=300$ and 400 working digits:

γ	β	λ_n	κ_N	F_N
30.425	0.45	4.07×10^{-7}	2.98×10^{-7}	5.05×10^{-7}
49.774	0.45	5.79×10^{-8}	7.56×10^{-10}	4.76×10^{-10}
77.145	0.45	1.07×10^{-8}	3.00×10^{-11}	1.51×10^{-11}
143.112	0.45	9.29×10^{-10}	4.23×10^{-13}	3.65×10^{-13}
248.102	0.45	1.04×10^{-10}	3.85×10^{-14}	3.85×10^{-14}
248.102	0.49	4.15×10^{-12}	1.54×10^{-15}	1.54×10^{-15}

A least-squares fit in the logarithms gives exponent 2.000 in $(\frac12-\beta)$ and -3.938 in γ. Over the range tested, the data are therefore consistent with the empirical scaling

$$\text{signal of order } \left(\tfrac12-\beta\right)^2\gamma^{-4}. \tag{25.16}$$

The symmetry $\beta\mapsto1-\beta$ forces an even expansion in $\beta-\frac12$; the observed quadratic leading term and the fourth-power height law are empirical, not proved asymptotics. The numerical attenuation is strongly height dependent. For the five $\beta=0.45$ rows, the base-10 logarithms of the ratio between the largest raw λ_n signal and the κ_N signal are, respectively,

$$0.14,\quad 1.88,\quad 2.55,\quad 3.34,\quad 3.43. \tag{25.17}$$

At the lowest tested height F_N actually has a slightly larger relative signal than the raw coefficients: 5.05×10^{-7} versus 4.07×10^{-7}. Only at the higher heights does a multi-order attenuation appear. These numbers quantify numerical conditioning; they do not imply that forming determinants destroys mathematical information, nor that an exact matrix identity, recurrence, convexity argument or sign mechanism is weaker than the scalar sequence.

Experiment 25.10 (What the band detects, and what it does not)**.** Corollary 17.5 is a necessary condition; the question is whether it is sharp enough to be a diagnostic. At $\gamma=14.1347$, $\beta=0.45$, where the pivot first fails at $N^*=5$:

N	κ_N (model)	κ_N (on-line)
3	3.9982260960	3.9982215228
4	3.9989005054	3.9987896058
5	3.9846420023	3.9989285277
6	**4.0092903964**	3.9991406580
7	3.9991204664	3.9994048590
8	3.9993769054	3.9994491550

The upper bound $\kappa_N\le4$ is violated at $N=6$, one step after the pivot sign change, and the violation is by 2.3×10^{-3}, four orders above the working precision. The lower bound is never approached. In this deformation the band is therefore a nontrivial numerical detector rather than

merely a consistency check, but it detects strictly later than the pivot itself, so it adds robustness rather than sensitivity. Second, the proof of the bound $\kappa_N \le 4$ under RH uses $\mathcal{J}_O \preceq I$, equivalently the support condition for the positive RH reflection measure. The synthetic off-line deformation no longer belongs to that positive-measure setting, but it does violate the resulting determinant inequality one step after the pivot changes sign. A wider band of the shape $(1 - \operatorname{Tr} \mathsf{K})\,\kappa_N^{\mathrm{ref}}$, which is what a first-order estimate of F_N/F_{N-1} suggests, is satisfied by the model at every N including $N = 6$ and therefore does not detect this deformation. In this example the sharp constant is essential.

26. The archimedean–prime split in reflection and Laguerre coordinates

The reflection matrices admit a second exact realization which is independent of the zero-side moment representation. The scalar theta–Laguerre formulation of the Li/Keiper coefficients, and scalar decompositions into dominant and subdominant components, have classical precedents in Coffey's work [13, 14]. The purpose of this section is therefore not to claim the scalar Laguerre representation as new. What is specific to the present reflection pair is that the entries are coefficient differences: the pole counterterms required by the individual scalar coefficients cancel identically inside $\lambda_{m+n} - \lambda_{|m-n|}$ (and analogously in the odd family). This leaves a matrix pairing directly against the prime-number-theorem error before any estimate is made. The section records that cancellation, the resulting matrix cosine/positive-definite structure, and the scalar coefficient criterion only as a diagnostic comparison with the classical Keiper–Li mechanism.

26.1. **An analytic split of the Li coefficients.** Put

$$s(z) := \frac{1}{1-z}, \qquad |z| \ll 1, \tag{26.1}$$

and factor

$$\xi(s) = \underbrace{\frac{s}{2}\pi^{-s/2}\Gamma\!\left(\frac{s}{2}\right)}_{\Xi_\infty(s)}\underbrace{(s-1)\zeta(s)}_{\Xi_{\mathrm{ar}}(s)}. \tag{26.2}$$

Both factors are analytic and nonzero at $s = 1$. Choose the logarithms which are real there and define, for $k \ge 1$,

$$A_k := k[z^k]\log \Xi_\infty(s(z)), \tag{26.3}$$

$$Z_k := k[z^k]\log \Xi_{\mathrm{ar}}(s(z)). \tag{26.4}$$

Then

$$\lambda_k = A_k + Z_k. \tag{26.5}$$

The archimedean coefficient is explicit:

$$A_k = 1 - \frac{k}{2}(\gamma_E + \log 4\pi) + \sum_{j=2}^{k}(-1)^j\binom{k}{j}(1-2^{-j})\zeta(j), \tag{26.6}$$

where γ_E is Euler's constant.

Proposition 26.1 (Regularized von Mangoldt–Laguerre formula)**.** *For every fixed integer $k \ge 1$,*

$$Z_k = -\lim_{X\to\infty}\left[\sum_{n\le X}\frac{\Lambda(n)}{n}L_{k-1}^{(1)}(\log n) + L_k(\log X) - 1\right], \tag{26.7}$$

where $L_k = L_k^{(0)}$ and $L_{k-1}^{(1)}$ are Laguerre polynomials. Equivalently, with

$$E(t) := \psi(t) - t, \tag{26.8}$$

one has the Stieltjes identity

$$\boxed{Z_k = \int_{1^-}^{\infty} L_k'(\log t)\,\frac{\mathrm{d}E(t)}{t}.} \tag{26.9}$$

Proof. For $\Re s > 1$,

$$\log\zeta(s) = \sum_{n\geq 2} \frac{\Lambda(n)}{\log n}\, n^{-s}. \tag{26.10}$$

Since

$$n^{-s(z)} = n^{-1}\exp\!\left(-\frac{(\log n)z}{1-z}\right) \tag{26.11}$$

and

$$\sum_{r\geq 0} L_r^{(1)}(u) z^r = \frac{1}{(1-z)^2}\exp\!\left(-\frac{uz}{1-z}\right), \tag{26.12}$$

coefficient extraction gives

$$k[z^k]n^{-s(z)} = -\frac{\log n}{n} L_{k-1}^{(1)}(\log n). \tag{26.13}$$

The factor $s(z)-1 = z/(1-z)$ supplies the canonical pole counterterm. Using

$$L_k'(u) = -L_{k-1}^{(1)}(u) \tag{26.14}$$

and hence

$$\int_0^{\log X} L_{k-1}^{(1)}(u)\ \mathrm{d}u = 1 - L_k(\log X), \tag{26.15}$$

we obtain (26.7). The bracket in (26.7) is exactly

$$\int_{1^-}^{X} L_{k-1}^{(1)}(\log t)\,\frac{\mathrm{d}(\psi(t)-t)}{t}. \tag{26.16}$$

Applying (26.14) gives (26.9). For fixed k, passage to infinity follows from the classical prime-number-theorem error bound, since the Laguerre factor has fixed polynomial degree. □

The first value is

$$Z_1 = \gamma_E, \tag{26.17}$$

so $A_1 + Z_1 = \lambda_1$ gives the familiar numerical value of the first Li coefficient.

26.2. **Arithmetic reconstruction of the Li coefficients without zeros.** The split (26.5) is not only structural. Together with the closed form (26.6) for A_k it yields a construction of $\lambda_1, \dots, \lambda_K$ from the Stieltjes constants alone, with no zero of ζ entering at any stage. This matters here for a specific reason: every numerical statement in this paper concerns positivity of matrices built from λ, and a λ computed by summing over a truncated list of ordinates would make those statements circular. The construction below removes that circularity.

Proposition 26.2 (Stieltjes recursion for the arithmetic coefficients)**.** *Let γ_m denote the Stieltjes constants, so that*

$$\zeta(s) = \frac{1}{s-1} + \sum_{m\geq 0} \frac{(-1)^m}{m!}\gamma_m (s-1)^m. \tag{26.18}$$

Put $b_0 := 1$ and, for $k \geq 1$,

$$\boxed{b_k := \sum_{m=0}^{k-1} (-1)^m \binom{k-1}{m} \frac{\gamma_m}{m!}} \tag{26.19}$$

Then $\Xi_{\mathrm{ar}}(s(z)) = \sum_{k\geq 0} b_k z^k$ for $|z|$ small, and the coefficients Z_k of (26.4) satisfy the recursion

$$\boxed{Z_k = k\, b_k - \sum_{j=1}^{k-1} Z_j\, b_{k-j}} \qquad (k \geq 1). \tag{26.20}$$

Consequently $\lambda_k = A_k + Z_k$ is determined by $\gamma_0, \ldots, \gamma_{k-1}$ and $\zeta(2), \ldots, \zeta(k)$.

Proof. Multiplying (26.18) by $s-1$ gives

$$\Xi_{\mathrm{ar}}(s) = (s-1)\zeta(s) = 1 + \sum_{m\geq 0} \frac{(-1)^m}{m!} \gamma_m (s-1)^{m+1}. \tag{26.21}$$

Under $s = s(z) = 1/(1-z)$ one has $s - 1 = z/(1-z)$, and

$$\left(\frac{z}{1-z}\right)^{m+1} = \sum_{k\geq m+1} \binom{k-1}{m} z^k, \tag{26.22}$$

since $[z^k]\, z^{m+1}(1-z)^{-(m+1)} = \binom{k-m-1+m}{m} = \binom{k-1}{m}$. Collecting the coefficient of z^k gives (26.19), and $b_0 = 1$ because $\Xi_{\mathrm{ar}}(1) = 1$.

For (26.20), write $F(z) := \log \Xi_{\mathrm{ar}}(s(z)) = \sum_{k\geq 1} (Z_k/k) z^k$, which is legitimate because Ξ_{ar} is analytic and nonvanishing at $s = 1$ and $F(0) = 0$. Then $F'(z)\, \Xi_{\mathrm{ar}}(s(z)) = \frac{\mathrm{d}}{\mathrm{d}z} \Xi_{\mathrm{ar}}(s(z))$, that is

$$\Big(\sum_{k\geq 1} Z_k z^{k-1}\Big)\Big(\sum_{j\geq 0} b_j z^j\Big) = \sum_{k\geq 1} k\, b_k z^{k-1}. \tag{26.23}$$

Comparing the coefficient of z^{k-1} and using $b_0 = 1$ gives $\sum_{j=0}^{k-1} Z_{k-j} b_j = k\, b_k$, which is (26.20). □

Remark 26.3 (Why this is the right normalization). The pole of ζ has been absorbed once and for all by the factor $s-1$, so (26.19)–(26.20) involve no regularization and no principal value. This is the coefficient-side counterpart of the counterterm cancellation in Proposition 26.1: the same pole is removed analytically there and combinatorially here.

Experiment 26.4 (Arithmetic λ and its independent audit)**.** Proposition 26.2 and (26.6) were evaluated at 300 decimal working digits for $k \leq 50$. Representative values:

k	Z_k	A_k	$\lambda_k = A_k + Z_k$
1	0.577215664901533	−0.554119955935412	0.0230957089661210
2	0.966885096962700	−0.874539361734654	0.0923457352280467
3	1.22069692821670	−1.01305800766237	0.207638920554325
4	1.37558813187126	−1.00679765237902	0.368790479492242
5	1.45826850019717	−0.882725785735990	0.575542714461177
10	1.32380368369629	0.955535679496864	2.27933936319316
20	0.670652812798991	8.09862405929422	8.76927687209322
40	1.16405623139036	29.3133191923903	30.4773754237807

Two independent audits were run. First, λ_k was recomputed as $k\,[z^k] \log \xi(s(z))$ by a trapezoidal Cauchy integral on the circle $|z| = 0.4$; for $k \leq 20$ the worst relative discrepancy against the Stieltjes route is 6.4×10^{-54} at 60 working digits. Second, Z_k alone was recomputed by the same method applied to $\log\big((s-1)\zeta(s)\big)$ on $|z| = 0.55$ at 120 digits; the discrepancies are 2.2×10^{-110} at $k = 40$ and 4.3×10^{-107} at $k = 50$.

For comparison, the zero-side representation $\lambda_k = \sum_{\gamma>0} 2(1-\cos k\theta_\gamma)$ truncated to the first K ordinates has the relative deficits

K	$k=1$	$k=2$	$k=4$	$k=20$
300	6.93%	6.93%	6.94%	7.30%
600	4.41%	4.41%	4.42%	4.64%
900	3.35%	3.35%	3.36%	3.53%
1200	2.75%	2.75%	2.75%	2.89%
1500	2.35%	2.35%	2.35%	2.47%

The deficit is the truncated zero tail and decreases like $1/K$ up to logarithms, consistently with the tail scale

$$\sum_{\gamma>T} \frac{2}{4\gamma^2+1}.$$

It is not an error of either computation.

Remark 26.5 (Removal of circularity). Every numerical experiment in Sections 15, 17 and 25 that reports positivity, pivots, determinants, F_N or κ_N uses the arithmetic λ of Experiment 26.4. No zero of ζ is used as input there. The zero list is used only where a *model* measure is deliberately constructed, namely in the off-line detection experiments of Subsection 25.1, where the atoms are the defining data of the model and their arithmetic origin is irrelevant.

26.3. **Exact Fourier–Laguerre transform of the Li test function.** For $r \in \mathbb{R}$, write

$$\theta(r) := 2\arctan\frac{1}{2r}, \qquad H_k(r) := 2\big(1-\cos(k\theta(r))\big). \tag{26.24}$$

Use the Fourier convention

$$\widehat{f}(u) := \frac{1}{2\pi}\int_{\mathbb{R}} f(r)\mathrm{e}^{-iru}\,\mathrm{d}r. \tag{26.25}$$

Theorem 26.6 (Exact Fourier–Laguerre identity)**.** *For every $k \geq 1$ and $u \in \mathbb{R}$,*

$$\boxed{\widehat{H}_k(u) = \mathrm{e}^{-|u|/2} L_{k-1}^{(1)}(|u|).} \tag{26.26}$$

Proof. For $|z|<1$, summing the elementary cosine series and using

$$\cos\theta(r) = \frac{4r^2-1}{4r^2+1} \tag{26.27}$$

gives

$$\sum_{k\geq 1} H_k(r) z^{k-1} = \frac{4(1+z)}{(1-z)\,(4r^2(1-z)^2+(1+z)^2)}. \tag{26.28}$$

Set

$$a(z) := \frac{1+z}{2(1-z)}. \tag{26.29}$$

Since

$$\frac{1}{2\pi}\int_{\mathbb{R}} \frac{\mathrm{e}^{-iru}}{r^2+a^2}\,\mathrm{d}r = \frac{1}{2a}\mathrm{e}^{-a|u|} \qquad (\Re a>0), \tag{26.30}$$

Fourier transformation of (26.28) yields

$$\sum_{k\geq 1} \widehat{H}_k(u) z^{k-1} = \frac{\mathrm{e}^{-|u|/2}}{(1-z)^2}\exp\!\left(-\frac{|u|z}{1-z}\right). \tag{26.31}$$

Comparison with (26.12) proves (26.26). □

Thus the Cayley reflection system on the critical line and the associated Laguerre system on the logarithmic prime variable are Fourier dual. If

$$G_k(u) := \mathrm{e}^{-u/2} L_{k-1}^{(1)}(u), \qquad u > 0, \tag{26.32}$$

then the standard Laguerre orthogonality gives

$$\int_0^\infty G_k(u) G_\ell(u)\, u \, \mathrm{d}u = k\, \delta_{k\ell}. \tag{26.33}$$

26.4. **Reflection cancellation of the prime main term.** Set $A_0 = Z_0 = 0$ and define

$$(\mathsf{A}_N^E)_{mn} := A_{m+n} - A_{|m-n|}, \qquad (\mathsf{Z}_N^E)_{mn} := Z_{m+n} - Z_{|m-n|}, \tag{26.34}$$

$$(\mathsf{A}_N^O)_{mn} := A_{m+n-1} - A_{|m-n|}, \qquad (\mathsf{Z}_N^O)_{mn} := Z_{m+n-1} - Z_{|m-n|}. \tag{26.35}$$

Then, unconditionally,

$$\mathsf{E}_N = \mathsf{A}_N^E + \mathsf{Z}_N^E, \qquad \mathsf{O}_N = \mathsf{A}_N^O + \mathsf{Z}_N^O. \tag{26.36}$$

For the even family put

$$\Phi_{mn}(u) := L_{m+n}(u) - L_{|m-n|}(u). \tag{26.37}$$

Since $\Phi_{mn}(0) = 0$, Proposition 26.1 gives the exact cancellation

$$\boxed{(\mathsf{Z}_N^E)_{mn} = \int_{1^-}^\infty \Phi'_{mn}(\log t)\, \frac{\mathrm{d}(\psi(t) - t)}{t}.} \tag{26.38}$$

No separate counterterm remains after reflection: the prime matrix pairs the Laguerre-difference kernel directly with the error $\psi(t) - t$ rather than with the main prime density.

The same construction applies to the odd family after replacing $m + n$ by $m + n - 1$ in the first Laguerre index. Moreover, because the displacement identities of Section 7 are algebraic identities for reflection differences of a scalar sequence, each fixed-u Laguerre kernel inherits the same rank-at-most-two displacement structure. Likewise, the Chebyshev changes of basis of Section 9 convert each such elementary reflection kernel into a Hankel matrix. Hence the prime side is simultaneously a Laguerre superposition in the logarithmic variable and a low-displacement/Hankel object in the finite reflection coordinates.

There is an additional positivity property at the level of quadratic test functions. For $c = (c_1, \dots, c_N)^T$ define

$$\Phi_c(u) := \sum_{m,n=1}^N c_m c_n \Phi_{mn}(u). \tag{26.39}$$

The trigonometric identity

$$H_{m+n}(r) - H_{|m-n|}(r) = 4 \sin(m\theta(r)) \sin(n\theta(r)) \tag{26.40}$$

shows that

$$h_c(r) := 4 \left| \sum_{m=1}^N c_m \sin(m\theta(r)) \right|^2 \geq 0. \tag{26.41}$$

By Theorem 26.6, its Fourier transform is

$$q_c(u) := \widehat{h}_c(u) = -\mathrm{e}^{-|u|/2} \Phi_c'(|u|). \tag{26.42}$$

Consequently q_c is a continuous positive-definite function on $\mathbb{R}$. In terms of this cone,

$$\boxed{c^T \mathsf{Z}_N^E c = -\int_{0^-}^\infty \mathrm{e}^{-u/2} q_c(u)\, \mathrm{d}E(\mathrm{e}^u).} \tag{26.43}$$

Thus the arithmetic obstruction is not pairing the prime-number-theorem error against arbitrary Laguerre polynomials: it is pairing against the Fourier transforms of the nonnegative reflection squares (26.41).

26.5. **First differences: a contractive matrix cosine family and an explicit archimedean lower decomposition.** The even reflection sector admits a further normalization which makes the prime kernel uniformly bounded before the prime summation is performed. Let

$$D_N = \begin{pmatrix} 1 & & & \\ -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix} \tag{26.44}$$

be the lower first-difference matrix. For $u \geq 0$ put

$$K_0(u) := 0, \qquad K_r(u) := L_{r-1}^{(1)}(u) \quad (r \geq 1), \tag{26.45}$$

and define the elementary Laguerre reflection matrix

$$\mathsf{M}_N(u) := \big(K_{m+n}(u) - K_{|m-n|}(u)\big)_{m,n=1}^{N}. \tag{26.46}$$

Finally set

$$\mathsf{C}_N(u) := \frac{1}{2}\mathrm{e}^{-u/2} D_N \mathsf{M}_N(u) D_N^T. \tag{26.47}$$

Proposition 26.7 (Contractive matrix-valued cosine representation)**.** *For every N there is a positive $N \times N$ matrix-valued measure Σ_N on $\mathbb{R}$, with total mass I_N, such that*

$$\boxed{\mathsf{C}_N(u) = \int_{\mathbb{R}} \mathrm{e}^{-iru}\, \mathrm{d}\Sigma_N(r).} \tag{26.48}$$

Consequently

$$\mathsf{C}_N(0) = I_N, \qquad \|\mathsf{C}_N(u)\|_2 \leq 1 \quad (u \in \mathbb{R}), \tag{26.49}$$

and $u \mapsto \mathsf{C}_N(u)$ is matrix-valued positive definite.

Proof. Let

$$s(r) := (\sin\theta(r), \dots, \sin(N\theta(r)))^T.$$

Theorem 26.6 and (26.40) imply

$$\mathrm{e}^{-u/2}\mathsf{M}_N(u) = \frac{1}{2\pi}\int_{\mathbb{R}} 4 s(r) s(r)^T \mathrm{e}^{-iru}\, \mathrm{d}r. \tag{26.50}$$

If

$$c(r) := \big(\cos(\theta(r)/2), \cos(3\theta(r)/2), \dots, \cos((N-\tfrac{1}{2})\theta(r))\big)^T,$$

then

$$D_N s(r) = 2\sin(\theta(r)/2) c(r).$$

Hence (26.48) holds with

$$\mathrm{d}\Sigma_N(r) = \frac{4}{\pi}\sin^2(\theta(r)/2) c(r) c(r)^T\, \mathrm{d}r. \tag{26.51}$$

This density is positive semidefinite. Since $L_{r-1}^{(1)}(0) = r$, one has $\mathsf{M}_N(0) = (2\min(m,n))_{m,n}$, and the first-difference identity gives $\mathsf{C}_N(0) = I_N$. Thus $\Sigma_N(\mathbb{R}) = I_N$. For a unit vector v, the scalar measure $v^T\Sigma_N v$ is positive of mass 1, so $|v^T\mathsf{C}_N(u)v| \leq 1$. Symmetry of $\mathsf{C}_N(u)$ yields (26.49). $\square$

Corollary 26.8 (Prime term as a matrix prime–continuum discrepancy)**.** *Put*

$$\widehat{\mathsf{Z}}_N^E := \frac{1}{2} D_N \mathsf{Z}_N^E D_N^T. \tag{26.52}$$

Then, entrywise and therefore in every norm on the fixed finite-dimensional matrix space,

$$\boxed{\widehat{\mathsf{Z}}_N^E = \lim_{X\to\infty} \left[\int_0^{\log X} \mathrm{e}^{u/2} \mathsf{C}_N(u)\, \mathrm{d}u - \sum_{n\le X} \frac{\Lambda(n)}{\sqrt{n}} \mathsf{C}_N(\log n) \right].} \tag{26.53}$$

Thus the first-difference prime matrix is a regularized discrepancy between the continuum measure and the prime powers, sampled against a single contractive matrix-valued cosine family.

Proof. Reflection in (26.7) gives

$$\mathsf{Z}_N^E = \lim_{X\to\infty} \left[\int_0^{\log X} \mathsf{M}_N(u)\, \mathrm{d}u - \sum_{n\le X} \frac{\Lambda(n)}{n} \mathsf{M}_N(\log n) \right].$$

Apply $D_N/\sqrt{2}$ on the left and right and use (26.47). □

The archimedean term admits a complementary positive decomposition in the same coordinates. Set $A_0 := 0$ and

$$\alpha_n := A_{n+1} - 2A_n + A_{n-1}, \qquad n \ge 1. \tag{26.54}$$

Proposition 26.9 (Positive moment formula for the archimedean second differences)**.** *For every* $n \ge 1$,

$$\boxed{\alpha_n = \sum_{j=1}^{\infty} \frac{1}{(2j+1)^2} \left(\frac{2j}{2j+1} \right)^{n-1} > 0.} \tag{26.55}$$

Proof. From (26.6) and $(1-2^{-r})\zeta(r) = \sum_{\substack{q\ge 1\\ q\,\mathrm{odd}}} q^{-r}$, the centered second difference gives, for $n \ge 2$,

$$\begin{aligned} \alpha_n &= \sum_{\substack{q\ge 1\\ q\,\mathrm{odd}}} q^{-2} \sum_{r=2}^{n+1} \binom{n-1}{r-2} (-q^{-1})^{r-2} \\ &= \sum_{\substack{q\ge 1\\ q\,\mathrm{odd}}} q^{-2} (1-q^{-1})^{n-1}. \end{aligned}$$

The $q = 1$ term vanishes. For $n = 1$, the convention $A_0 = 0$ subtracts exactly that $q = 1$ term, and the same formula results. Writing $q = 2j+1$ proves (26.55). □

Define

$$\widehat{\mathsf{A}}_N^E := \frac{1}{2} D_N \mathsf{A}_N^E D_N^T \tag{26.56}$$

and index its rows and columns by $0 \le i, j \le N-1$. A direct first-difference calculation gives

$$(\widehat{\mathsf{A}}_N^E)_{ij} = A_1 \delta_{ij} + \frac{1}{2}(1-\delta_{ij}) \alpha_{|i-j|} + \frac{1}{2} \alpha_{i+j+1}. \tag{26.57}$$

Put

$$\rho_j := \frac{2j}{2j+1}, \qquad w_j := \frac{1}{(2j+1)^2}, \tag{26.58}$$

and

$$T_N(\rho) := (\rho^{|i-j|})_{i,j=0}^{N-1}, \qquad v_N(\rho) := (1, \rho, \dots, \rho^{N-1})^T. \tag{26.59}$$

Theorem 26.10 (Archimedean positive decomposition after first differences)**.** *For every N,*

$$\boxed{\widehat{\mathsf{A}}_N^E = c_\infty I_N + \frac{1}{2}\sum_{j\geq 1}\frac{w_j}{\rho_j}T_N(\rho_j) + \frac{1}{2}\sum_{j\geq 1} w_j v_N(\rho_j)v_N(\rho_j)^T,} \tag{26.60}$$

where

$$c_\infty := \frac{1-\gamma_E-\log(2\pi)}{2} = -0.7075463656\ldots. \tag{26.61}$$

In particular

$$\widehat{\mathsf{A}}_N^E \succeq c_\infty I_N. \tag{26.62}$$

Proof. Proposition 26.9 gives

$$\alpha_n = \sum_{j\geq 1} w_j \rho_j^{n-1}.$$

Thus the Toeplitz part of (26.57) is

$$\frac{1}{2}\sum_j \frac{w_j}{\rho_j}T_N(\rho_j),$$

and the Hankel part is

$$\frac{1}{2}\sum_j w_j v_N(\rho_j)v_N(\rho_j)^T.$$

The diagonal correction is

$$\begin{aligned} A_1 - \frac{1}{2}\sum_{j\geq 1}\frac{w_j}{\rho_j} &= 1 - \frac{1}{2}(\gamma_E+\log 4\pi) - \frac{1}{2}\sum_{j\geq 1}\frac{1}{2j(2j+1)} \\ &= 1 - \frac{1}{2}(\gamma_E+\log 4\pi) - \frac{1}{2}(1-\log 2) = c_\infty. \end{aligned}$$

For $0<\rho<1$, $T_N(\rho)$ is positive definite: it is the Toeplitz moment matrix of the positive Poisson kernel $(1-\rho^2)/(1-2\rho\cos\vartheta+\rho^2)$. The remaining summands are rank-one positive semidefinite matrices, proving (26.62). □

Define the explicit positive matrix

$$\mathsf{S}_N := \frac{1}{2}\sum_{j\geq 1}\frac{w_j}{\rho_j}T_N(\rho_j) + \frac{1}{2}\sum_{j\geq 1} w_j v_N(\rho_j)v_N(\rho_j)^T \succeq 0. \tag{26.63}$$

Combining the two sides gives the exact normal form

$$\boxed{\frac{1}{2}D_N\mathsf{E}_N D_N^T = c_\infty I_N + \mathsf{S}_N + \widehat{\mathsf{Z}}_N^E.} \tag{26.64}$$

Since D_N is invertible, positivity of the even reflection block is equivalent to

$$\boxed{\mathsf{S}_N + \widehat{\mathsf{Z}}_N^E \succeq \frac{\gamma_E+\log(2\pi)-1}{2}\, I_N.} \tag{26.65}$$

The archimedean matrix itself is not positive definite: already $\mathsf{A}_1^E = (A_2) = (-0.8745393617\ldots)$. Thus a normalization based on a positive square root of A_N^E is unavailable; (26.64) instead isolates its negative scalar part and an explicit positive semidefinite remainder.

26.6. **A scalar prime-only coefficient criterion.** The arithmetic factor also gives a direct coefficient formulation. Define

$$\mathcal{F}_{\mathrm{ar}}(z) := \log\left[\left(\frac{1}{1-z}-1\right)\zeta\left(\frac{1}{1-z}\right)\right] = \mathcal{F}_{\mathrm{ar}}(0) + \sum_{k\geq 1}\frac{Z_k}{k}z^k, \tag{26.66}$$

where the logarithm denotes the branch analytic near $z=0$. The Cayley relation is

$$z = 1 - \frac{1}{s} = \frac{s-1}{s}. \tag{26.67}$$

Theorem 26.11 (Prime–Laguerre root-growth criterion)**.** *The following statements are equivalent:*

(i) RH *holds;*

(ii) $\mathcal{F}_{\mathrm{ar}}$ *has an analytic logarithmic branch throughout* $|z|<1$*;*

(iii)

$$\boxed{\limsup_{k\to\infty}|Z_k|^{1/k}\leq 1;} \tag{26.68}$$

(iv) for every $\varepsilon>0$ *there is a constant* C_ε *such that*

$$|Z_k| \leq C_\varepsilon(1+\varepsilon)^k \qquad (k\geq 1). \tag{26.69}$$

Equivalently, by (26.9)*,* RH *is equivalent to subexponential root growth of the Laguerre transform of* $\psi(t)-t$*.*

Proof. The map $z=(s-1)/s$ satisfies

$$|z|<1 \quad\Longleftrightarrow\quad |s-1|<|s| \quad\Longleftrightarrow\quad \Re s>\frac{1}{2}. \tag{26.70}$$

The function $(s-1)\zeta(s)$ is analytic and nonzero at $s=1$. On the simply connected disk $|z|<1$, it therefore admits an analytic logarithm exactly when it has no zero there. By (26.70), this is the absence of nontrivial zeros with $\Re s>1/2$; the functional equation then makes this equivalent to RH. Trivial zeros map outside the unit disk. This proves (i)$\Longleftrightarrow$(ii).

By Cauchy–Hadamard, the radius R of the Taylor series in (26.66) satisfies

$$R^{-1} = \limsup_{k\to\infty}\left|\frac{Z_k}{k}\right|^{1/k} = \limsup_{k\to\infty}|Z_k|^{1/k}. \tag{26.71}$$

If (ii) holds, its Taylor series has radius at least one, so (iii) follows. Conversely, if (iii) holds, the series defines an analytic function $G(z)$ on $|z|<1$ which agrees near 0 with the chosen logarithm of the arithmetic factor. Hence $\exp G(z)$ agrees near 0, and therefore throughout the disk by the identity theorem, with the analytic arithmetic factor. In particular that factor has no zero in $|z|<1$, so G is its analytic logarithmic branch there and (ii) follows. The equivalence of (iii) and (iv) is the elementary root-test formulation of subexponential growth. □

The criterion admits a quantitative form. If R_Z denotes the radius of convergence in (26.66), then

$$\boxed{R_Z = \min\left\{1, \inf_{\rho}\left|1-\frac{1}{\rho}\right|\right\},} \tag{26.72}$$

where the infimum runs over the nontrivial zeros. Indeed, their Cayley images are precisely the zeros of the analytic arithmetic factor, trivial zeros map outside $|z|=1$, and $z=1$ is itself a singular point because $s(z)\to\infty$ and $\log((s-1)\zeta(s))\sim\log s$. In particular, a zero $\rho=\beta+i\gamma$ with $\beta>1/2$ satisfies

$$\left|1-\frac{1}{\rho}\right|^2 = \frac{(\beta-1)^2+\gamma^2}{\beta^2+\gamma^2} < 1 \tag{26.73}$$

and forces exponential coefficient growth at the reciprocal Cayley scale.

Proposition 26.12 (Dominant off-line Cayley modes)**.** *Suppose $R_Z < 1$, and let $z_1, \dots, z_M$ be the Cayley images of the nontrivial zeros lying on $|z| = R_Z$, with multiplicities $m_1, \dots, m_M$. Then there exists $R_2 > R_Z$ such that*

$$Z_k = -\sum_{j=1}^{M} m_j z_j^{-k} + O(R_2^{-k}). \tag{26.74}$$

In particular,

$$\limsup_{k\to\infty} |Z_k|^{1/k} = R_Z^{-1} > 1. \tag{26.75}$$

Proof. The closed disk $|z| \le r < 1$ corresponds under $s = 1/(1-z)$ to a compact subset of $\Re s > 1/2$, so it contains only finitely many zero images. Near a zero image z_j, the arithmetic generating function has the local form

$$\mathcal{F}_{\mathrm{ar}}(z) = m_j \log(1 - z/z_j) + \text{a function analytic near } z_j. \tag{26.76}$$

Choose $R_2 > R_Z$ smaller than the modulus of every other singularity. Subtracting the finitely many logarithmic singularities on $|z| = R_Z$ leaves a function analytic in $|z| < R_2$. Since

$$k[z^k] \log(1 - z/z_j) = -z_j^{-k}, \tag{26.77}$$

Cauchy estimates for the analytic remainder give (26.74). The root-growth statement follows from the radius formula (26.72). □

26.7. **The scalar L^2 route, its exact Bessel weight, and the return to the matrix cone.** The scalar criterion above is a useful diagnostic reduction, but by itself it discards the reflection structure. It is therefore important to record exactly what can and cannot be obtained from Laguerre orthogonality alone.

Proposition 26.13 (Second cancellation and the $L^{(2)}$ representation)**.** *Let $E(t) = \psi(t) - t$. For every $k \ge 1$,*

$$\boxed{Z_k = -\int_1^\infty \frac{\psi(t) - t + 1}{t^2} L_{k-1}^{(2)}(\log t)\, \mathrm{d}t.} \tag{26.78}$$

Equivalently, with $u = \log t$,

$$Z_k = -\int_0^\infty \mathrm{e}^{-u}(E(\mathrm{e}^u) + 1) L_{k-1}^{(2)}(u)\, \mathrm{d}u. \tag{26.79}$$

Proof. Using (26.9) and (26.14), first write

$$Z_k = -\int_{0^-}^\infty \mathrm{e}^{-u} L_{k-1}^{(1)}(u)\, \mathrm{d}E(\mathrm{e}^u). \tag{26.80}$$

Then put

$$R(u) := \mathrm{e}^{-u} E(\mathrm{e}^u). \tag{26.81}$$

Then

$$\mathrm{e}^{-u}\, \mathrm{d}E(\mathrm{e}^u) = \mathrm{d}R(u) + R(u)\, \mathrm{d}u. \tag{26.82}$$

The prime number theorem gives $R(u) \to 0$ as $u \to \infty$, while $R(0) = E(1^-) = -1$. Writing $n = k - 1$ and integrating the $\mathrm{d}R$ term by parts gives

$$Z_k = -k - \int_0^\infty R(u) \left[L_n^{(1)}(u) - (L_n^{(1)})'(u) \right] \mathrm{d}u. \tag{26.83}$$

The Laguerre identities

$$(L_n^{(1)})' = -L_{n-1}^{(2)}, \qquad L_n^{(1)} = L_n^{(2)} - L_{n-1}^{(2)} \tag{26.84}$$

give the exact cancellation

$$L_n^{(1)} - (L_n^{(1)})' = L_n^{(2)}. \tag{26.85}$$

Finally,

$$\int_0^\infty \mathrm{e}^{-u} L_n^{(2)}(u)\, \mathrm{d}u = n+1 = k, \tag{26.86}$$

which absorbs the boundary term $-k$ and proves (26.79); the change of variables $t = \mathrm{e}^u$ gives (26.78). □

The ordinary integral (26.78) removes the Stieltjes atoms, but it does not remove the critical-line difficulty. Indeed, the $\alpha = 2$ orthogonality relation is

$$\int_0^\infty L_n^{(2)}(u) L_m^{(2)}(u) u^2 \mathrm{e}^{-u}\, \mathrm{d}u = (n+1)(n+2)\delta_{nm}. \tag{26.87}$$

Set

$$f(u) := \frac{E(\mathrm{e}^u)+1}{u^2}. \tag{26.88}$$

Then

$$Z_{n+1} = -\left\langle f, L_n^{(2)} \right\rangle_{L^2(u^2\mathrm{e}^{-u}\,\mathrm{d}u)}. \tag{26.89}$$

Therefore Bessel's inequality has the direction and normalization

$$\boxed{\sum_{n\ge 0} \frac{|Z_{n+1}|^2}{(n+1)(n+2)} \le \int_0^\infty \frac{|E(\mathrm{e}^u)+1|^2}{u^2} \mathrm{e}^{-u}\, \mathrm{d}u,} \tag{26.90}$$

whenever the right-hand side is finite. Equivalently,

$$\boxed{\sum_{k\ge 1} \frac{|Z_k|^2}{k(k+1)} \le \int_1^\infty \frac{|\psi(t)-t+1|^2}{t^2(\log t)^2}\, \mathrm{d}t.} \tag{26.91}$$

The polynomial factor is in the denominator, not the numerator. Thus no estimate of the form

$$\sum k(k+1)|Z_k|^2 \le \|f\|_2^2 \tag{26.92}$$

follows from Bessel. If the right-hand side of (26.91) were finite, then one would obtain the much stronger individual estimate

$$|Z_k| \le \sqrt{k(k+1)}\, \|f\|_2, \tag{26.93}$$

and hence (26.69).

Remark 26.14 (What the standard RH pointwise bound gives in the global L^2 norm). The standard pointwise consequence of RH,

$$E(t) = O(t^{1/2}\log^2 t), \tag{26.94}$$

inserted into the right-hand side of (26.91) gives only

$$\frac{|E(t)+1|^2}{t^2(\log t)^2} = O\!\left(\frac{\log^2 t}{t}\right), \tag{26.95}$$

whose integral diverges. Thus the standard pointwise RH estimate does not establish the global L^2 integrability required on the right-hand side. This calculation does not prove that RH itself forces divergence of that norm; it shows only that the standard pointwise bound is insufficient.

Any successful Laguerre argument along this line would therefore need additional cancellation, for example inside finite projections.

Warning 26.15 (Abel deformation does not bound the undeformed coefficients). For $\delta > 0$ one may insert an extra factor $\mathrm{e}^{-\delta u}$ in (26.79) and obtain deformed coefficients $Z_k(\delta)$ for which global Bessel estimates are easier. These coefficients form a different sequence. Bounds for fixed $\delta > 0$ do not imply bounds for $Z_k = Z_k(0)$ unless one also proves uniform control as $\delta \downarrow 0$ on the k-dependent scale under consideration. Thus Abel regularization alone does not cross the critical barrier.

There is also an elementary one-sided sign implication. If the coefficients Z_k are eventually all nonnegative, or eventually all nonpositive, then Pringsheim's theorem applied after removing finitely many initial terms forces the radius of convergence of (26.66) to be at least one, and hence RH follows from Theorem 26.11. This implication is logically correct but is not used as a proposed proof route. High-precision numerical exploration through $k = 80$ gives positive values but a pronounced oscillation about a baseline near 1; representative values are

$$Z_5 \approx 1.4583, \qquad Z_{20} \approx 0.6707, \qquad Z_{50} \approx 1.4979, \qquad Z_{75} \approx 0.5882, \qquad Z_{80} \approx 0.3576. \tag{26.96}$$

Two independent evaluations were used, the Stieltjes recursion of Proposition 26.2 for $k \le 50$ and a Cauchy integral on $\log\big((s-1)\zeta(s)\big)$ for $k \le 80$, agreeing to 4.3×10^{-107} at $k = 50$. All eighty values are strictly positive and lie in $[0.3576, 1.5128]$, with the minimum attained at the right endpoint and the sequence still falling there. The observed excursion amplitude does not show stabilization. These data neither prove nor disprove eventual one-sidedness, but they provide no numerical basis for treating it as the main analytic mechanism.

Remark 26.16 (Classical scalar mechanism versus the reflection compression). The root-growth criterion (26.68) is a factorwise version of the classical Keiper–Li singularity mechanism [22, 26, 3]: after the Cayley map, an off-line zero becomes a singularity inside the unit disk and therefore produces exponentially growing coefficients. The role of the scalar sequence Z_k in the present paper is consequently diagnostic. The structure specific to the reflection construction is lost upon scalarization.

Before scalarization, the even arithmetic matrix satisfies (26.38), and every quadratic test has the form

$$c^T \mathsf{Z}_N^E c = -\int_{0^-}^{\infty} \mathrm{e}^{-u/2} q_c(u)\, \mathrm{d}E(\mathrm{e}^u), \tag{26.97}$$

where q_c is the Fourier transform of the nonnegative sine square (26.41) and is therefore positive-definite. At the same time each fixed-u prime atom inherits rank-at-most-two displacement and becomes Hankel after the Chebyshev change of basis. The first-difference normal form further represents the prime side by the contractive matrix-valued cosine family (26.47), while the archimedean side has the positive decomposition (26.64). These simultaneous constraints are absent from the scalar criterion. They do not, however, weaken the logical content of the problem: positivity of all even reflection truncations is itself equivalent to RH by Theorem 5.5. The fact that the admissible quadratic tests form a smaller structured cone is therefore not by itself an analytic gain. A matrix proof requires an additional mechanism that converts displacement, nesting, or the prime–continuum representation into a sign estimate; without such a mechanism the cone description is only a structural reformulation.

Remark 26.17 (Strategic status of the matrix route). The matrix formulation is retained not because it is logically weaker, but because it carries operations unavailable to a scalar sequence. The most concrete non-circular target is to combine the nesting of the principal reflection blocks with their rank-two displacement structure. For a positive definite truncation write the last Schur complement

$$p_N := \frac{\det \mathsf{E}_N}{\det \mathsf{E}_{N-1}} = e_{NN} - b_N^T \mathsf{E}_{N-1}^{-1} b_N, \tag{26.98}$$

where $b_N = (e_{1N}, \dots, e_{N-1,N})^T$. Positivity of all truncations is equivalent, away from singular boundary cases, to positivity of the successive pivots. The research question is therefore whether the rank-two displacement identities can turn (26.98) into a recurrence or comparison principle whose sign is visible directly on the prime–archimedean side. This would be an actual mechanism; merely restating positivity on the reflection cone would not.

The preceding discussion identifies the current boundary of this scalar estimate: the standard pointwise RH error bound does not by itself establish the global Laguerre L^2 norm required by Bessel. This leaves finite-projection cancellation open and motivates the matrix route, where Schur complements, displacement generators, and critical-vector geometry provide additional exact structure on which a sign mechanism could act.

26.8. **Exact action of the Laguerre transform on exponential modes.** The preceding singularity statement has a direct integral counterpart.

Lemma 26.18 (Laguerre–Laplace transform)**.** *For $\Re a > 0$ and $k \geq 1$,*

$$\boxed{\int_0^\infty \mathrm{e}^{-ax} L_{k-1}^{(1)}(x)\, \mathrm{d}x = 1 - \left(1 - \frac{1}{a}\right)^k.} \tag{26.99}$$

Proof. Integrating the generating function (26.12) termwise gives

$$\sum_{k\geq 1} z^{k-1} \int_0^\infty \mathrm{e}^{-ax} L_{k-1}^{(1)}(x)\, \mathrm{d}x = \frac{1}{(1-z)^2} \frac{1}{a + z/(1-z)} \tag{26.100}$$

$$= \frac{1}{(1-z)(a-(a-1)z)}. \tag{26.101}$$

Coefficient extraction yields (26.99). □

For a nontrivial zero ρ one may take $a = 1 - \rho$, since $0 < \Re\rho < 1$. With

$$z_\rho := 1 - \frac{1}{\rho} \tag{26.102}$$

one obtains

$$\int_0^\infty \mathrm{e}^{-(1-\rho)x} L_{k-1}^{(1)}(x)\, \mathrm{d}x = 1 - z_\rho^{-k}. \tag{26.103}$$

Thus the Laguerre transform sends the exponential mode generated by a zero directly to a geometric sequence in the Cayley coordinate. The critical line is exactly the unit-modulus threshold $|z_\rho| = 1$.

Remark 26.19 (Correct large-k formulation). A large-k proof of (26.69) must be applied to the weighted Stieltjes form

$$Z_k = -\int_{0^-}^\infty \mathrm{e}^{-u} L_{k-1}^{(1)}(u)\, \mathrm{d}E(\mathrm{e}^u) = -\int_{0^-}^\infty \mathrm{e}^{-u/2} G_k(u)\, \mathrm{d}E(\mathrm{e}^u), \tag{26.104}$$

not to an unweighted integral in the variable u. The standard turning-point structure also belongs to the weighted Laguerre functions rather than to a bare assertion that $L_k(k\xi)$ decays for $\xi < 1$. Finally, the prime-side Stieltjes representation and the zero-side explicit formula are alternative representations of the same functional; they cannot be added as independent contributions. Equation (26.103) identifies the exact phenomenon that any uniform asymptotic argument must suppress: a zero with $\Re\rho > 1/2$ produces a geometric mode with modulus greater than one.

The scalar identity therefore remains a precise diagnostic of exponentially growing Cayley modes, but the Bessel analysis in Subsection 26.7 shows why scalar orthogonality alone does not supply the missing estimate. The principal analytic target retained here is the reflection-matrix inequality (26.65), where the arithmetic discrepancy is tested only against the structured positive-definite cone (26.43) and the contractive cosine family (26.47).

27. Route A: the even reflection matrix and the exact RH target

The proof problem can now be stated without the odd matrix. By Theorem 5.5,

$$\boxed{\mathrm{RH} \iff \mathsf{E}_N \succeq 0 \ \forall N \iff \mathsf{E}_N \succ 0 \ \forall N.} \tag{27.1}$$

The role of O_N in the preceding structural sections is comparative; it is not needed in the implication to RH.

27.1. **A1. The exact Schur induction.** Write

$$\mathsf{E}_N = \begin{pmatrix} \mathsf{E}_{N-1} & b_N \\ b_N^T & \lambda_{2N} \end{pmatrix}, \qquad p_N := \lambda_{2N} - b_N^T \mathsf{E}_{N-1}^{-1} b_N. \tag{27.2}$$

Whenever $\mathsf{E}_{N-1} \succ 0$,

$$\boxed{\mathsf{E}_N \succ 0 \iff p_N > 0.} \tag{27.3}$$

Therefore the single inductive implication

$$\boxed{\mathsf{E}_{N-1} \succ 0 \quad \Longrightarrow \quad \lambda_{2N} > b_N^T \mathsf{E}_{N-1}^{-1} b_N \qquad (N \geq 2)} \tag{27.4}$$

would prove $\mathsf{E}_N \succ 0$ for every N, and hence RH.

27.2. **A2. The local LDL^T form.** The recurrence part is already exact. For the even boundary condition $\ell_0 = 0$, Theorem 15.6 and Corollary 15.7 give

$$\frac{p_{n+1}}{p_n} = 1 + (\tau_n - \tau_{n-1}) - \ell_n(\ell_n - \ell_{n-1}) = V_n - V_{n-1} - \frac{1}{2}(d_n - d_{n-1})^2. \tag{27.5}$$

Hence

$$\boxed{V_n - V_{n-1} > \frac{1}{2}(d_n - d_{n-1})^2 \quad \forall n} \tag{27.6}$$

would imply $p_{n+1}/p_n > 0$ for every n. Since $p_1^E = \lambda_2 > 0$, induction would give every even pivot positive and prove RH. No odd pivot is required.

27.3. **A3. Exact prime–continuum scalarization of the matrix target.** The first-difference normal form

$$\frac{1}{2} D_N \mathsf{E}_N D_N^T = c_\infty I_N + \mathsf{S}_N + \widehat{\mathsf{Z}}_N^E \tag{27.7}$$

gives a second exact formulation. Fix a unit vector $v \in \mathbb{R}^N$, put

$$c := D_N^T v, \qquad p_v(u) := \frac{1}{2} c^T \mathsf{M}_N(u) c, \qquad g_v(u) := p_v'(u) - p_v(u), \tag{27.8}$$

and set

$$s_v := v^T \mathsf{S}_N v, \qquad P_v := v^T \widehat{\mathsf{Z}}_N^E v. \tag{27.9}$$

Proposition 27.1 (Exact signed PNT discrepancy)**.** *For every fixed N and every unit v,*

$$p_v(0) = 1 \tag{27.10}$$

and

$$\boxed{P_v = -1 + \int_1^\infty (\psi(t) - t)\, g_v(\log t)\, \frac{\mathrm{d}t}{t^2}.} \tag{27.11}$$

Consequently,

$$\boxed{\mathsf{E}_N \succ 0 \iff \int_1^\infty (\psi(t) - t)\, g_v(\log t)\, \frac{\mathrm{d}t}{t^2} > 1 - c_\infty - s_v \quad \textit{for every unit } v.} \tag{27.12}$$

Proof. The identity $\mathsf{M}_N(0) = (2\min(m,n))_{m,n}$, together with $D_N(\min(m,n))D_N^T = I_N$, gives

$$p_v(0) = c^T(\min(m,n))c = v^T v = 1.$$

Since $\Phi'_{mn} = -\mathsf{M}_{N,mn}$, the prime matrix formula (26.38) yields

$$P_v = -\int_{1^-}^{\infty} p_v(\log t)\,\frac{\mathrm{d}(\psi(t)-t)}{t}.$$

Integration by parts is legitimate for fixed N: p_v is a polynomial in u, while the classical zero-free region for ζ gives, for some $c > 0$,

$$\psi(t) - t = O\Big(t\exp(-c\sqrt{\log t})\Big),$$

which is more than sufficient to make the boundary term vanish and the resulting integral converge; see, for example, [39]. The lower boundary is $(\psi(1^-) - 1)p_v(0) = -1$, and differentiation of $p_v(\log t)/t$ gives $g_v(\log t)\,\mathrm{d}t/t^2$. This proves (27.11). Substitution into

$$v^T\left(\frac{1}{2}D_N\mathsf{E}_N D_N^T\right)v = c_\infty + s_v + P_v$$

gives (27.12). □

Remark 27.2 (Why magnitude bounds do not close the sign). Equation (27.12) is a *lower* bound on a signed oscillatory integral. A pointwise estimate for $|\psi(t) - t|$, including the classical Schoenfeld bound under RH [46], yields only an upper bound for the absolute value of that integral. Accordingly, no argument that discards the sign correlation between $\psi(t) - t$ and the structured test g_v can prove (27.12).

Experiment 27.3 (Finite evaluation of the signed discrepancy and the magnitude barrier)**.** For the finite critical directions used in the multiprecision computation, the target values $1 - c_\infty - s_v$ were

$$1.67748,\ 1.68054,\ 1.68200,\ 1.6834,\ 1.6843,\ 1.6848$$

for $N = 2, 3, 4, 6, 9, 12$, approaching the observed level 1.6860917. For $N = 2, 3$, direct prime summation and piecewise integration up to $X = 10^7$ gave

N	$P_v + 1$	finite-X integral at $X = 10^7$
2	1.6774829902	1.67851
3	1.6805385806	1.68211

with the finite-X values decreasing over the tested truncations. No monotonicity or asymptotic convergence claim is inferred from those finite values.

Using the Schoenfeld-type RH pointwise majorant only as an absolute-value diagnostic gives the following much wider bounds:

N	required lower level	absolute-value majorant	ratio
2	1.67748	16.72	9.97
4	1.68200	39.54	23.5
9	1.68432	116.68	69.3
12	1.68477	173.66	103.1

These are finite numerical diagnostics, not theorem statements. Their analytic meaning is already contained in Remark 27.2: the missing information is signed prime cancellation against the reflection-selected test.

28. Route B: inverse dynamics and the adaptive Schur direction

This route keeps the same target $p_N > 0$ but treats E_N^{-1} as the object that selects the only new direction at each bordering step. The purpose is not to estimate $\|\mathsf{E}_N^{-1}\|_2$. Under RH the norm necessarily diverges at least at the rate proved in Corollary 12.10; by itself this divergence contains no sign information. The useful information is the structured *directional* action of the inverse.

28.1. **B1. Two generators determine the inverse.** Let $\mathcal{G}_N := \mathsf{E}_N^{-1}$, whenever the inverse exists, and write

$$g_N := \mathcal{G}_N e_N, \qquad h_N := \mathcal{G}_N r_E^{(N)}.$$

Corollary 7.4 gives

$$\boxed{\mathsf{J}_E^{(N)}\mathcal{G}_N - \mathcal{G}_N\mathsf{J}_E^{(N)} = \frac{1}{2}(g_N h_N^T - h_N g_N^T).} \tag{28.1}$$

Thus the inverse has the same exact displacement rank two as the original reflection matrix.

Let

$$\mathsf{J}_E^{(N)} = S_N^{(J)}\,\mathrm{diag}(x_1,\dots,x_N)(S_N^{(J)})^T, \qquad x_k = \cos\frac{k\pi}{N+1},$$

be the Dirichlet sine diagonalization, and put hats on vectors and matrices in that basis. For $i \neq j$, (28.1) gives the explicit off-diagonal formula

$$\boxed{\widehat{\mathcal{G}}_{ij} = \frac{\widehat{g}_i\widehat{h}_j - \widehat{h}_i\widehat{g}_j}{2(x_i - x_j)}.} \tag{28.2}$$

The diagonal is then fixed by $g_N = \mathcal{G}_N e_N$, because every component of $(S_N^{(J)})^T e_N$ is nonzero. Hence the pair (g_N, h_N), together with the known Dirichlet spectrum, determines the entire inverse.

28.2. **B2. Nesting is a rank-one inverse update.** With

$$G := \mathsf{E}_{N-1}^{-1}, \qquad q_N := Gb_N,$$

the exact block inverse of Proposition 15.4 can be written

$$\boxed{\mathsf{E}_N^{-1} = \begin{pmatrix} G & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{p_N}\begin{pmatrix} q_N \\ -1 \end{pmatrix}\begin{pmatrix} q_N \\ -1 \end{pmatrix}^T.} \tag{28.3}$$

In particular,

$$\boxed{(\mathsf{E}_N^{-1})_{NN} = \frac{1}{p_N}.} \tag{28.4}$$

The size of this entry measures the inverse Schur margin, but its sign is what matters: under the inductive hypothesis $\mathsf{E}_{N-1} \succ 0$,

$$(\mathsf{E}_N^{-1})_{NN} > 0 \iff p_N > 0 \iff \mathsf{E}_N \succ 0.$$

This is an exact induction criterion, not a norm criterion.

Experiment 28.1 (Arithmetic growth of a fixed inverse entry)**.** Using the arithmetic reconstruction of the Li coefficients and no zero ordinates as input, direct multiprecision inversion gives

N	$(\mathsf{E}_N^{-1})_{11}$	$(\mathsf{E}_N^{-1})_{NN} = 1/p_N$
2	$1.1884763894 \times 10^{7}$	$2.9759641880 \times 10^{6}$
3	$5.6137194399 \times 10^{13}$	$2.2546793098 \times 10^{12}$
4	$9.0498470627 \times 10^{20}$	$4.6400962986 \times 10^{18}$
5	$2.6501319296 \times 10^{28}$	$1.5104983683 \times 10^{25}$
6	$1.0568460170 \times 10^{36}$	$6.1010537580 \times 10^{31}$
7	$8.9652480957 \times 10^{43}$	$4.9015837096 \times 10^{38}$
8	$1.3441381914 \times 10^{52}$	$6.6152967546 \times 10^{45}$
9	$1.8902274295 \times 10^{60}$	$8.0490977053 \times 10^{52}$

The first column gives the fixed-entry values. No separate "deficit" notation is introduced here, because the symbol d_n is already reserved for the local LDL^T defect variables. The computation verifies rapid finite growth; the asymptotic divergence itself is supplied independently by Corollary 12.10 and is not inferred from this table.

28.3. **B3. The inverse selects the residual polynomial.** Define

$$\chi_N := \begin{pmatrix} -q_N \\ 1 \end{pmatrix}. \tag{28.5}$$

Then

$$\boxed{\mathsf{E}_N \chi_N = p_N e_N, \qquad \chi_N^T \mathsf{E}_N \chi_N = p_N, \qquad \mathsf{E}_N^{-1} e_N = \frac{\chi_N}{p_N}.} \tag{28.6}$$

Thus χ_N is the unique Schur direction orthogonal, in the Gram geometry, to all preceding modes.

In the Chebyshev basis $u_N(x) = (U_0(x), \dots, U_{N-1}(x))^T$, set

$$R_{N-1}(x) := \chi_N^T u_N(x). \tag{28.7}$$

Under RH this is the residual obtained by projecting U_{N-1} onto $\operatorname{span}\{U_0, \dots, U_{N-2}\}$, and

$$\boxed{\|R_{N-1}\|^2_{L^2(\mu_E)} = p_N.} \tag{28.8}$$

Moreover,

$$u_N(x)^T \mathsf{E}_N^{-1} e_N = \frac{R_{N-1}(x)}{p_N}.$$

28.4. **B4. Christoffel–Darboux kernel and leverage.** Theorem 14.1 identifies

$$\mathcal{K}_N^E(x,y) = u_N(x)^T \mathsf{E}_N^{-1} u_N(y) \tag{28.9}$$

with the degree-$< N$ reproducing kernel. Equation (28.3) becomes

$$\boxed{\mathcal{K}_N^E(x,y) = \mathcal{K}_{N-1}^E(x,y) + \frac{R_{N-1}(x) R_{N-1}(y)}{p_N}.} \tag{28.10}$$

This is the kernel form of the same Schur step.

For a critical-line zero,

$$a_{N,\gamma}^E = 2 \sin\theta_\gamma \, u_N(x_\gamma),$$

hence

$$\boxed{\ell_{N,\gamma}^E = 4 \sin^2 \theta_\gamma \, \mathcal{K}_N^E(x_\gamma, x_\gamma).} \tag{28.11}$$

At the accumulation point $x = 1$,

$$u_N(1) = (1, 2, \dots, N)^T,$$

so the high-zero constant in Theorem 11.8 is exactly

$$c_N^E = \mathcal{K}_N^E(1,1). \tag{28.12}$$

The same formula complexifies. For $q \in \mathbb{C} \setminus \{0, \pm 1\}$, put $x(q) = (q + q^{-1})/2$. Since

$$q^m - q^{-m} = (q - q^{-1})U_{m-1}(x(q)),$$

the zero-orbit vector

$$v_N(q) := (q - q^{-1}, q^2 - q^{-2}, \dots, q^N - q^{-N})^T$$

satisfies

$$\boxed{v_N(q)^T \mathsf{E}_N^{-1} v_N(r) = (q - q^{-1})(r - r^{-1})\mathcal{K}_N^E(x(q), x(r)).} \tag{28.13}$$

Thus every rank-two secular determinant associated with a hypothetical off-line orbit can be written using only complex evaluations of the same inverse kernel.

28.5. **B5. The adaptive prime–continuum inequality.** The inverse becomes potentially useful for RH because it selects one structured test rather than all vectors. Define χ_N by (28.5) and

$$y_N := D_N^{-T} \chi_N. \tag{28.14}$$

The first-difference normal form gives exactly

$$\boxed{\frac{p_N}{2} = y_N^T \left(c_\infty I_N + \mathsf{S}_N + \widehat{\mathsf{Z}}_N^E \right) y_N.} \tag{28.15}$$

Set

$$a := -c_\infty = \frac{\gamma_E + \log(2\pi) - 1}{2}.$$

Then the single inequality

$$\boxed{y_N^T (\mathsf{S}_N + \widehat{\mathsf{Z}}_N^E) y_N > a \|y_N\|^2 \qquad (N \geq 2)} \tag{28.16}$$

under the inductive hypothesis $\mathsf{E}_{N-1} \succ 0$ implies $p_N > 0$, hence propagates positivity and proves RH.

The arithmetic term is tested against

$$\phi_N(u) := y_N^T \mathsf{C}_N(u) y_N. \tag{28.17}$$

Since C_N is a contractive matrix-valued positive-definite cosine family,

$$\phi_N(0) = \|y_N\|^2, \qquad |\phi_N(u)| \leq \|y_N\|^2. \tag{28.18}$$

The exact prime–continuum formula yields

$$y_N^T \widehat{\mathsf{Z}}_N^E y_N = \lim_{X \to \infty} \left[\int_0^{\log X} \mathrm{e}^{u/2} \phi_N(u) \, \mathrm{d}u - \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \phi_N(\log n) \right]. \tag{28.19}$$

Unlike the full matrix inequality $\mathsf{S}_N + \widehat{\mathsf{Z}}_N^E \succeq aI$, the induction needs (28.16) only on the recursively generated sequence y_N. This is the precise open target of the inverse route.

28.6. **B6. Finite numerical status and shift nonmonotonicity.** The following numerical statements are recorded at their verified finite level:

- arithmetic Li reconstruction gives positive even reflection pivots through $N = 22$; the reported values include

$$p_2 = 3.360255490 \times 10^{-7}, \quad p_{12} = 1.214721178 \times 10^{-75}, \quad p_{22} = 2.854130423 \times 10^{-154};$$

- for the first-difference congruence $\widehat{E}_N = \frac{1}{2} D_N \mathsf{E}_N D_N^T$, the computed smallest eigenvalues at $N = 4, 8, 12, 16, 20$ are respectively

$$N = 4: \quad 8.19930569 \times 10^{-22}, \qquad N = 8: \quad 2.84288630 \times 10^{-53},$$
$$N = 12: \quad 6.68407776 \times 10^{-87}, \qquad N = 16: \quad 3.22962551 \times 10^{-122},$$
$$N = 20: \quad 1.30149539 \times 10^{-158}.$$

  all positive in this finite computation; these are eigenvalues of $\widehat{E}_N$, not of E_N itself;
- the Christoffel pivot ratio $\kappa_N = p_N^E / p_N^O$ is numerically inside the interval $[4 - \lambda_1, 4]$ through the tested orders; it begins near 3.99650 and moves toward 4 over that finite range;
- zero leverage satisfies the exact identities of Section 11. At $N = 10$, the first five positive zeros carry about 49.6% of total leverage, the first ten about 67.3%, and the first two hundred about 92.7%;
- the arithmetic coefficients Z_k, computed through $k = 80$, are all positive in the tested range. Their observed range is $[0.3576, 1.5128]$; see Section 26.

The shift-index sequence is not monotone even at fixed matrix order. For $k \geq 0$, define the shifted even block

$$\mathsf{E}_N^{[k]} := \left(\lambda_{2k+m+n} - \lambda_{|m-n|}\right)_{m,n=1}^{N}. \tag{28.20}$$

Under RH it has the exact Gram representation

$$(\mathsf{E}_N^{[k]})_{mn} = 4 \sum_{\gamma > 0} \sin((k+m)\theta_\gamma) \sin((k+n)\theta_\gamma). \tag{28.21}$$

Whenever the block is invertible, set

$$\delta_k^{(N)} := \frac{1}{((\mathsf{E}_N^{[k]})^{-1})_{11}}, \qquad R_k^{(N)} := \lambda_{2k+2} - \delta_k^{(N)}. \tag{28.22}$$

Under RH, $\delta_k^{(N)}$ is the squared residual norm of the mode $k+1$ after projection onto the modes $k+2, \ldots, k+N$. Therefore, for each fixed k,

$$\delta_k^{(N+1)} \leq \delta_k^{(N)}, \qquad R_k^{(N+1)} \geq R_k^{(N)}, \qquad 0 \leq R_k^{(N)} \leq \lambda_{2k+2}. \tag{28.23}$$

These finite projection inequalities do *not* by themselves identify the limit with λ_{2k+2}; that equality would require the corresponding tail-completeness statement.

By contrast, monotonicity in the shift k is false even at $N = 2$. Here

$$\delta_k^{(2)} = \lambda_{2k+2} - \frac{(\lambda_{2k+3} - \lambda_1)^2}{\lambda_{2k+4}}. \tag{28.24}$$

Arithmetic reconstruction of the Li coefficients gives

k	$\delta_k^{(2)}$	$\delta_k^{(2)} - \delta_{k-1}^{(2)}$
40	0.02985633140654687	$+5.782 \times 10^{-4}$
41	0.03028625109019011	$+4.299 \times 10^{-4}$
42	0.03056677743905413	$+2.805 \times 10^{-4}$
43	0.03069982127912338	$+1.330 \times 10^{-4}$
44	0.03069030817860707	-9.513×10^{-6}
45	0.03054607923982695	-1.442×10^{-4}
46	0.03027771991886325	-2.684×10^{-4}
47	0.02989832437134847	-3.794×10^{-4}
48	0.02942320442607452	-4.751×10^{-4}

Thus the finite sequence rises through $k = 43$ and decreases from $k = 44$ onward. No proof route in this paper uses monotonicity in the shift index.

29. What would prove RH in the present framework

For clarity, the proof obligations supplied by the preceding sections are listed below; in the local line, V_n, d_n are the even-family quantities (boundary parameter $\sigma = 0$):

$$\text{local route:} \qquad V_n - V_{n-1} > \frac{1}{2}(d_n - d_{n-1})^2 \quad \forall n, \tag{29.1}$$

$$\text{Schur/inverse route:} \qquad \lambda_{2N} > b_N^T \mathsf{E}_{N-1}^{-1} b_N \quad \forall N \geq 2, \tag{29.2}$$

$$\text{adaptive prime route:} \qquad y_N^T(\mathsf{S}_N + \widehat{\mathsf{Z}}_N^E)y_N > a\|y_N\|^2 \quad \forall N \geq 2. \tag{29.3}$$

Each one, proved under the inductive positivity of the preceding even block, implies all $p_N^E > 0$, hence all $\mathsf{E}_N \succ 0$, hence RH by Theorem 5.5. The odd family does not occur in this implication chain.

The de Bruijn–Newman filter of Section 24 further constrains the form of such a proof. Reflection, displacement, Schur, and inverse identities alone cannot provide the sign, because the same algebra survives negative heat time while positivity fails. Within the present coordinate system, the zeta-specific ingredient that does not survive is the Euler-product/prime representation (26.53). Hence the natural remaining task is to combine the adaptive inverse direction y_N with prime cancellation strongly enough to prove (29.3).

30. Conclusion

The input kernel is Lagarias's published identity

$$\langle G_n, G_m \rangle_W = \lambda_n + \lambda_{-m} - \lambda_{n-m},$$

taken from Lagarias [25, Section 3, Theorem 3.1]; see the exact source ledger in Remark 1.1. The matrices E_N and O_N are obtained from that kernel by the explicit antisymmetric basis changes recorded in Remark 1.1. The finite reflection transformation and the identities derived from it are the subject of this paper. The even matrix alone is sufficient for RH:

$$\text{RH} \iff \mathsf{E}_N \succeq 0 \ \forall N \iff \mathsf{E}_N \succ 0 \ \forall N.$$

The odd family remains structurally useful but is not required in the proof chain.

For E_N, the algebraic part of the matrix route is now closed to a single sign step. Nesting, rank-two displacement, the column wave equation, and the closed two-generator scheme give an exact Schur recurrence; the local LDL^T identity reduces the next pivot to a 2×2 cell. Independently, the prime–archimedean decomposition converts the same positivity question into a signed prime-number-theorem discrepancy against explicit reflection-selected tests.

The inverse route identifies a sharper directional formulation. The inverse retains displacement rank two, is updated by rank one under nesting, and is the coefficient matrix of the Christoffel–Darboux kernel. Its last column selects the Schur residual polynomial and therefore the unique new direction that needs to be controlled. In first-difference coordinates this produces the adaptive test y_N and the sufficient inequality

$$y_N^T(\mathsf{S}_N + \widehat{\mathsf{Z}}_N^E)y_N > a\|y_N\|^2.$$

Proving that inequality for every N, or equivalently proving every even Schur pivot positive, would prove RH.

The de Bruijn–Newman deformation shows why the missing mechanism cannot be purely structural. Negative heat time preserves the coefficient/reflection algebra while leaving the all-real-zero region. Therefore a successful proof must use information not invariant under that deformation. In the present paper that information is carried by the Euler-product/prime side of the prime–continuum discrepancy. This isolates the next analytic task without replacing it by a finite numerical observation.

Appendix A. Lateral questions

The following problems are retained because they may reveal additional structure or useful asymptotics, but they are not steps in the logical chain (29.1)–(29.3). In particular, failure to solve any of them does not block the local pivot route described in Section 27.

A.1. **Rate of the Christoffel factor.** By Theorem 17.4 the sequence F_N decreases to $\det(I - \frac{1}{2}\mathcal{D}) = \prod_\gamma 4\gamma^2/(4\gamma^2+1)$. Determine the rate. The numerical evidence of Experiment 17.8 suggests that each Jacobi level resolves approximately one further Cayley angle, which would give $F_N - \det(I - \frac{1}{2}\mathcal{D}) \asymp \sum_{j>N}(4\gamma_j^2+1)^{-1} \asymp N^{-1}\log N$ after the Riemann–von Mangoldt count. A proof would convert the band (17.8) into a quantitative statement and, by Corollary 17.5, into an effective test.

A.2. **Sharpness of the detection threshold.** Experiment 25.7 finds $N^*\theta_\gamma$ between approximately 0.48 and 0.56 over the tested range, with N^* increasing as β approaches $1/2$ in the reported data. Determine whether this product has a limit as $\beta \uparrow 1/2$ and $\gamma \to \infty$, and, if so, identify it. A natural guess is that such a limit is determined by the first zero of a limiting sine-polynomial extremal problem associated with (25.7).

A.3. **Prime–archimedean reflection inequality.** Establish the matrix inequality

$$\mathsf{S}_N + \widehat{\mathsf{Z}}_N^E \succeq \frac{\gamma_E + \log(2\pi) - 1}{2}\, I_N \qquad (N \geq 1) \tag{A.1}$$

directly from the prime–continuum representation (26.53). This is an alternative equivalent form of RH, not a prerequisite for the local route. The scalar estimate (26.69) remains an equivalent diagnostic, while Remark 26.14 records only the insufficiency of the standard RH pointwise estimate for the global Bessel norm.

A.4. **Critical sine-polynomial flatness.** Let $v^{(N)}$ be the normalized smallest-eigenvalue direction for the even reflection problem, with a consistent sign convention, and let $P_N = P_{v^{(N)}}$ be defined by (25.6). Determine the largest scale $r = r(N)$ for which the odd moments

$$\sum_{m=1}^{N} m^{2j+1} v_m^{(N)}, \qquad 0 \leq j < r, \tag{A.2}$$

are asymptotically negligible in a normalization intrinsic to the eigenvalue problem. Equivalently, determine the growth of the order of flatness of P_N at $\theta = 0$. The present computations show low-index concentration but do not establish saturation at the scale θ_1^{-1}; that scale remains a numerical comparison until larger-N moment data are available.

A.5. **Joint leverage profile.** For fixed N, Theorem 11.8 gives a summable high-zero tail. Determine the scaling limit of

$$\ell^{E,O}_{N,\gamma_j} \tag{A.3}$$

when $N \to \infty$ and $j = j(N)$ grows simultaneously. Determine whether the canonical Cauchy–Binet ensemble has a macroscopic selection profile in the zero index and whether its transition region is governed only by the Riemann–von Mangoldt density or also by finer zero statistics.

A.6. **Asymptotics of the smallest eigenvalue.** Determine the asymptotic behavior of

$$\sigma_{\min}(\mathsf{E}_N), \qquad \sigma_{\min}(\mathsf{O}_N) \tag{A.4}$$

under RH, and relate it to approximation of the zero-angle support by finite sine polynomials.

A.7. **Asymptotics of determinant ratios.** Determine asymptotics for

$$\frac{\Delta_N^E}{\Delta_{N-1}^E}, \qquad \frac{\Delta_N^O}{\Delta_{N-1}^O}. \tag{A.5}$$

These are residual Gram energies and squared leading Cholesky coefficients.

A.8. **Jacobi parameters.** Find direct formulas or asymptotics for

$$\{a_n^E, b_n^E\}_{n\geq 0}, \qquad \{a_n^O, b_n^O\}_{n\geq 0} \tag{A.6}$$

in terms of Li coefficients or classical zeta data, and quantify the Christoffel/Darboux relation induced by (8.9).

A.9. **Inverse-entry structure.** Determine whether the inverse entries possess asymptotic sign, decay, or localization patterns that are not forced by generic positive Gram matrices. The rank-at-most-two inverse displacement identity (7.8)–(7.9) provides a starting point.

A.10. **Automorphic reflection systems below the universal exponent.** Lagarias defines generalized Li coefficients for principal automorphic L-functions [25]. Determine which functional-equation symmetries are sufficient to reproduce the pair of reflection matrices, their sine/half-sine Gram systems, and the Christoffel relation between the two associated measures. Once such a positive Cayley reflection system is available under GRH, Corollary 12.7 fixes its leading determinant coefficient at -2; the genuinely automorphic information must therefore be sought in lower-order terms, leverage profiles, or other matrix invariants.

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